

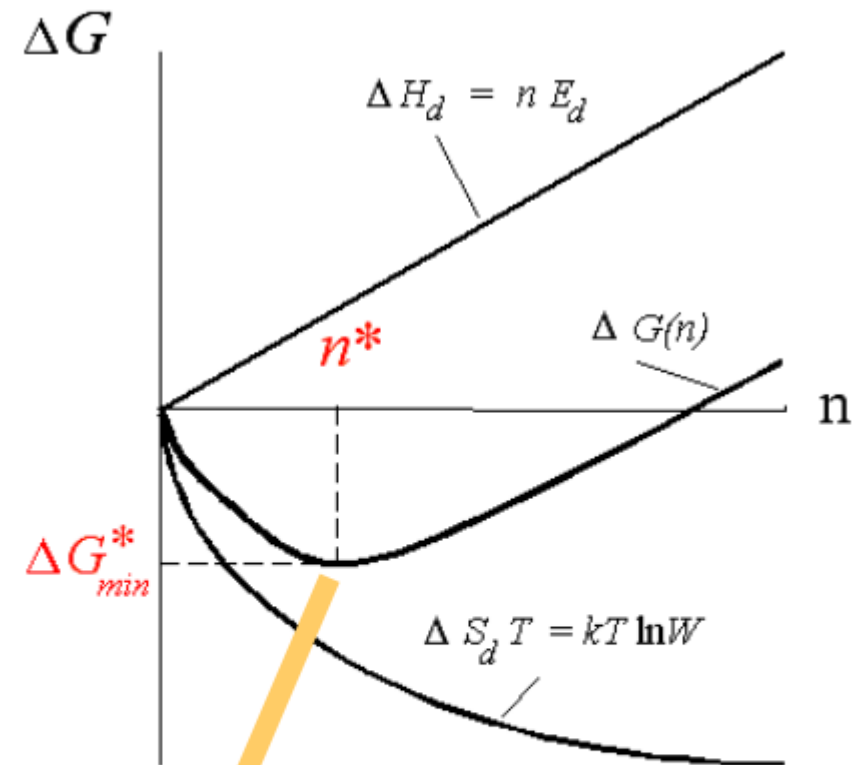
Imperfections in Solids

A certain point defect content increases the entropy and, hence, decreases the Gibbs potential !

$$\Delta G = \Delta H_d - \Delta S_d T$$

$\Delta H_d = n E_d$ - defect enthalpy (n - number of defects)
 $\Delta S_d = k \ln W$ - configurational entropy, $W = N! / n!(N-n)!$

$$n^* = N \exp\left(\frac{-E_d}{kT}\right) \quad N \gg n^*$$



$$\frac{\partial \Delta G}{\partial n} = E_d - kT \ln \frac{N - n^*}{n^*} = 0$$

Imperfections in Solids

Generally, an idealized solid (perfect order exists throughout crystalline material) i.e., real perfect crystal does not exist; all contain large numbers of various **defects or imperfections**. Many of the important properties of materials are due to the presence of these imperfections.

Defects may be classified into four categories

➤ **Point defects:**

atoms missing or in irregular places in the lattice (vacancies, interstitials, impurities)

➤ **Linear defects**

groups of atoms in irregular positions (e.g. screw and edge dislocations)

➤ **Planar defects**

the interfaces between homogeneous regions of the material (grain boundaries, external surfaces).

➤ **Volume defects**

extended defects (pores, cracks)

Point Defects

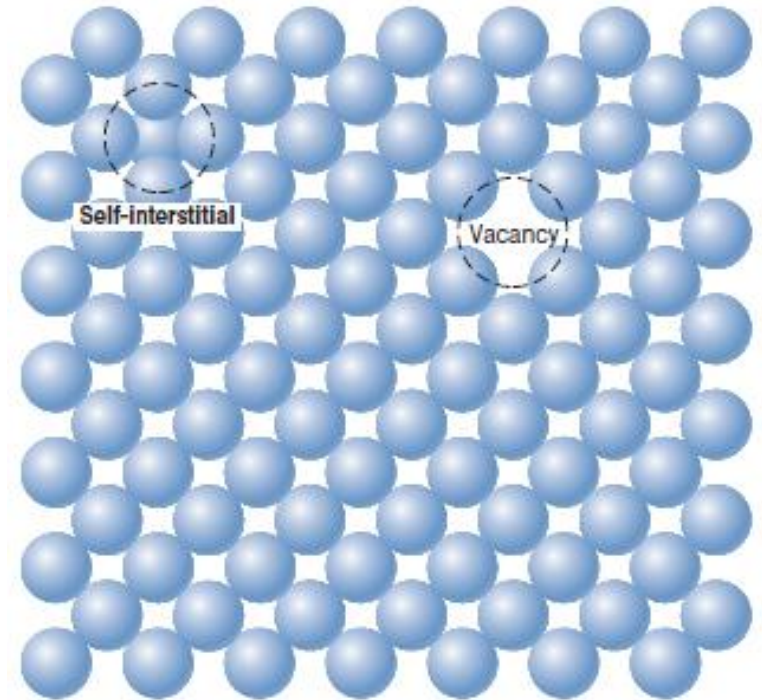
- **Vacancies**

vacant atomic sites in a structure.

- **Self-Interstitial atoms (SIA)**

"extra" atoms positioned between atomic sites.

Leads to distortion of planes



Point Defects - Equilibrium Concentration

- The *equilibrium number of vacancies* for a given quantity of material *depends on and increases with temperature* according to

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$$

N is the total number of atomic sites,

Q_v is the energy required for the formation of a vacancy,

T is the absolute temperature in kelvins, and

k is the gas or Boltzmann's constant, where k is 1.38×10^{-23} J/atom·K, or 8.62×10^{-5} eV/atom·K, depending on the units of Q_v

Point Defects - Equilibrium Concentration

EXAMPLE PROBLEM 4.1

Number-of-Vacancies Computation at a Specified Temperature

Calculate the equilibrium number of vacancies per cubic meter for copper at 1000°C. The energy for vacancy formation is 0.9 eV/atom; the atomic weight and density (at 1000°C) for copper are 63.5 g/mol and 8.4 g/cm³, respectively.

Solution

This problem may be solved by using Equation 4.1; it is first necessary, however, to determine the value of N , the number of atomic sites per cubic meter for copper, from its atomic weight A_{Cu} , its density ρ , and Avogadro's number N_A , according to

$$\begin{aligned} N &= \frac{N_A \rho}{A_{\text{Cu}}} \\ &= \frac{(6.022 \times 10^{23} \text{ atoms/mol})(8.4 \text{ g/cm}^3)(10^6 \text{ cm}^3/\text{m}^3)}{63.5 \text{ g/mol}} \\ &= 8.0 \times 10^{28} \text{ atoms/m}^3 \end{aligned} \quad (4.2)$$

Thus, the number of vacancies at 1000°C (1273 K) is equal to

$$\begin{aligned} N_v &= N \exp \left(-\frac{Q_v}{kT} \right) \\ &= (8.0 \times 10^{28} \text{ atoms/m}^3) \exp \left[-\frac{(0.9 \text{ eV})}{(8.62 \times 10^{-5} \text{ eV/K})(1273 \text{ K})} \right] \\ &= 2.2 \times 10^{25} \text{ vacancies/m}^3 \end{aligned}$$

Point Defects - Impurities in Solids

Most familiar metals are not highly pure;

rather, they are **alloys**, in which impurity atoms have been added intentionally to impart specific characteristics to the material.

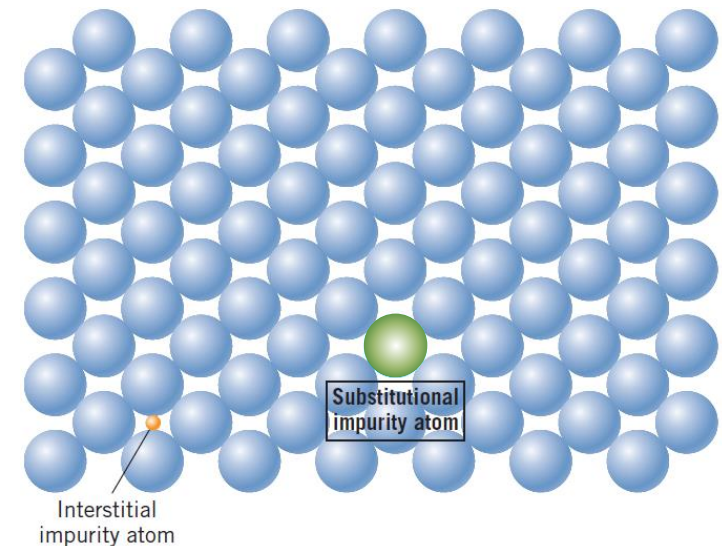
The addition of impurity atoms to a metal will result in the formation of a **solid solution** and/or a **new second phase**

Impurity point defects are found in **solid solutions**, of which there are two types:

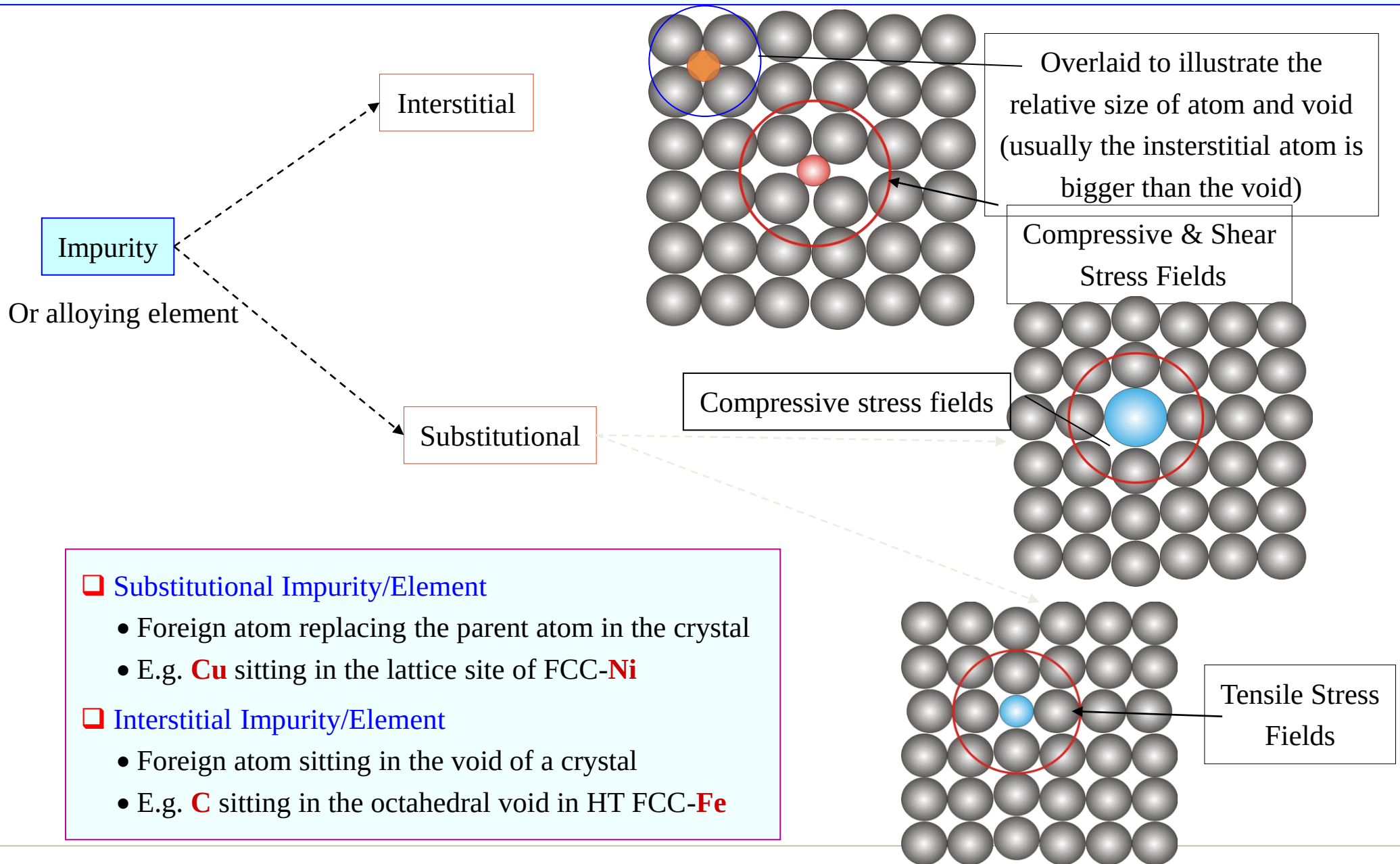
substitutional solid solutions: solute or impurity atoms replace or substitute for the host atoms. Eg. **Cu** in **Ni**

interstitial solid solutions: impurity atoms fill the voids or interstices among the host atoms. Eg. **C** in **Fe**

If the substitutional atom is smaller than the original atom then the lattice is in **tension**. If the substitutional atom is larger than the original atom then the lattice is in **compression**



- ❑ A 'foreign' element added (called as impurity/alloying element/dopant based on the context) can go to an interstitial site (between atoms) or may substitute for an atom of the host.



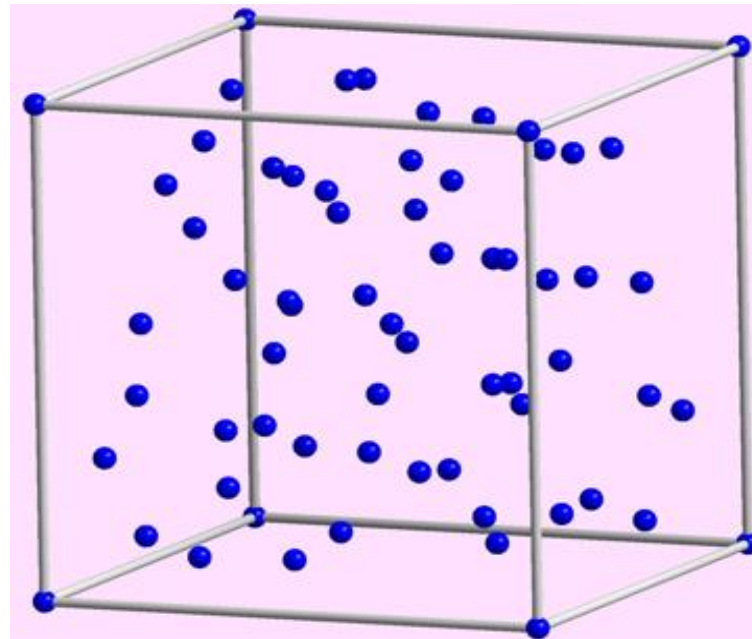
Interstitial C sitting in the octahedral void in HT FCC-Fe

- $r_{\text{Octahedral void}} / r_{\text{FCC atom}} = 0.414$
- $r_{\text{Fe-FCC}} = 1.29 \text{ \AA} \Rightarrow r_{\text{Octahedral void}} = 0.414 \times 1.29 = 0.53 \text{ \AA}$
- $r_{\text{C}} = 0.71 \text{ \AA}$
- $\Rightarrow$ **Compressive strains** around the C atom
- Solubility limited to **2 wt%** (9.3 at%)

Interstitial C sitting in the octahedral void in LT BCC-Fe

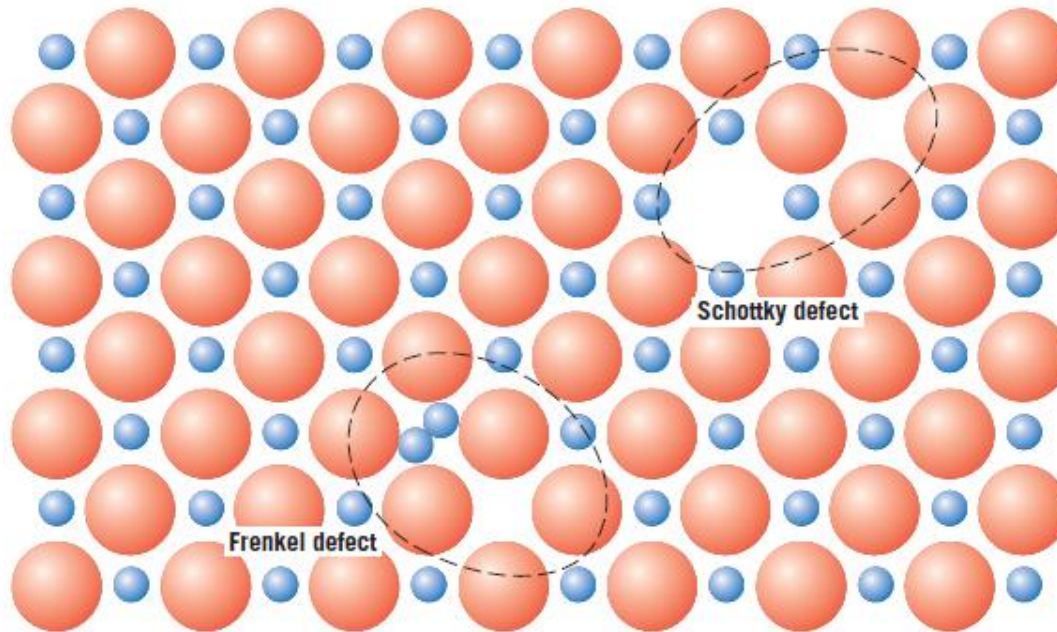
- $r_{\text{Tetrahedral void}} / r_{\text{BCC atom}} = 0.29 \bullet r_{\text{C}} = 0.71 \text{ \AA}$
- $r_{\text{Fe-BCC}} = 1.258 \text{ \AA} \Rightarrow r_{\text{Tetrahedral void}} = 0.29 \times 1.258 = 0.364 \text{ \AA}$
- ► But C sits in smaller octahedral void- displaces fewer atoms
- $\Rightarrow$ **Severe compressive strains** around the C atom
- Solubility limited to **0.008 wt%** (0.037 at%)

-
- ❑ In simple crystals with monoatomic decoration of lattices like SC, BCC or FCC, these two terms are equivalent (i.e. 'atomic site' $\equiv$ 'lattice site').
 - ❑ In a crystal where many atoms are associated with the lattice, any of the missing atoms can be considered a vacancy. E.g. in α -Mn (BCC lattice, cI58) with 58 atoms in a unit cell, any of the missing atoms can be considered a vacancy.



Point Defects – Ionic Solids

- atoms exist as charged ions, when defect structures are considered, conditions of **electroneutrality** must be maintained.



Schottky Defect

cation vacancy and
anion vacancy pair

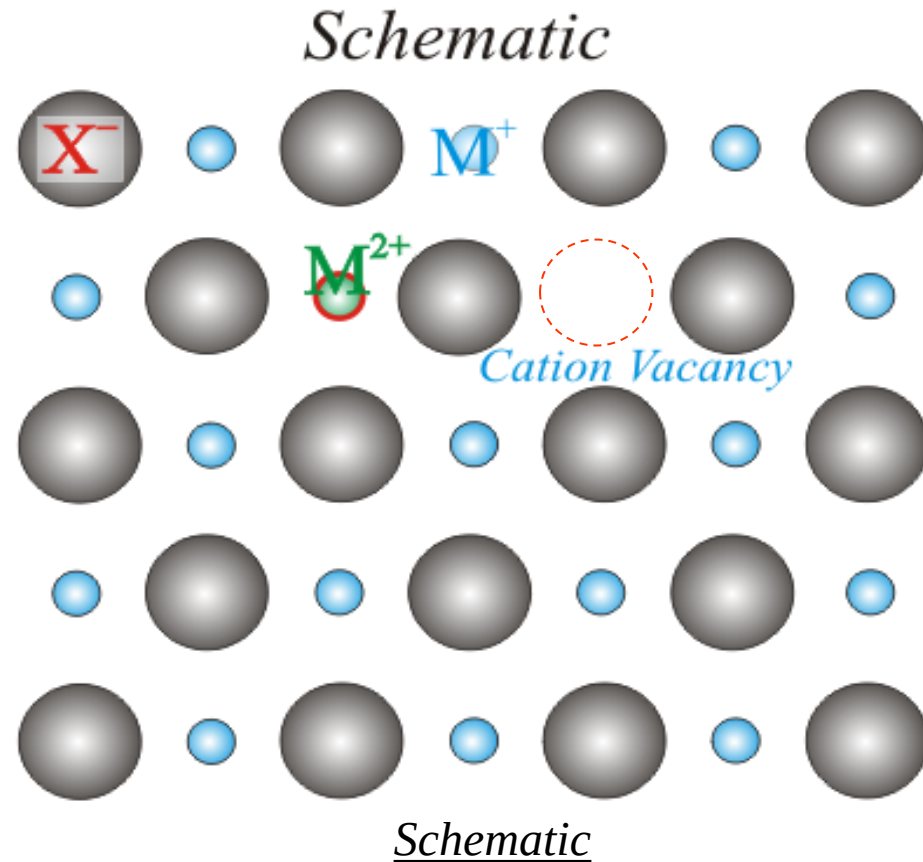
Frenkel Defect

cation–vacancy and
cation–interstitial pair

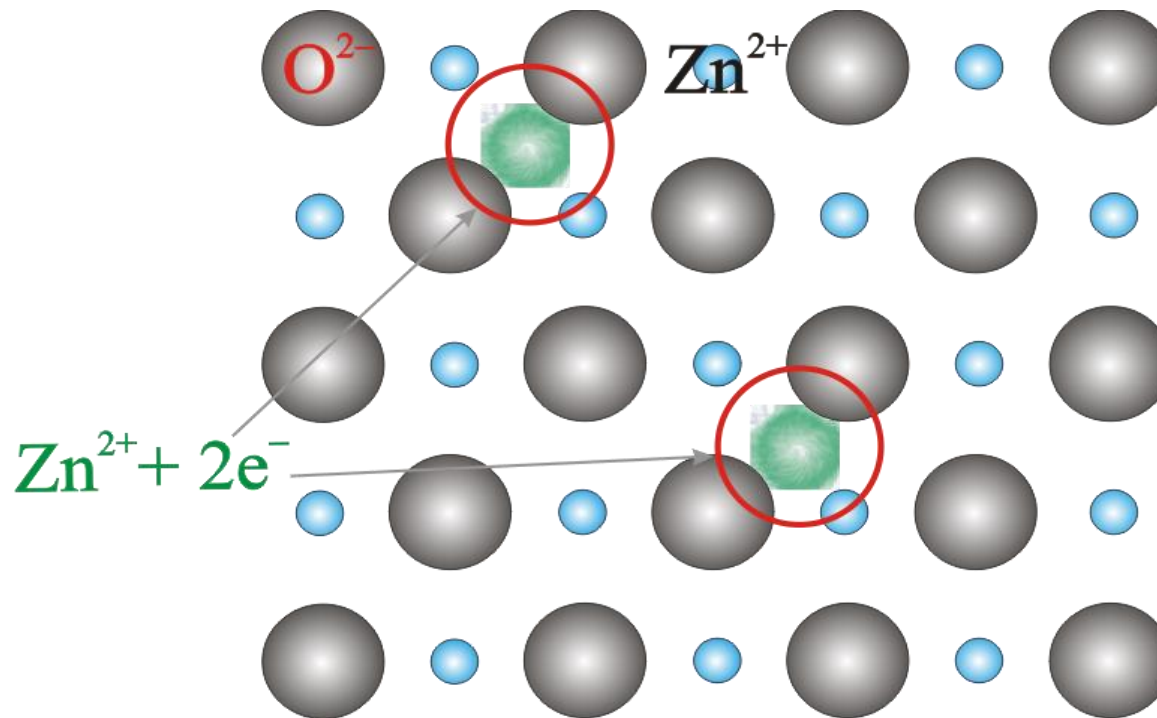
Electroneutrality is the state that exists when there are equal numbers of positive and negative charges from the ions

Other defects due to charge balance (/neutrality condition)

- If Cd^{2+} replaces Na^+ $\rightarrow$ one cation vacancy is created

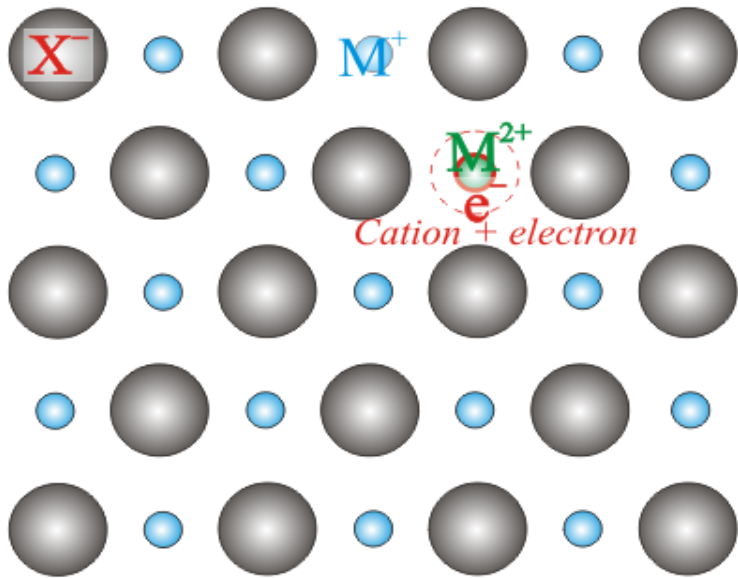


Defects due to off stoichiometry



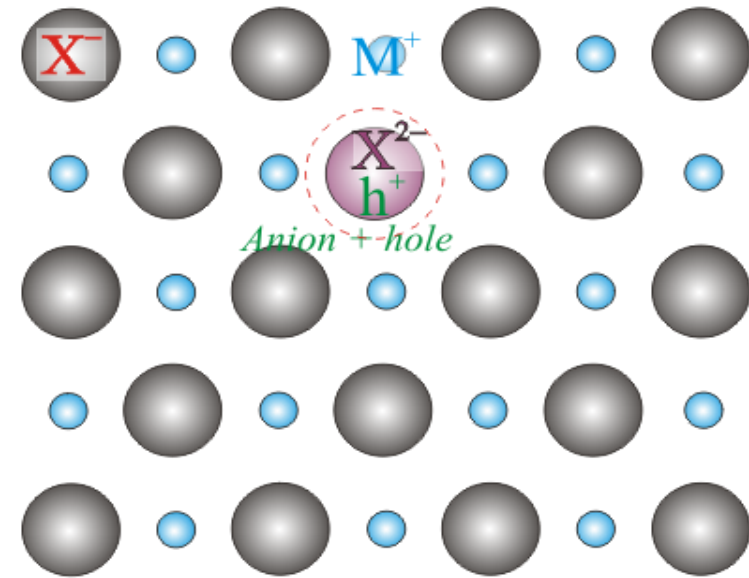
Other defect configurations: association of ions with electrons and holes

Schematic

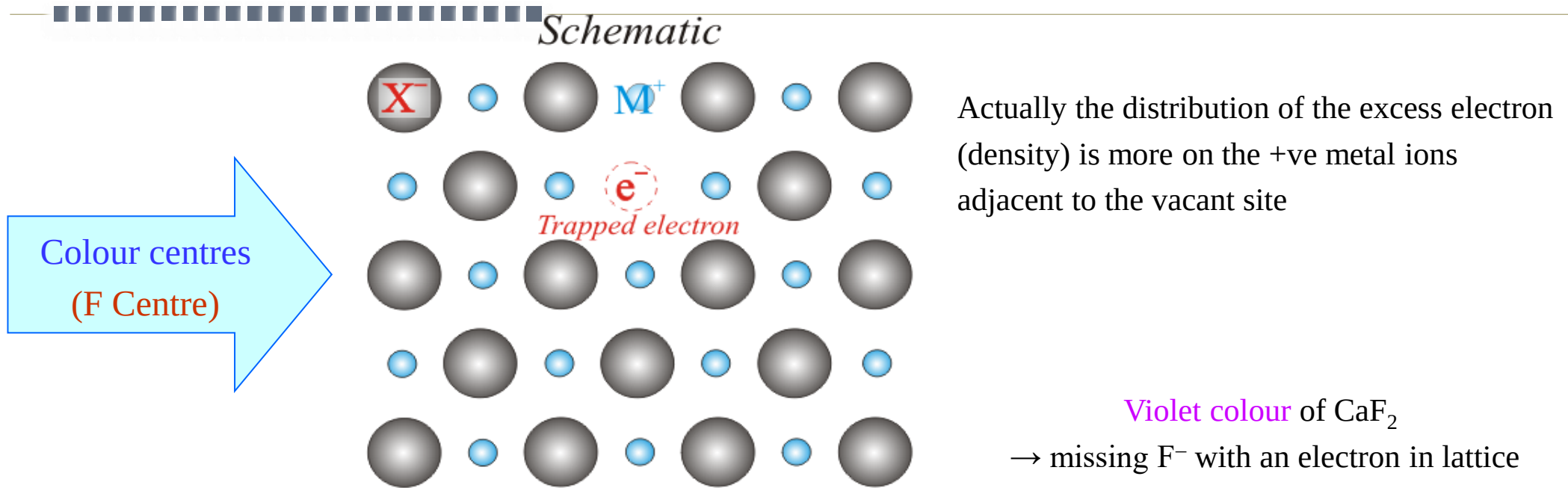


M^{2+} cation associated with an electron

Schematic



X^{2-} anion associated with a hole



Ionic Crystal	F centre absorption energy (eV)
LiCl	3.1
NaCl	2.7
KCl	2.2
CsCl	2.0
KBr	2.0
LiF	5.0

$$E = h\nu \quad E = h \frac{c}{\lambda}$$

$$E_{\text{KBr}} = 2 \text{ eV} = 2 \times (1.602 \times 10^{-19}) \text{ J}$$

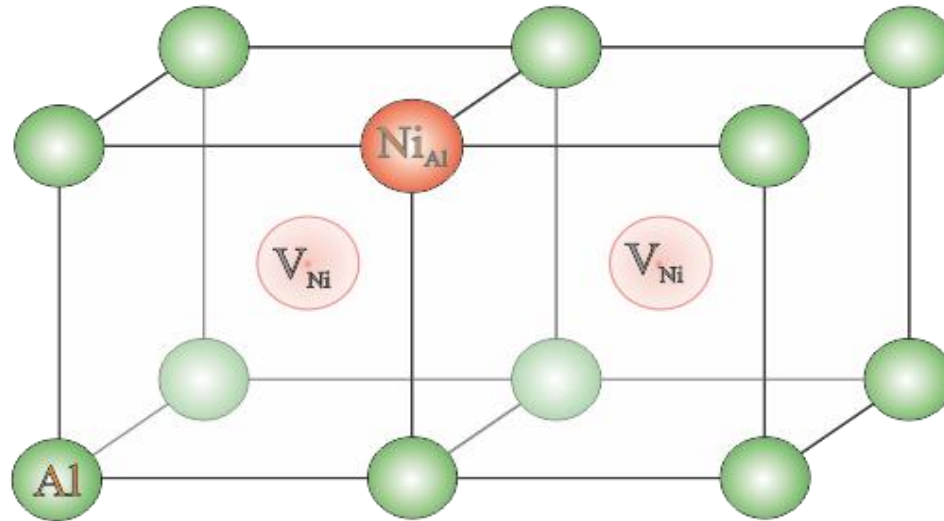
$$\lambda_{\text{KBr}}^{\text{absorption}} = \frac{(6.628 \times 10^{-34}) (3 \times 10^8)}{2 \times (1.602 \times 10^{-19})} = 6.2 \times 10^{-7} \text{ m} = 620 \text{ nm}$$

Visible spectrum: 390-750 nm

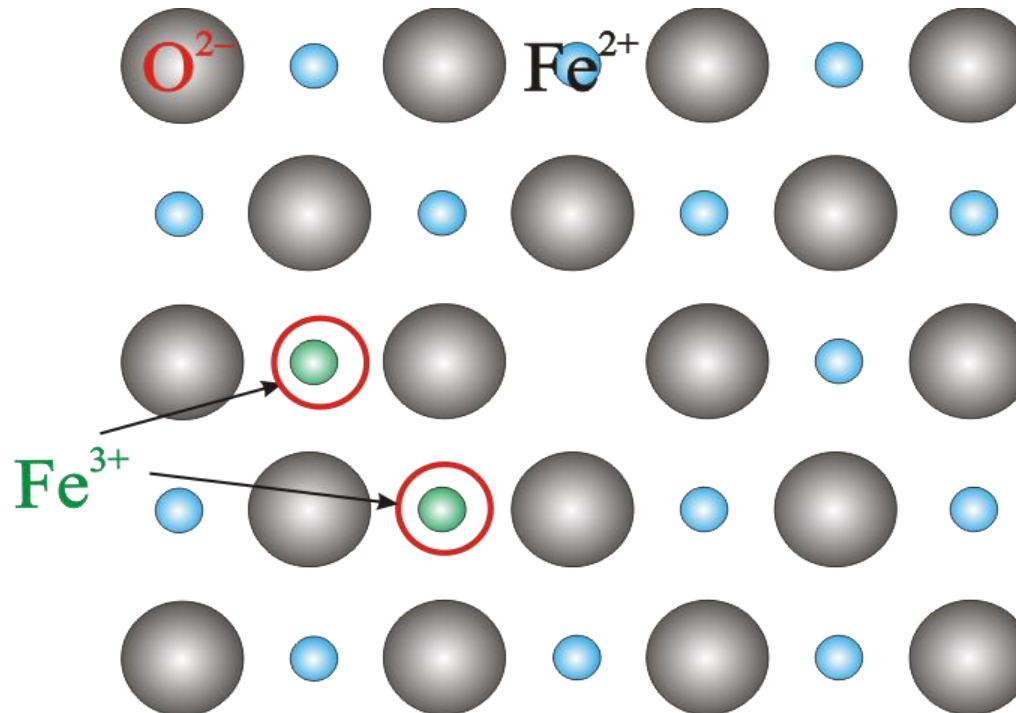


Vacancies in ordered structures

- ❑ With the example of NiAl, let us get a glimpse of some interesting possibilities which arise in ordered structures.
- ❑ Now let us consider thermal (equilibrium) vacancies in NiAl.



- FeO heated in oxygen atmosphere $\rightarrow \text{Fe}_x\text{O}$ ($x < 1$)
- Vacant cation sites are present
- Charge is compensated by conversion of ferrous to ferric ion:
$$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$$
- For every vacancy (of Fe cation) two ferrous ions are converted to ferric ions $\rightarrow$ provides the 2 electrons required by excess oxygen



Methods of producing point defects

❑ Growth and synthesis

Impurities may be added to the material during synthesis.

❑ Thermal & thermochemical treatments and other stimuli

- Heating to high temperature and quench
- Heating in reactive atmosphere
- Heating in vacuum → e.g. in oxides it may lead to loss of oxygen
- Etc.

❑ Plastic Deformation

❑ Ion implantation and irradiation

- Electron irradiation (typically $>1\text{MeV}$)
 - Direct momentum transfer or during relaxation of electronic excitations)
- Ion beam implantation (As, B etc.)
- Neutron irradiation.

What is the equilibrium concentration of vacancies at 800K in Cu

Data for Cu:

- Melting point = 1083 °C = 1356K
- ΔH_f (Cu vacancy) = 120×10^3 J/mole
- k (Boltzmann constant) = 1.38×10^{-23} J/K
- R (Gas constant) = 8.314 J/mole/K

First point we note is that we are below the melting point of Cu
 $800\text{K} \sim 0.59 T_m(\text{Cu})$

$$\frac{n}{N} = \exp\left(-\frac{\Delta H_f}{RT}\right)$$

$$\frac{n}{N} = \exp\left(-\frac{120 \times 10^3}{8.314 \times 800}\right) = \exp(-18.04) = 1.46 \times 10^{-8}$$

If we increase the temperature to 1350K (near MP of copper)

$$\frac{n}{N} = \exp\left(-\frac{120 \times 10^3}{8.314 \times 1350}\right) = \exp(-10.69) = 2.27 \times 10^{-5}$$

Point Defects - Specification of Composition

Express the composition of an alloy in terms of its constituent elements.

The two most common ways to specify composition are

weight (or mass) percent is the weight of a particular element relative to the total alloy weight.

For an alloy that contains two hypothetical atoms denoted by 1 and 2, the concentration of 1 in wt%, C_1 , is defined as

$$C_1 = \frac{m_1}{m_1 + m_2} \times 100$$

atom percent is the number of moles of an element in relation to the total moles of the elements in the alloy. Concentration in terms of atom percent of element 1 in an alloy containing element 1 and element 2 atoms, C_1 , is defined by

$$C_1 = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100$$

$$n_{m1} = \frac{m_1}{A_1}$$

mass (in grams)
atomic weight

Composition Conversions (for a two-element alloy)

- Conversion of weight percent to atom percent

$$C'_1 = \frac{C_1 A_2}{C_1 A_2 + C_2 A_1} \times 100 \qquad C'_2 = \frac{C_2 A_1}{C_1 A_2 + C_2 A_1} \times 100$$

- Conversion of atom percent to weight percent

$$C_1 = \frac{C'_1 A_1}{C'_1 A_1 + C'_2 A_2} \times 100 \qquad C_2 = \frac{C'_2 A_2}{C'_1 A_1 + C'_2 A_2} \times 100$$

$$C_1 + C_2 = 100$$

$$C'_1 + C'_2 = 100$$

Point Defects - Specification of Composition

EXAMPLE PROBLEM 4.3

Composition Conversion—From Weight Percent to Atom Percent

Determine the composition, in atom percent, of an alloy that consists of 97 wt% aluminum and 3 wt% copper.

Solution

If we denote the respective weight percent compositions as $C_{\text{Al}} = 97$ and $C_{\text{Cu}} = 3$, substitution into Equations 4.6a and 4.6b yields

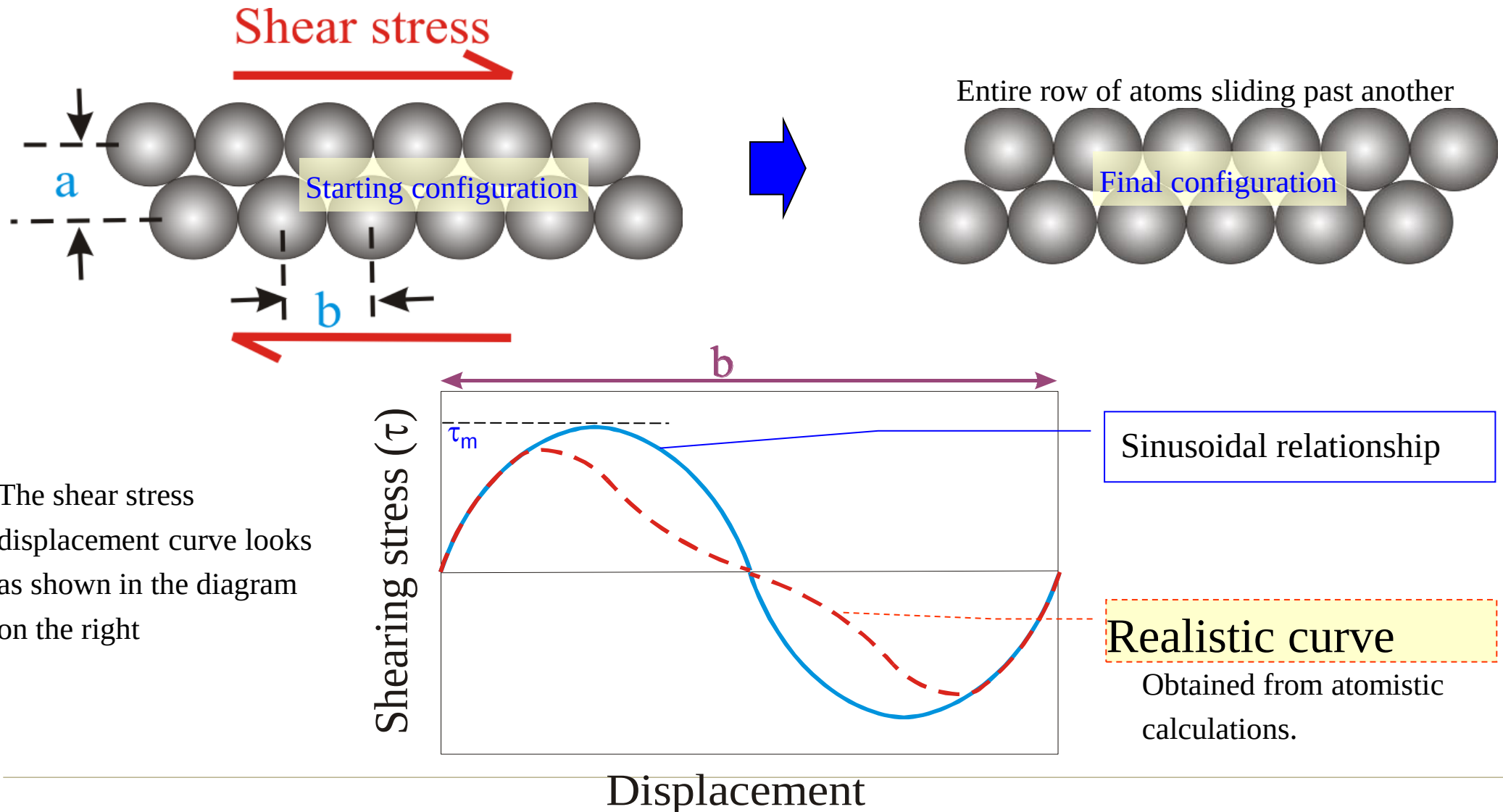
$$\begin{aligned} C'_{\text{Al}} &= \frac{C_{\text{Al}} A_{\text{Cu}}}{C_{\text{Al}} A_{\text{Cu}} + C_{\text{Cu}} A_{\text{Al}}} \times 100 \\ &= \frac{(97)(63.55 \text{ g/mol})}{(97)(63.55 \text{ g/mol}) + (3)(26.98 \text{ g/mol})} \times 100 \\ &= 98.7 \text{ at \%} \end{aligned}$$

and

$$\begin{aligned} C'_{\text{Cu}} &= \frac{C_{\text{Cu}} A_{\text{Al}}}{C_{\text{Cu}} A_{\text{Al}} + C_{\text{Al}} A_{\text{Cu}}} \times 100 \\ &= \frac{(3)(26.98 \text{ g/mol})}{(3)(26.98 \text{ g/mol}) + (97)(63.55 \text{ g/mol})} \times 100 \\ &= 1.30 \text{ at \%} \end{aligned}$$

Plastic deformation of a crystal by shear

- Let us consider the shearing of an entire plane of atoms over one another
→ causing plastic deformation by shear.
- If the atoms 'climb' only to the top of the peak (of another atom, corresponding to the peak in shear stress in the plot below), then it can 'slide' back to its original position and the deformation will be elastic.



$$\tau = \tau_m \sin\left(\frac{2\pi x}{b}\right)$$

$$\tau = G\gamma = G \frac{x}{a}$$

$$G \frac{x}{a} = \tau_m \left(\frac{2\pi x}{b}\right)$$

$$\tau = \tau_m \left(\frac{2\pi x}{b}\right)$$

$$\tau_m = \frac{G}{2\pi} \frac{b}{a}$$

If $b \sim a$

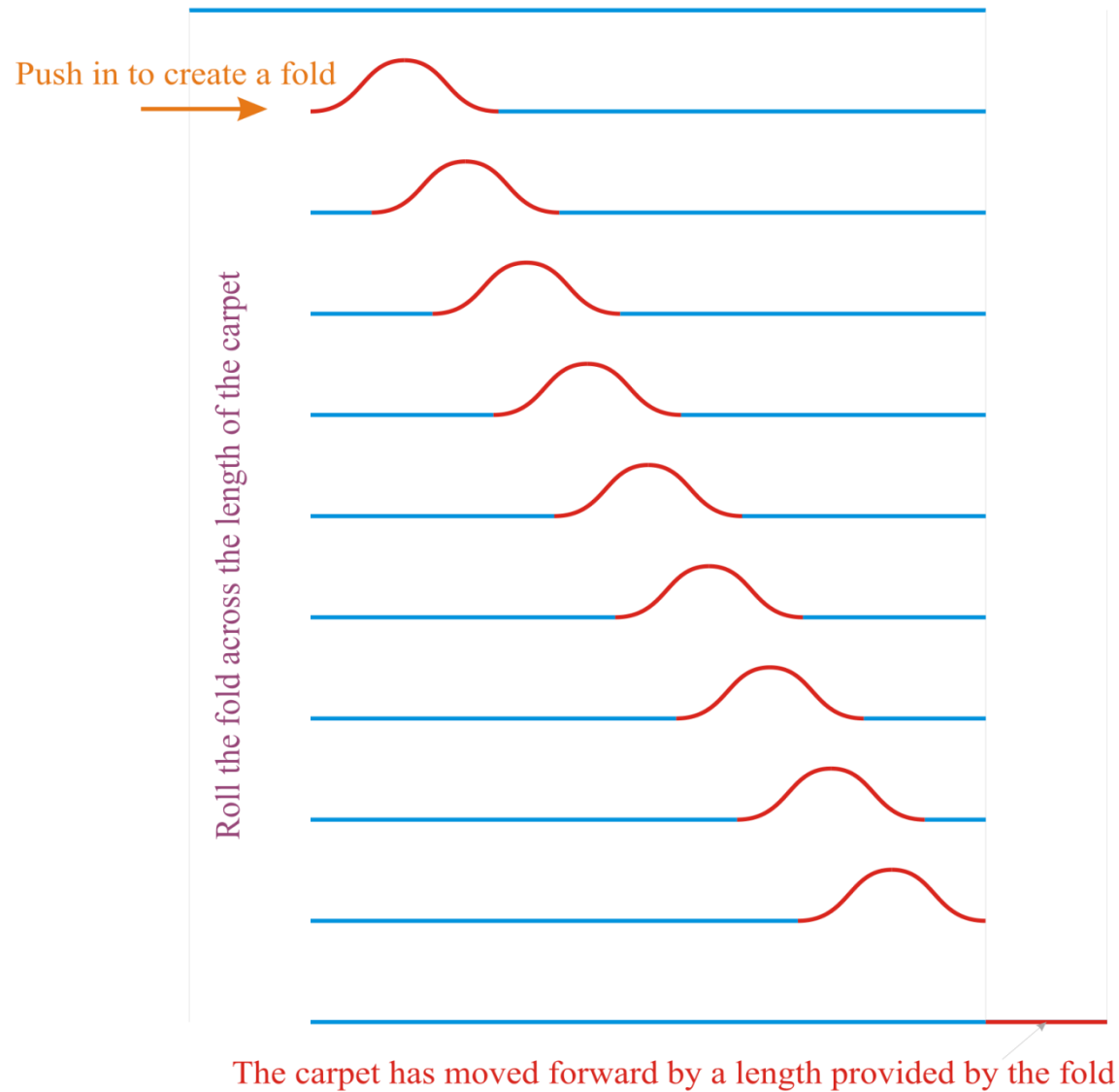
$$\tau_m \approx \frac{G}{2\pi}$$

$$\tau_m \approx \frac{G}{2\pi}$$

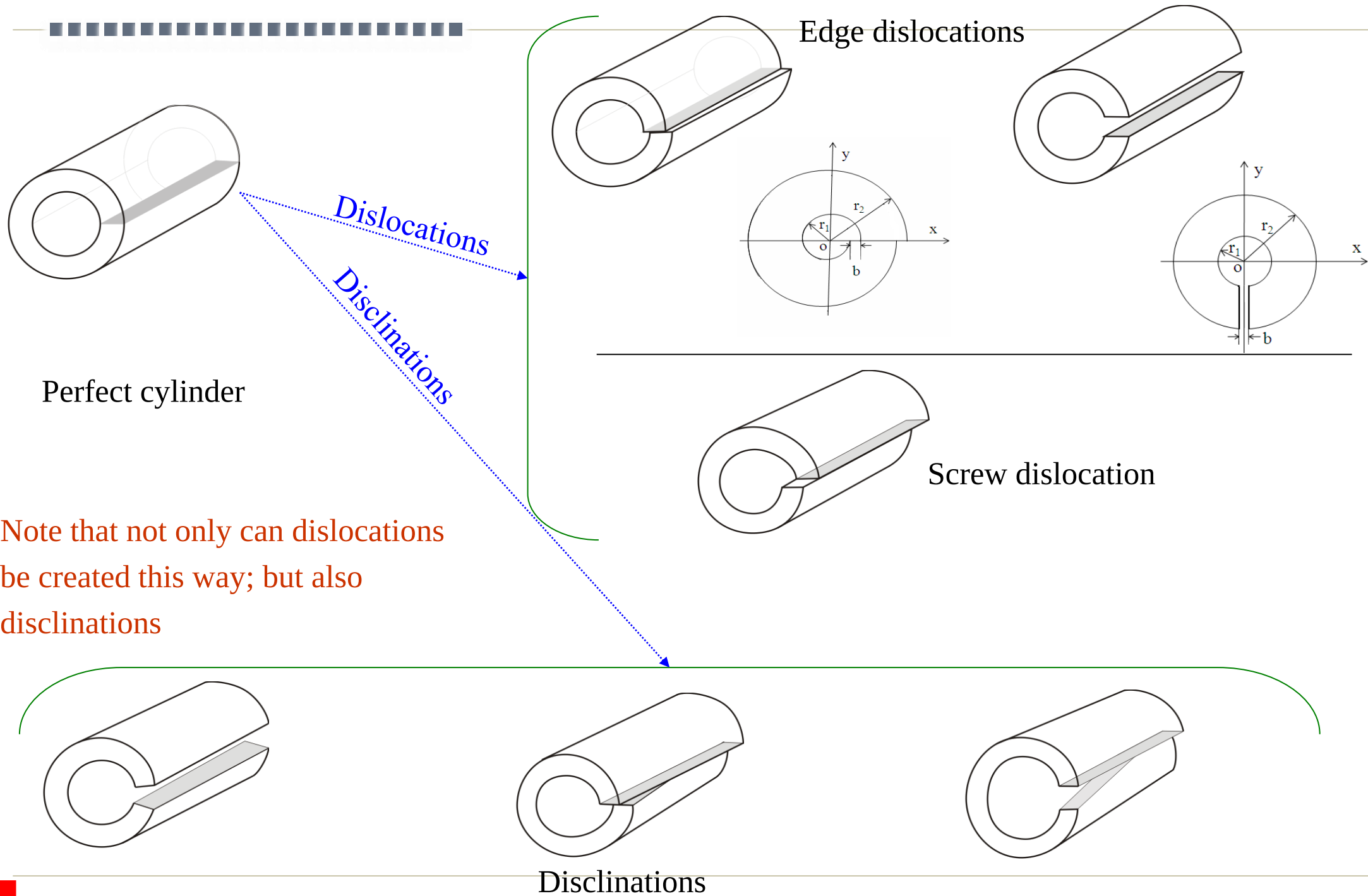
Material	Strength	
	Theoretical [MPa] $G/2\pi$	Experimental [MPa]
Al	4200	0.7-0.8
Cu	7700	0.5-3
Fe	13000	25-30
Al ₂ O ₃	20000	19000
Diamond	46000	21000

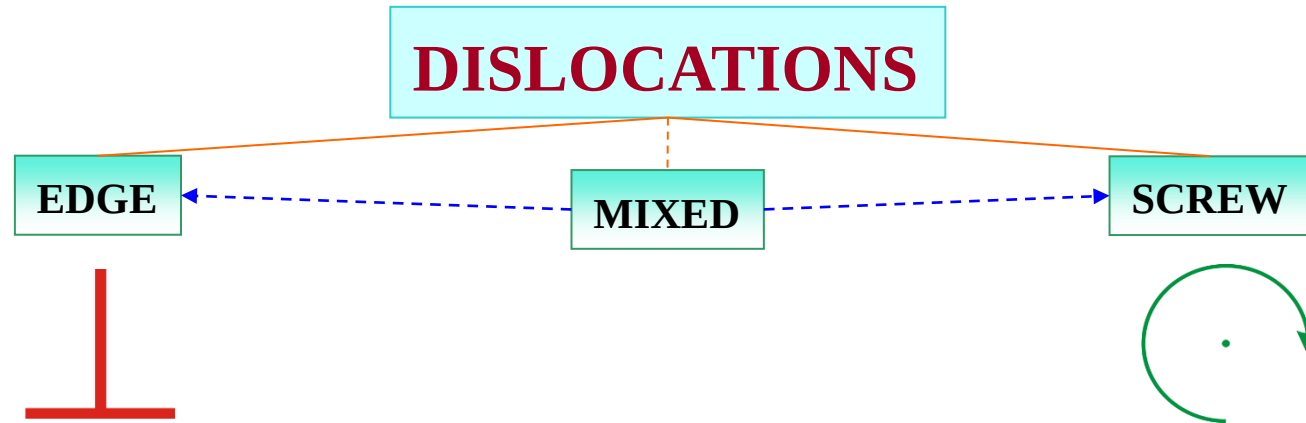
□ The theoretical shear stress will be in the range 3-30 GPa

The 'carpet-pulling' analogy



Deformations of a hollow cylinder showing the formation of various defects (Volterra constructions)

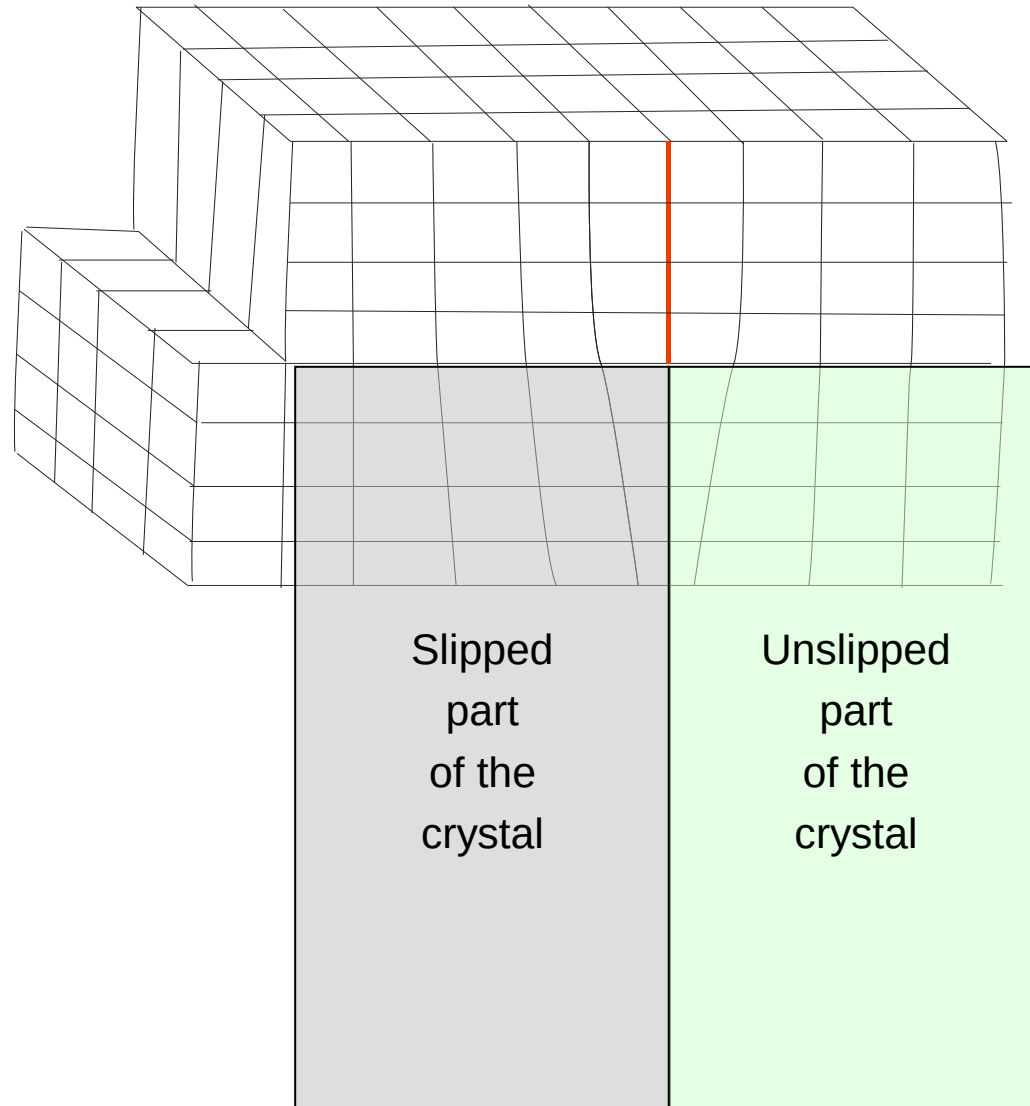




TEM micrograph
showing dislocation lines

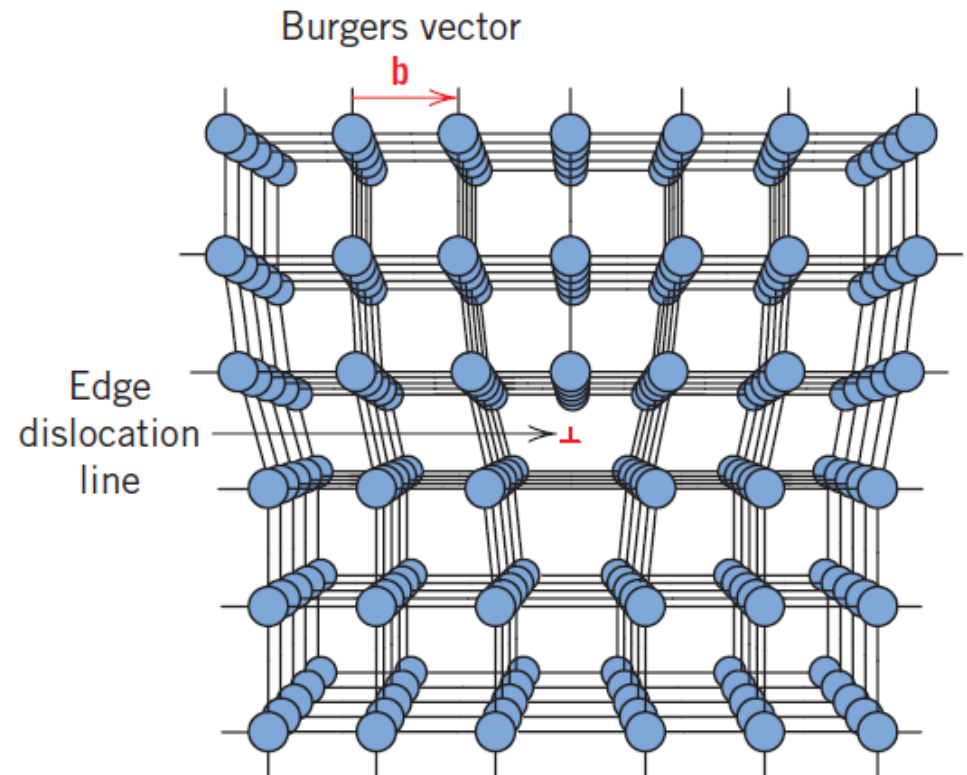


Dislocation lines

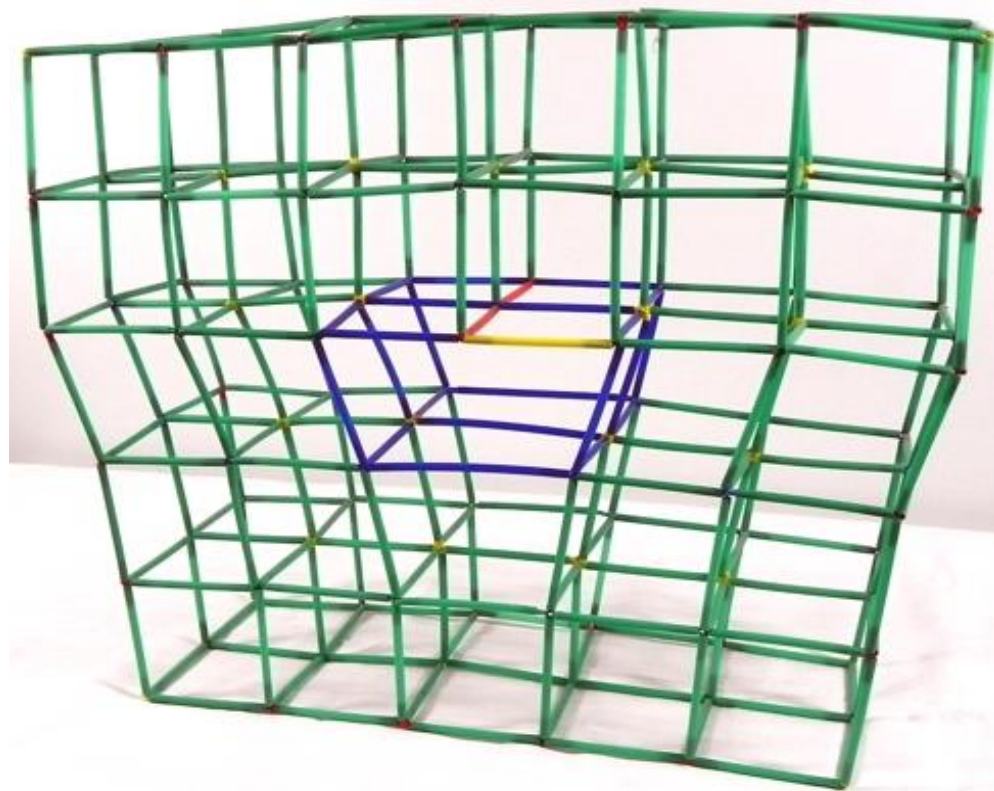
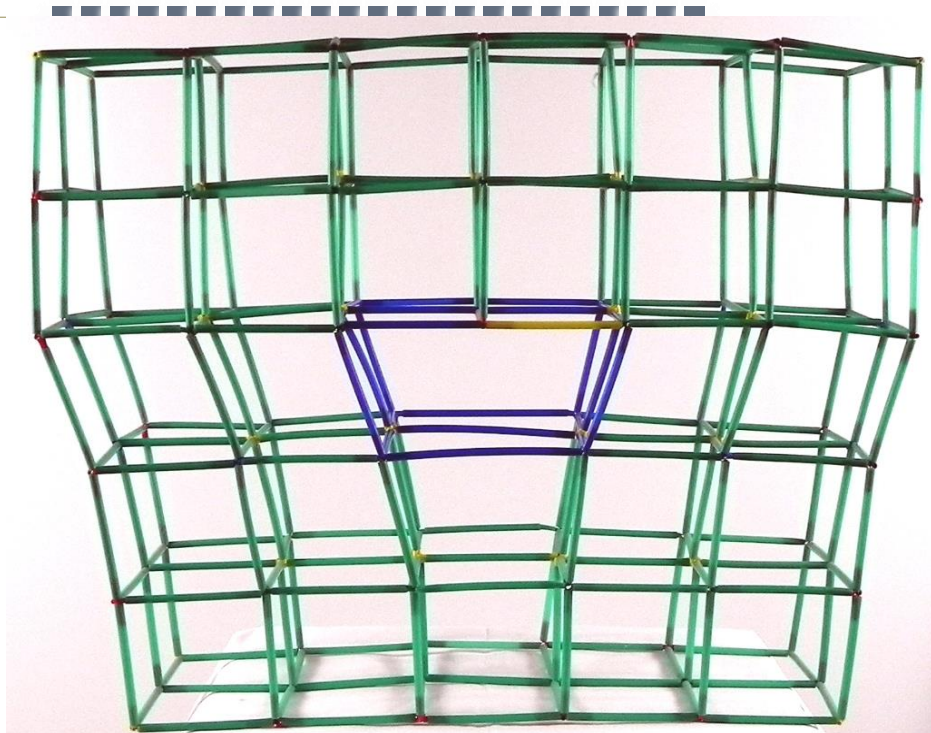


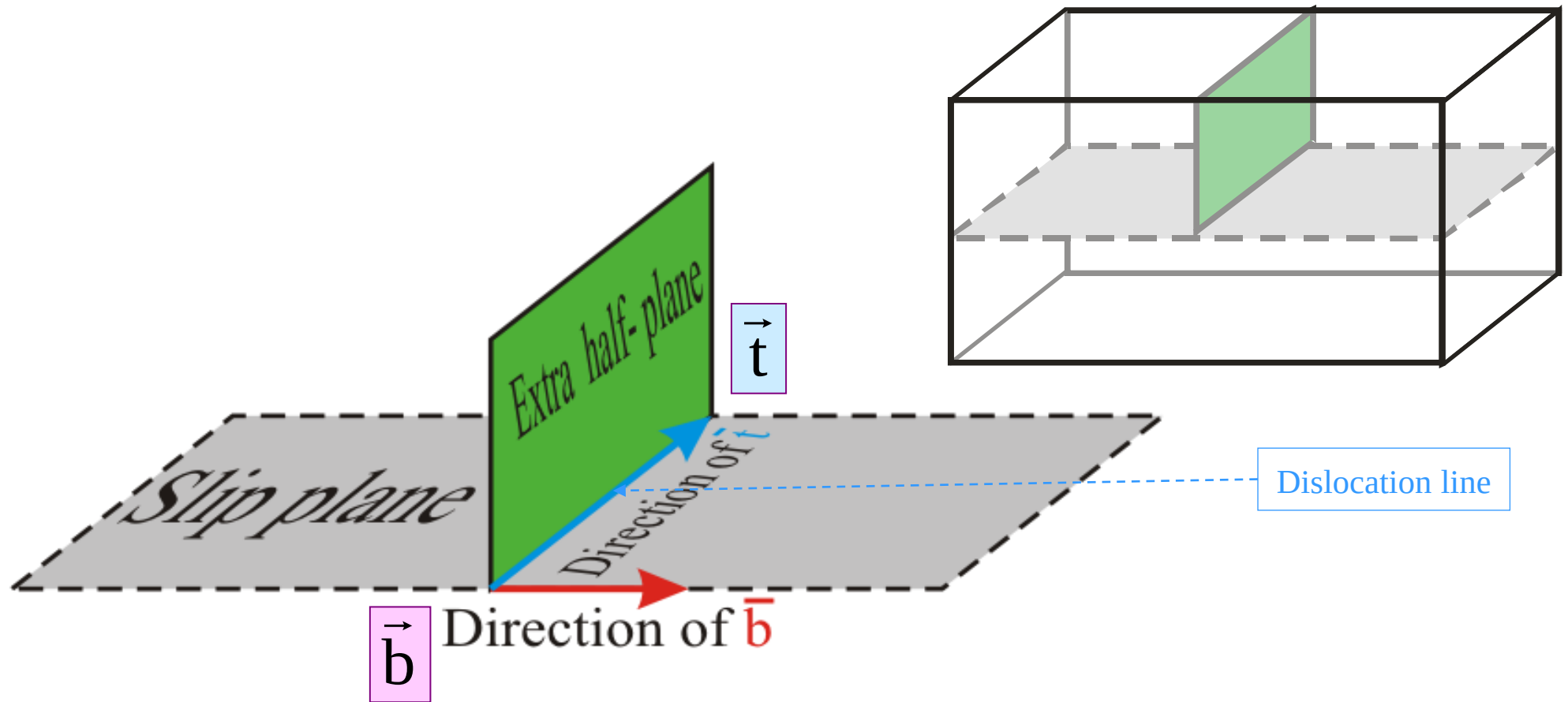
Linear Defects - Edge Dislocations

- A **dislocation** is a linear or one-dimensional defect around which some of the atoms are misaligned. characterized by a vector in the lattice called the **Burger's vector b** , a crystal vector, $[uvw]$, which is a measure of the distortion of the lattice around the defect.
- An extra portion of a plane of atoms, or half-plane, the edge of which terminates within the crystal. This is termed an **edge dislocation**.
- the line that is defined along the end of the extra half-plane of atoms is sometimes termed the **dislocation line**, which, for the edge dislocation in Figure, is perpendicular to the plane of the slide.



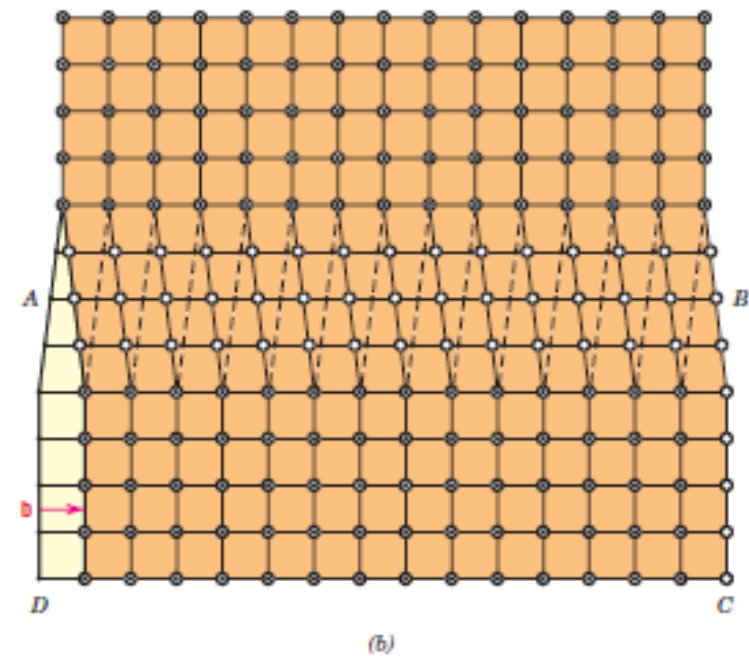
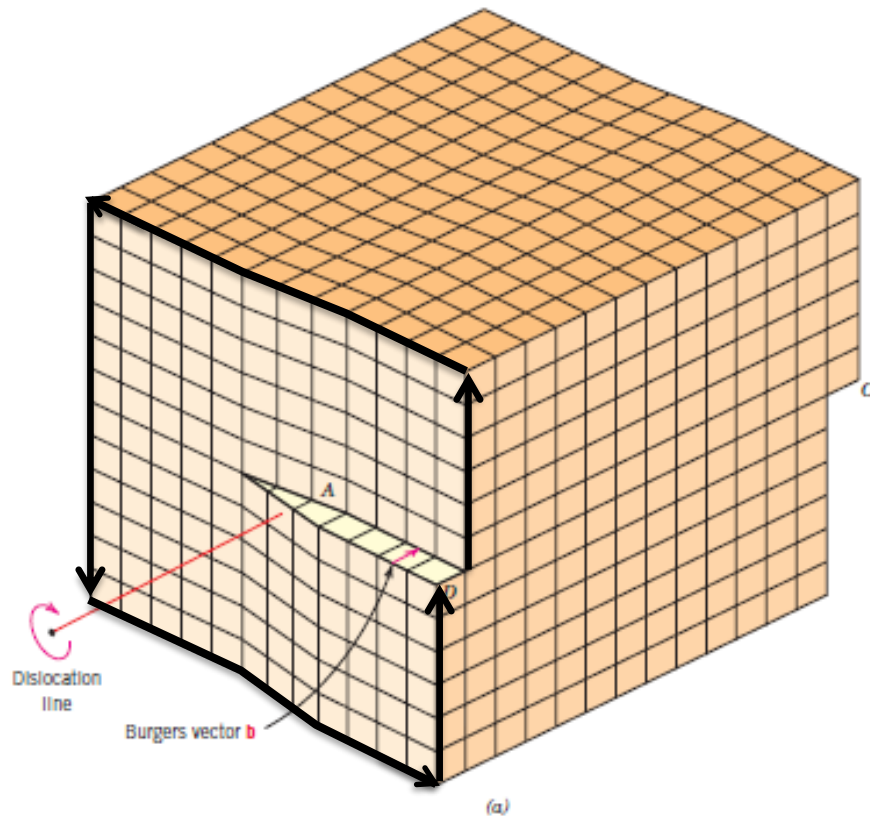
$b \perp$ to dislocation line





Linear Defects - Screw Dislocations

Screw Dislocation: The trace of the atomic planes around the screw dislocation makes a spiral or helical path (shade) like a screw and hence, the name.



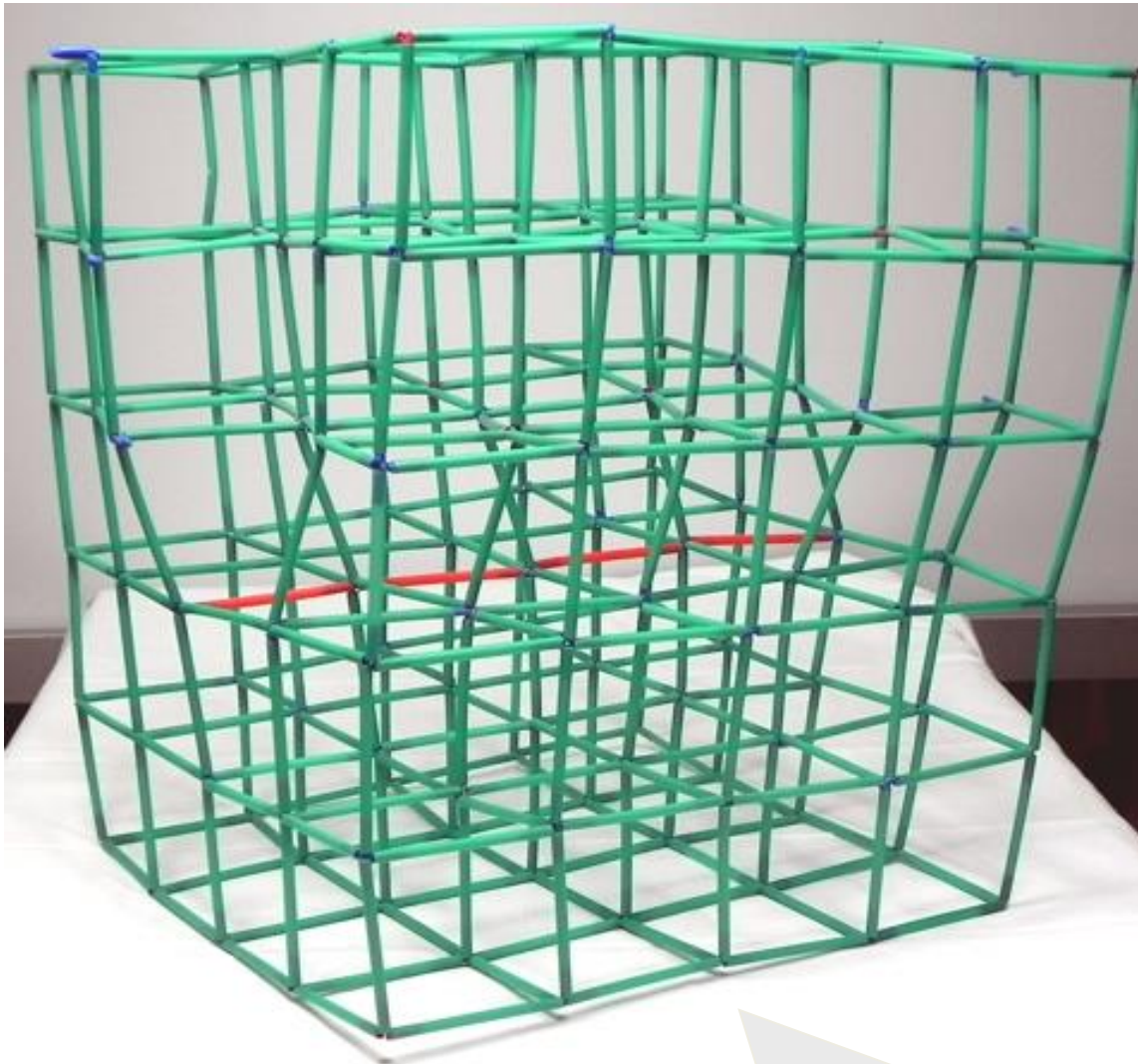
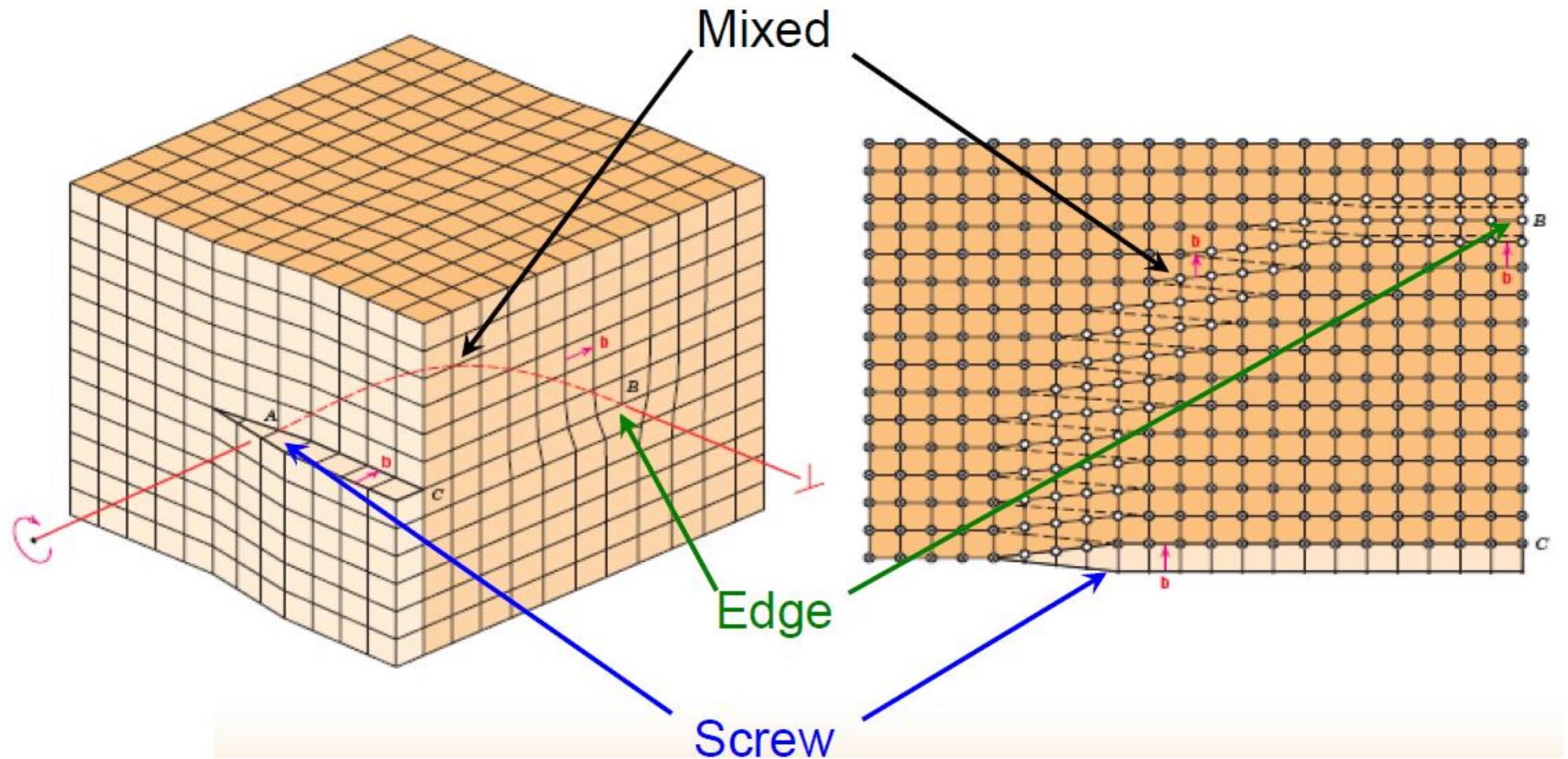


Figure 1.1: A 3D model of a crystal lattice structure, showing the arrangement of atoms in a solid material.

Linear Defects - Mixed Dislocations



Burger vector and dislocation line neither parallel, antiparallel, or perpendicular

Linear Defects - Dislocations



one-dimensional defects

- Edge dislocation

b $\perp$ to dislocation line

- Screw dislocation

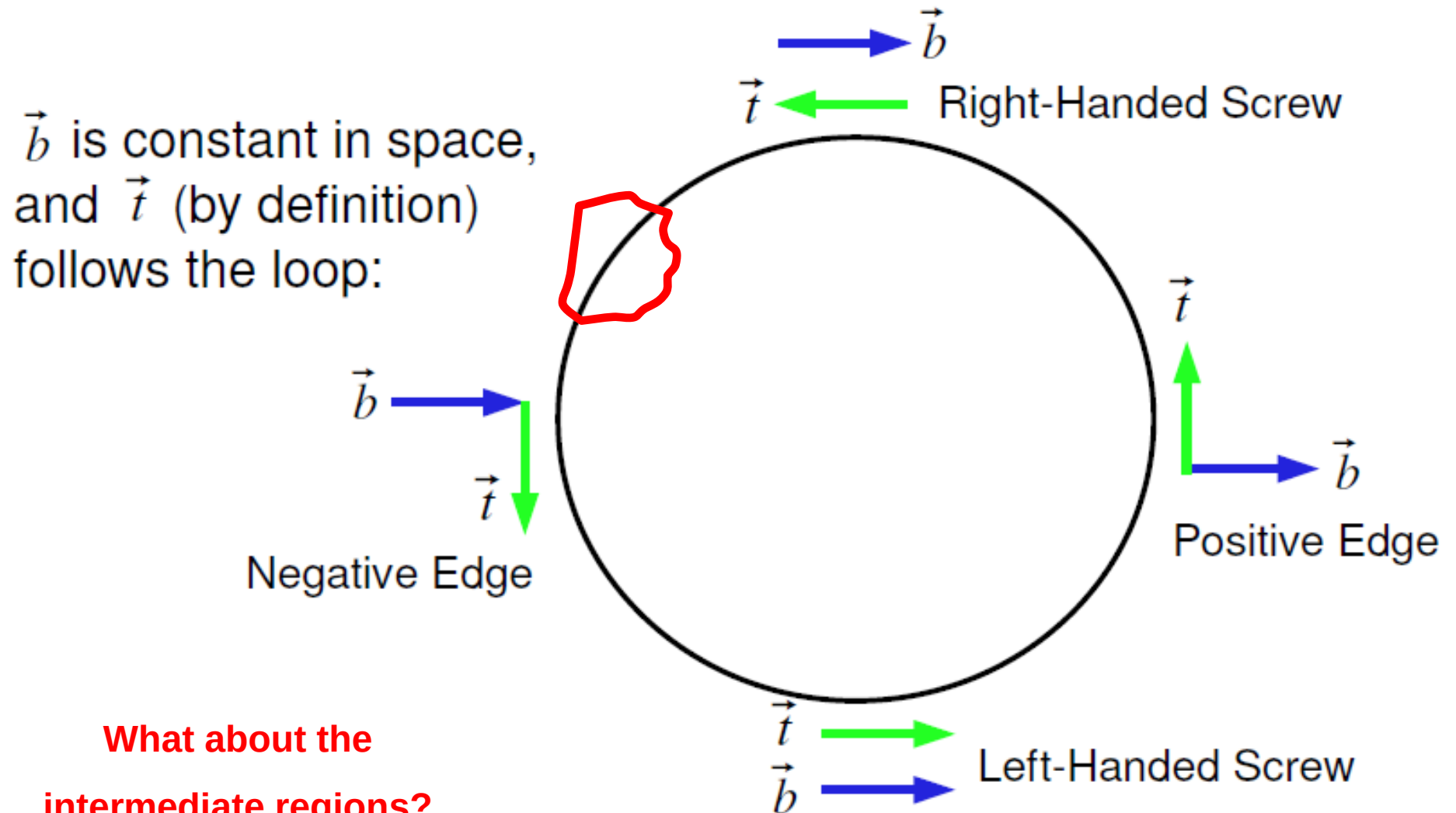
b $\parallel$ to dislocation line

- Mixed dislocation

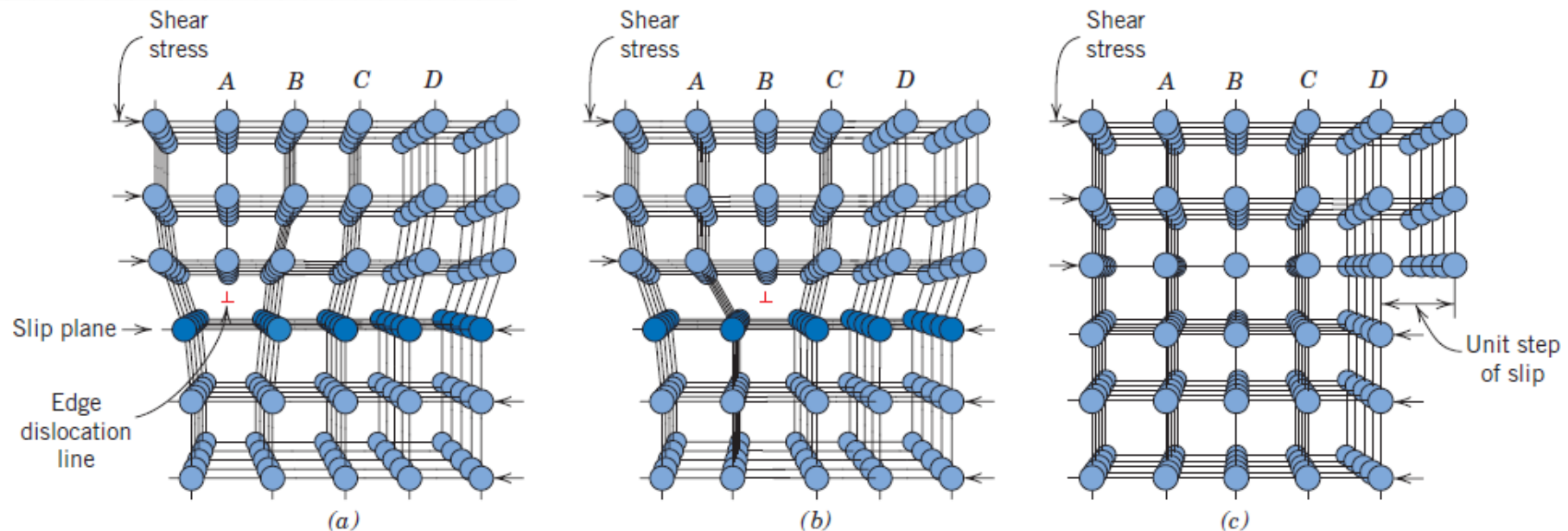
b $\perp$ to dislocation line

b $\parallel$ to dislocation line

Dislocation LOOP



motion of an edge dislocation - shear stress



Atomic rearrangements that accompany the motion of an edge dislocation as it moves in response to an applied shear stress.

- a) The extra half-plane of atoms is labeled A.
- b) The dislocation moves one atomic distance to the right as A links up to the lower portion of plane B; in the process, the upper portion of B becomes the extra half-plane.
- c) A step forms on the surface of the crystal as the extra half-plane exits.
- d) The process by which plastic deformation is produced by dislocation motion is termed **slip**; the crystallographic plane along which the dislocation line traverses is the **slip plane**

Macroscopic deformation

- The formation of a step on the surface of a crystal by the motion of

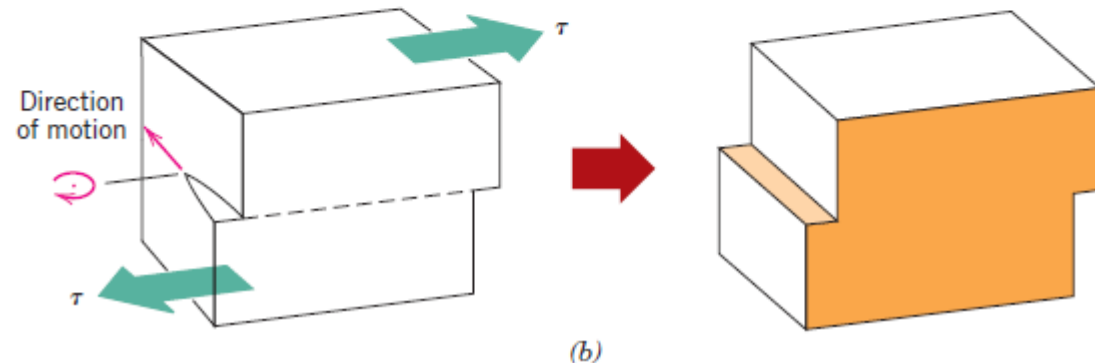
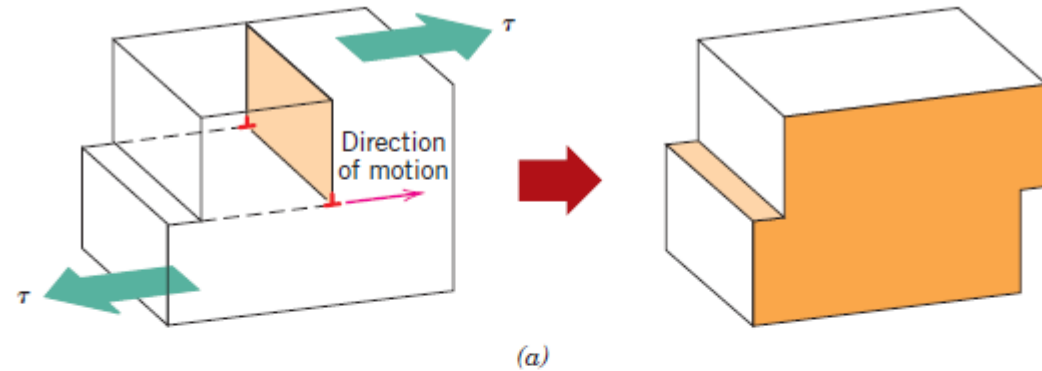
(a) an edge dislocation and

(b) a screw dislocation.

for an edge, the dislocation line moves in the direction of the applied shear stress;

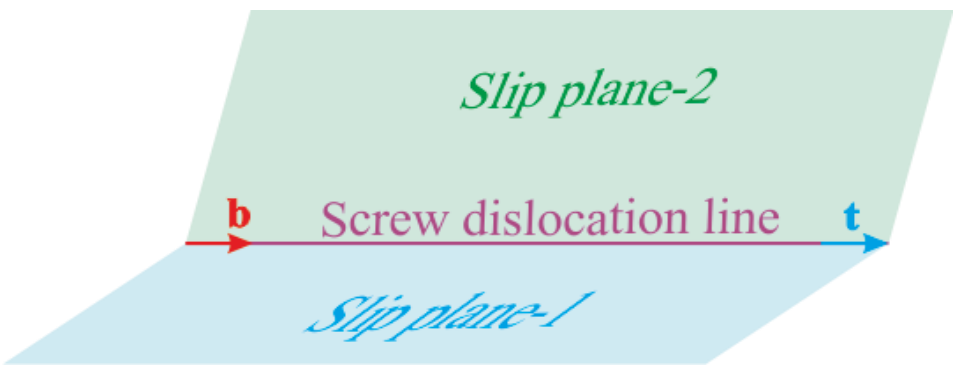
for a screw, the dislocation line motion is perpendicular to the stress direction.

However, the net plastic deformation for the motion of both dislocation types is the same



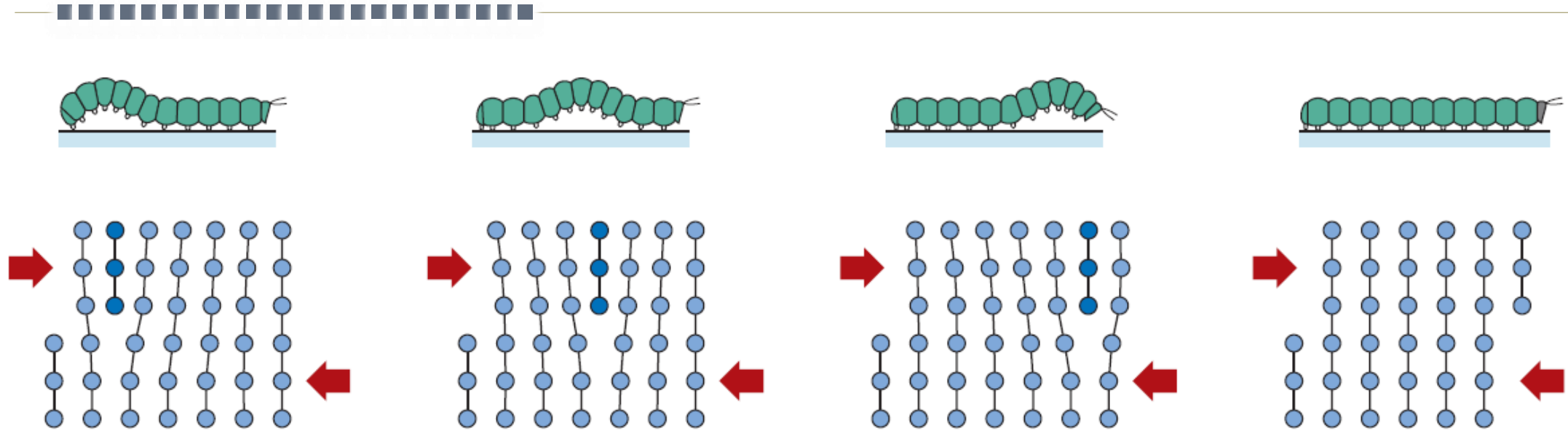
The **number of dislocations**, or **dislocation density** in a material, is expressed as the total dislocation length per unit volume or, equivalently, the number of dislocations that intersect a unit area of a random section

Geometric properties of dislocations



Dislocation Property	Type of dislocation	
	Edge	Screw
Relation between dislocation line (t) and b	⊥	
Slip direction** (& the ‘direction’ of step created when dislocation leaves the crystal)	to b	to b
Direction of dislocation line movement relative to b		⊥
Process by which dislocation may leave slip plane*	climb	Cross-slip

Dislocation motion



Dislocation motion is analogous to the mode of locomotion employed by a caterpillar

- Incrementally breaking bonds
- If dislocations don't move, deformation doesn't happen! (But fracture will, like in ceramic)

Dislocation leaving/changing the slip plane

Edge dislocation

Climb

Non-conservative*:
involves mass transport

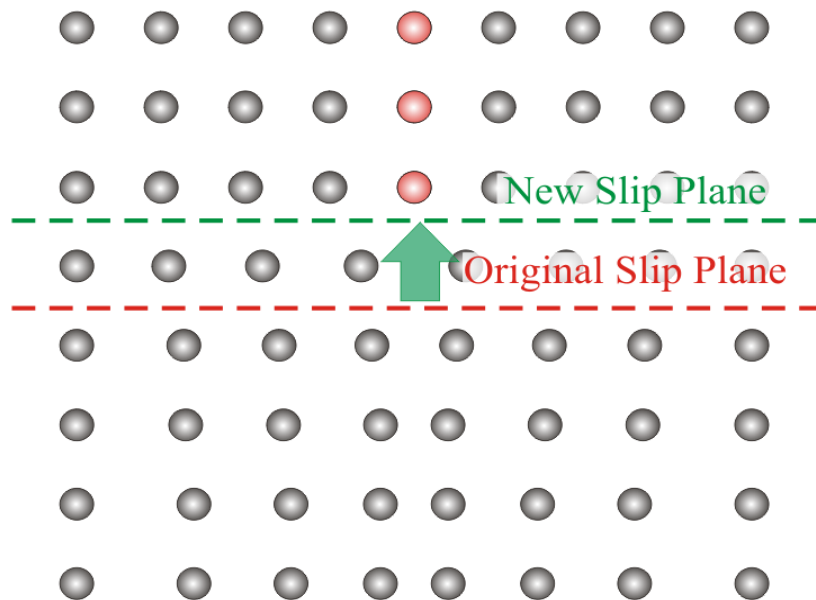
Screw dislocation

Cross Slip

Conservative

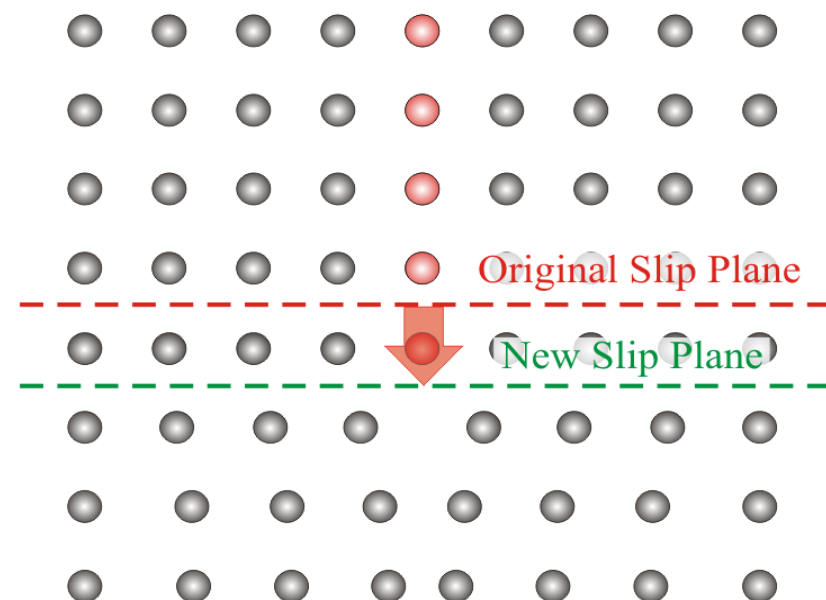
In climb an edge dislocation moves to an adjacent parallel plane, but in cross slip a screw dislocation moves to a plane inclined to the original plane.

Climb of an Edge Dislocation



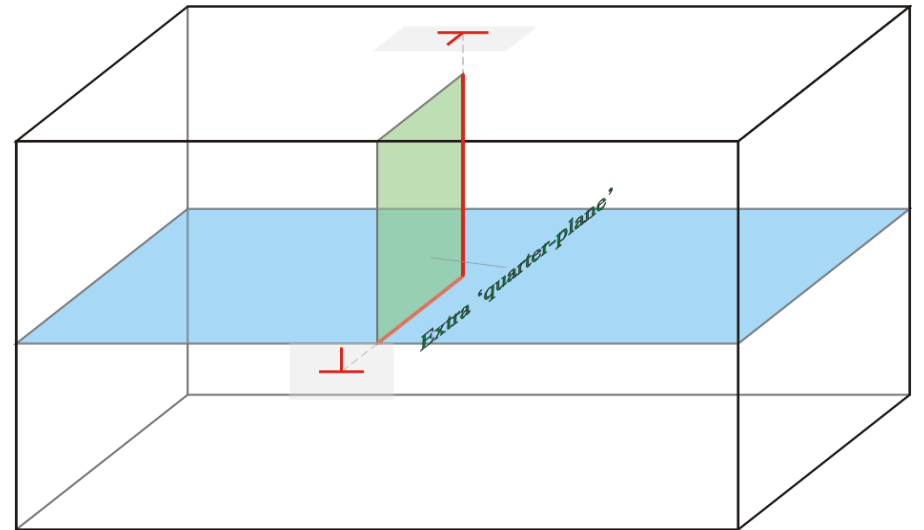
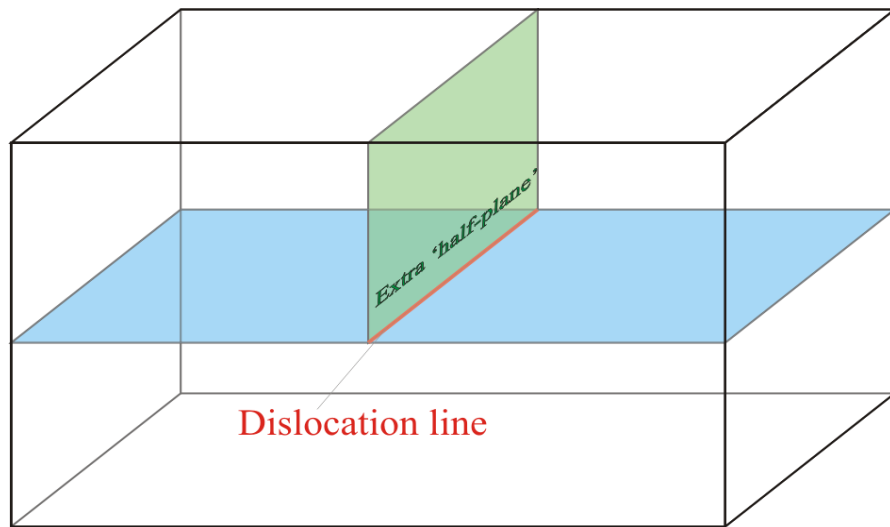
Positive climb

Removal of a row of atoms



Negative climb

Addition of a row of atoms



Energy of dislocations

Energy of dislocation

E

Elastic

Non-elastic (Core)

Elastic Energy of a dislocation / unit length

$$E_d \sim \frac{1}{2} G b^2$$

$G \rightarrow (\mu)$ shear modulus

$b \rightarrow |\mathbf{b}|$

Dislocations
(in terms of lattice translation)

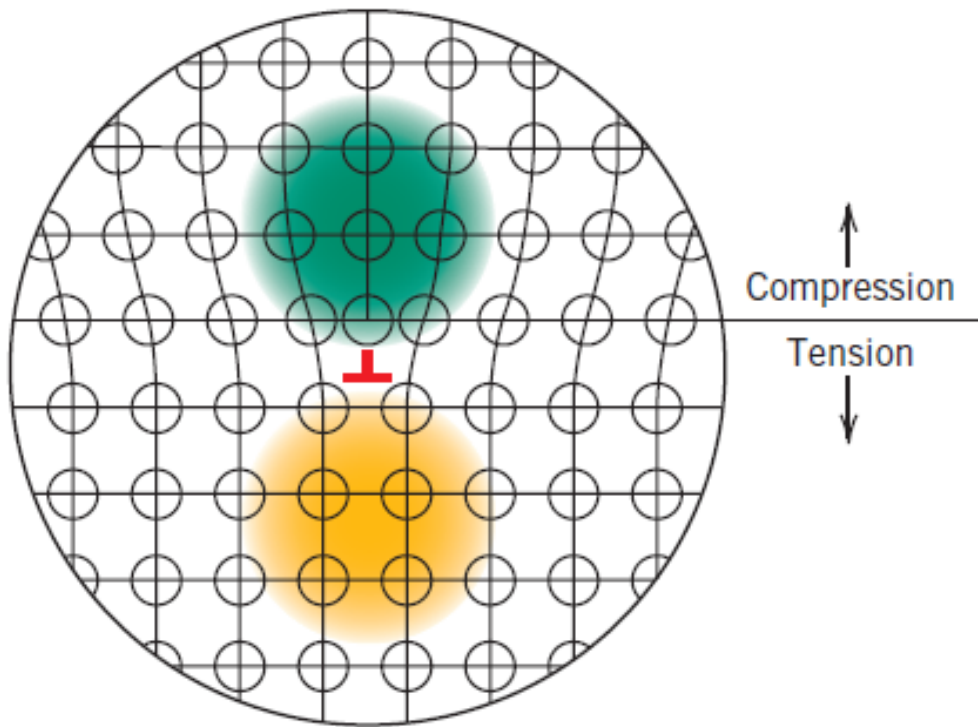
Full

$\mathbf{b} \rightarrow$ Full lattice translation

Partial

$\mathbf{b} \rightarrow$ Fraction of lattice translation

Characteristics of Dislocations



- Some atomic lattice distortion exists around the dislocation line because of the presence of the extra half-plane of atoms.
- As a consequence, there are regions in which compressive, tensile, and shear **lattice strains** are imposed on the neighboring atoms

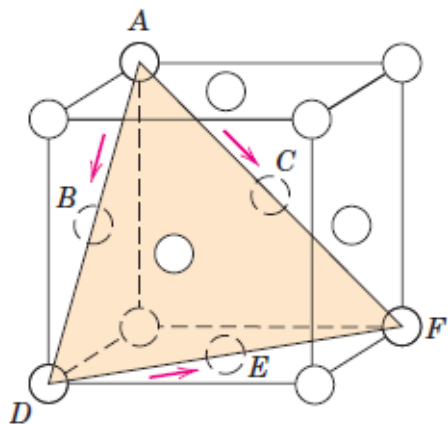
Dissociation of dislocations

$$2b \rightarrow b + b$$

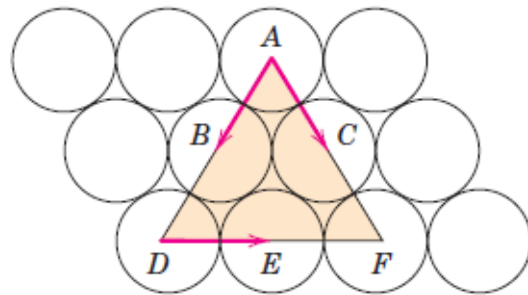


Slip Systems

- Dislocations do not move with the same degree of ease on all crystallographic planes of atoms and in all crystallographic directions.
- Ordinarily there is a preferred plane, and in that plane there are specific directions along which dislocation motion occurs. This plane is called the **slip plane**; it follows that the direction of movement is called the **slip direction**.
- This combination of the slip plane and the slip direction is termed the **slip system**.



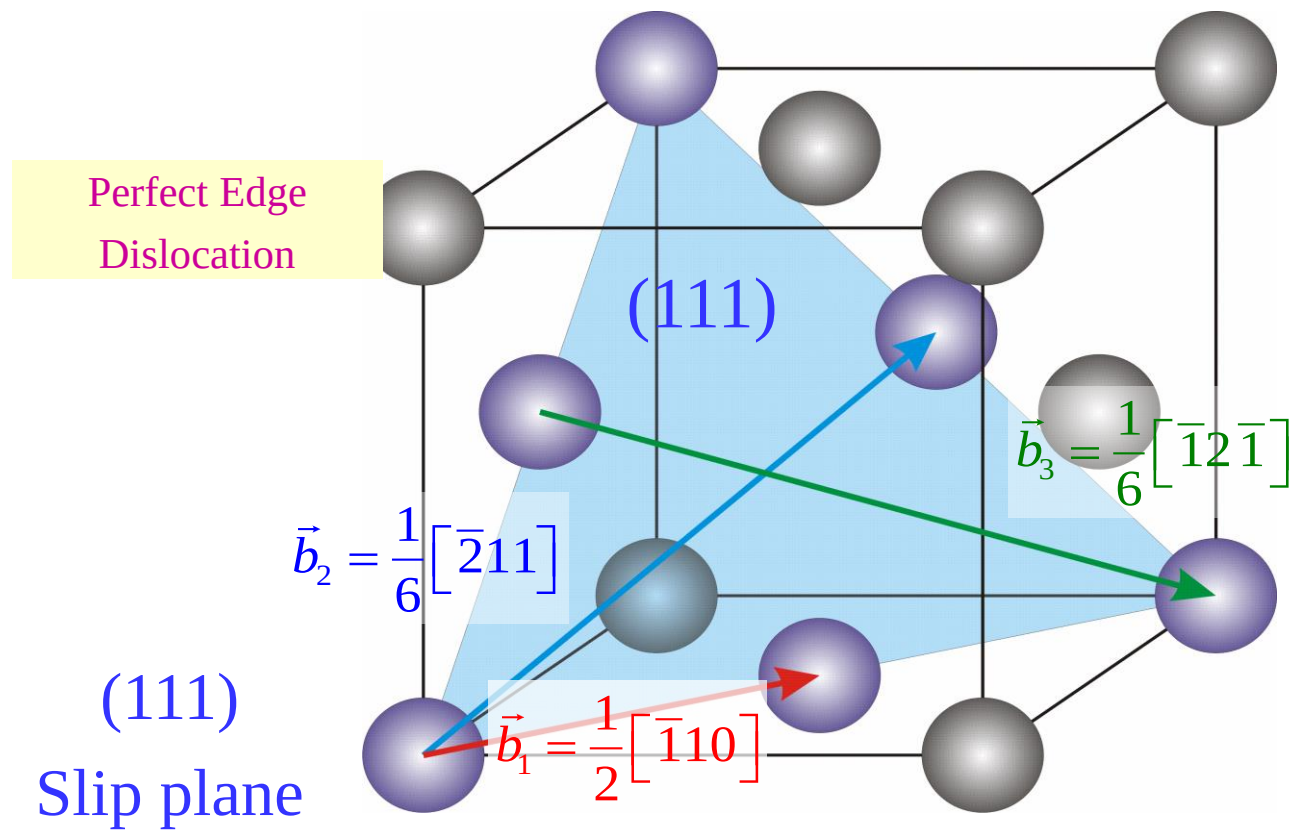
(a)



(b)

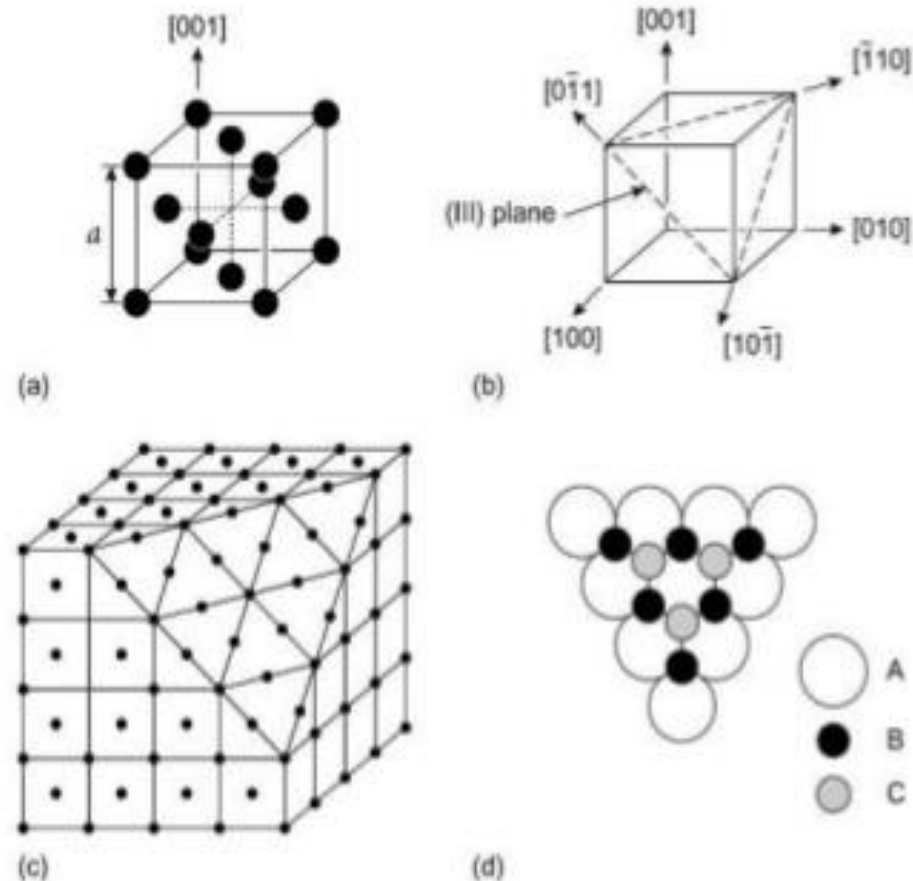
For face-centered cubic, there are 12 slip systems: four unique $\{111\}$ planes and, within each plane, three independent directions.

- Slip occurs along $\langle 110 \rangle$ type directions within the $\{111\}$ planes, as indicated by arrows in Figure.
- $\{111\}$ and $\langle 110 \rangle$ represents the slip system for FCC.
- A given slip plane may contain more than a single slip direction.
- Thus, several slip systems may exist for a particular crystal structure; the number of independent slip systems represents the different possible combinations of slip planes and directions.



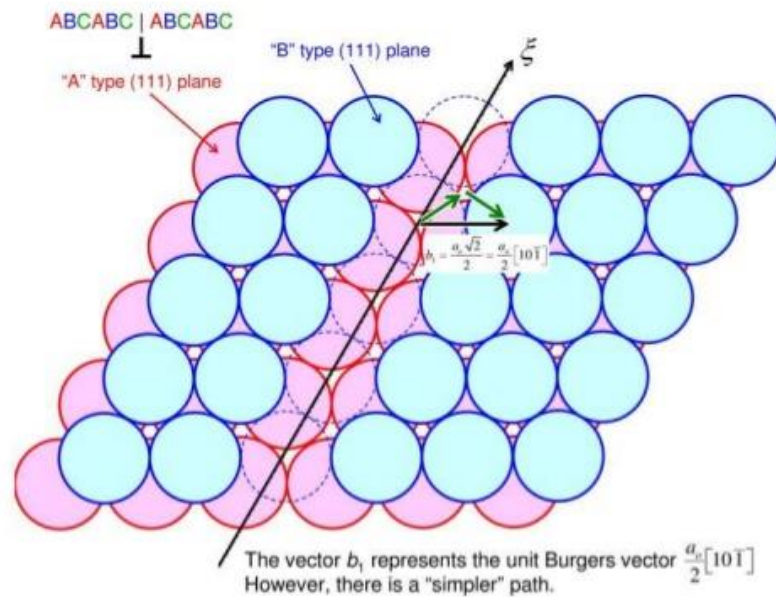
Introduction

- Each unit cell contains **4 {111} planes**
- Each {111} plane contains **3 <110> directions**
- Thus, there are **12 slip systems** in an FCC unit cell
- For the face-centered crystal structure, the centers of the third plane are situated over the C sites of the first plane. This yields an **ABCABCABC** ...stacking sequence; that is, the **atomic alignment repeats every third plane**.

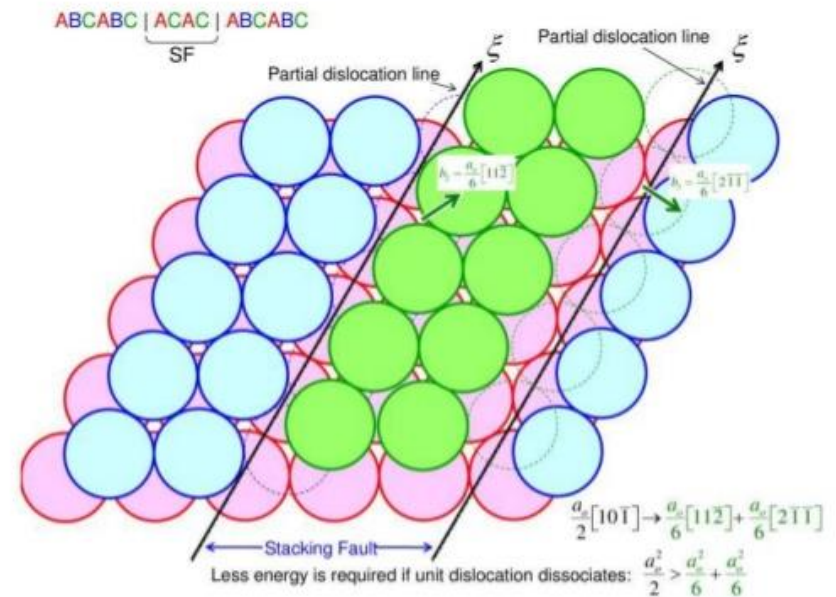


Hull and Bacon, 2011

Shockley Partial Dislocations



Shockley Partial Dislocations



Dislocation line vector

Extra half plane

Part of the plane shown

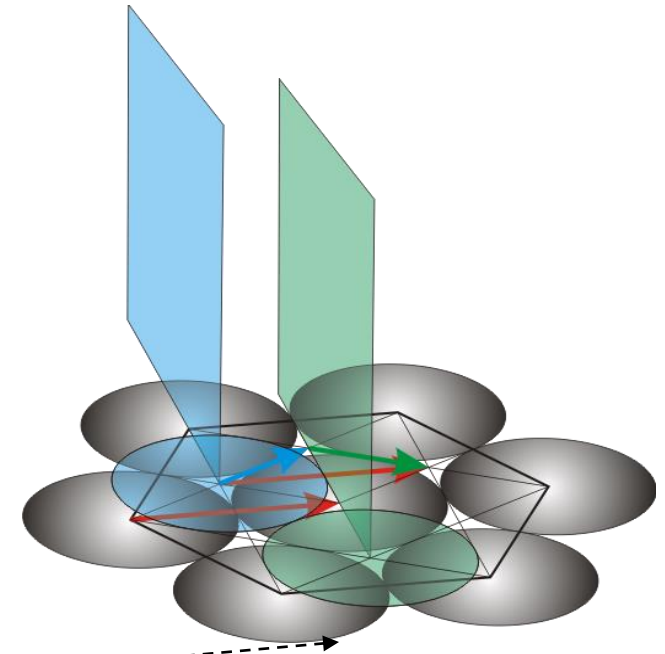
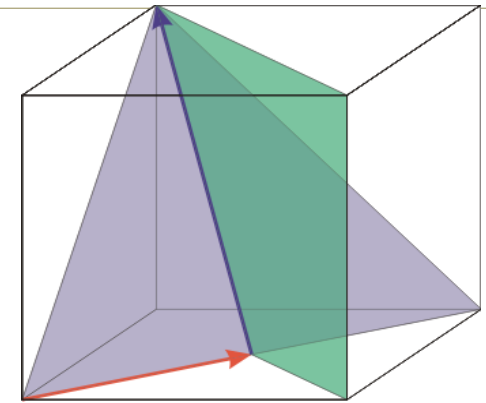
$$\left(\frac{1}{2} [\bar{1}\bar{1}2] \right)_{(111), (1\bar{1}0)}$$

$$(1\bar{1}0)$$

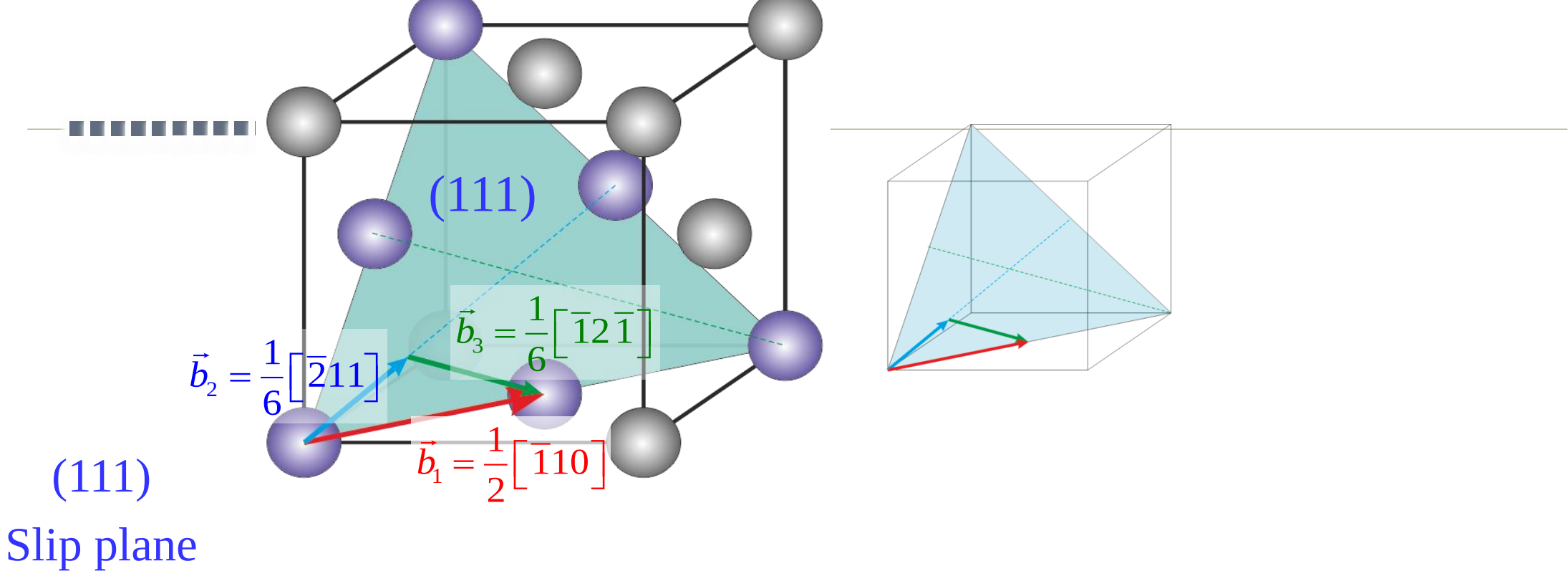
Slip plane (111)

Burger's vector

$$\left(\frac{1}{2} [\bar{1}10] \right)_{(111)}$$



The extra- "half plane" consists of two 'planes' of atoms!
One crystal plane is 'lattice plane' decorated with two atomic planes.



Some of the atoms are omitted for clarity, Full vectors $[\bar{2}11]$ (blue vector) & $[\bar{1}2\bar{1}]$ (green vector) shown in the left figure

$$\left(\frac{1}{2}[\bar{1}10]\right)_{(111)} \rightarrow \left(\frac{1}{6}[\bar{1}2\bar{1}]\right)_{(111)} + \left(\frac{1}{6}[\bar{2}11]\right)_{(111)}$$

Shockley Partial

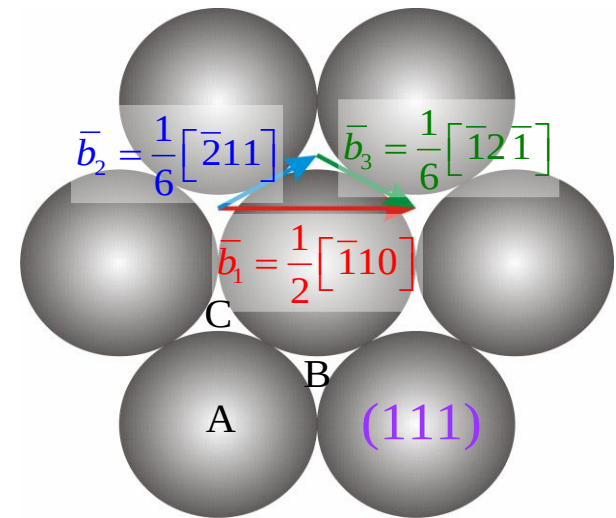
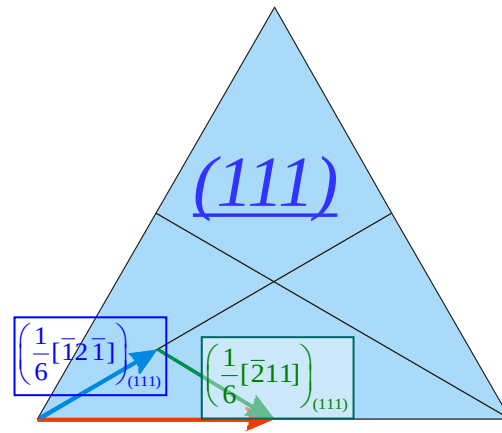
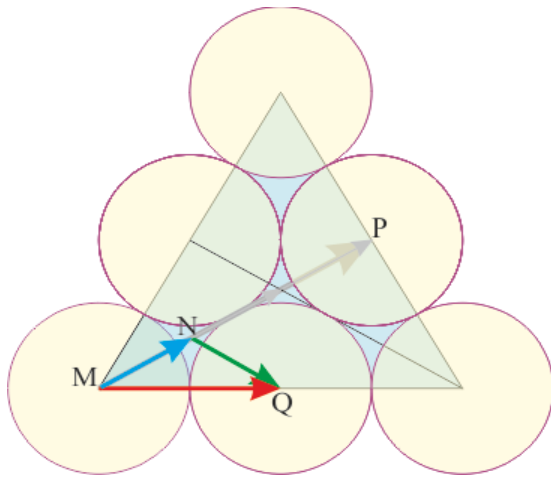
Reduction in the strain energy of the dislocation on splitting into Shockley partials

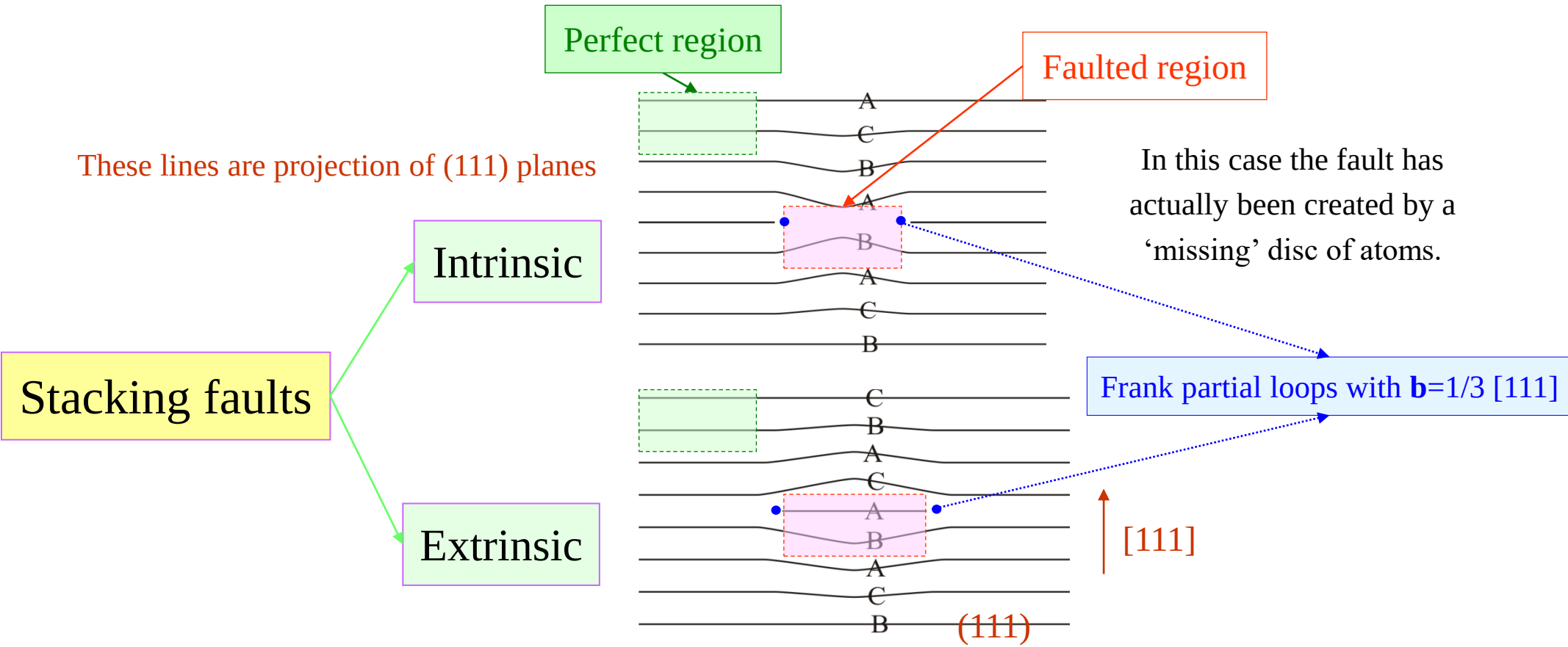
$$|\vec{b}_2| = \left(\frac{\sqrt{1^2 + 2^2 + 1^2}}{6} \right)^2 = \left(\frac{\sqrt{6}}{6} \right)^2 = \frac{1}{6}$$

$$(b_2^2 + b_3^2) = 1/6 + 1/6 = 1/3$$

$$b_1^2 > (b_2^2 + b_3^2)$$

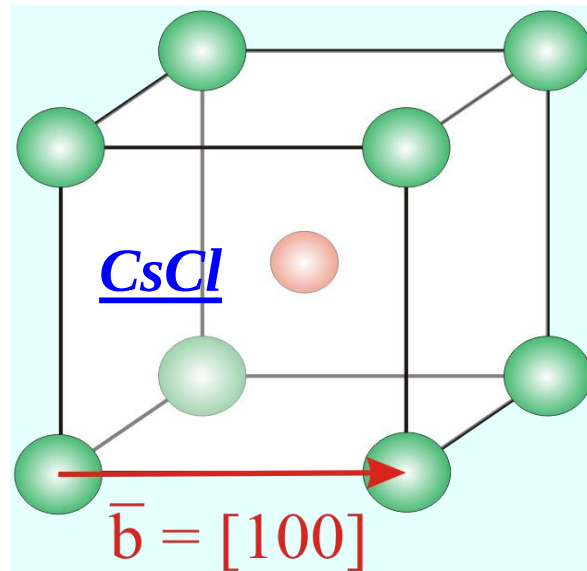
$$\frac{1}{2} > \frac{1}{3}$$



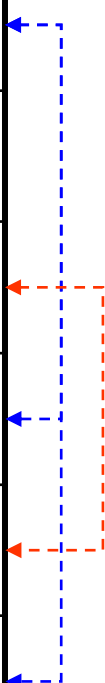


Two breaks introduced into the stacking sequence

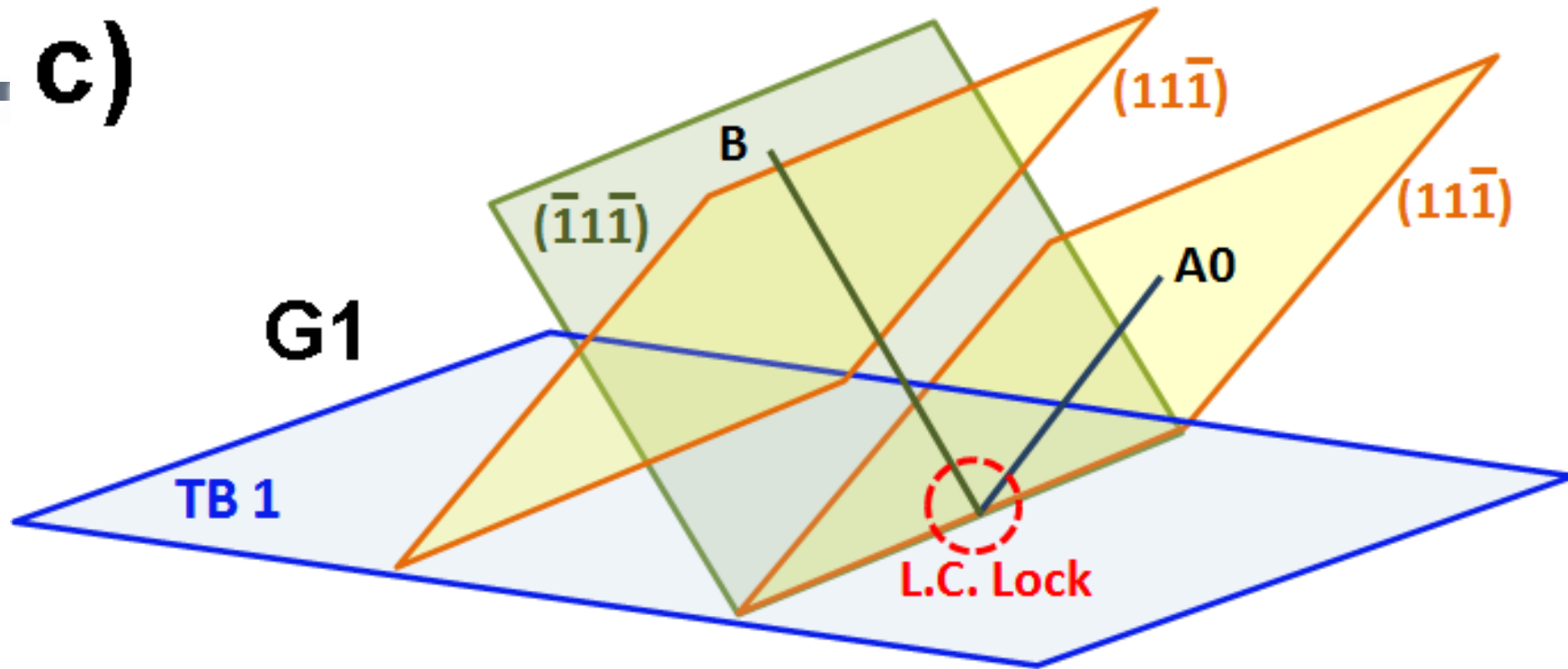
Dislocations in Ionic crystals



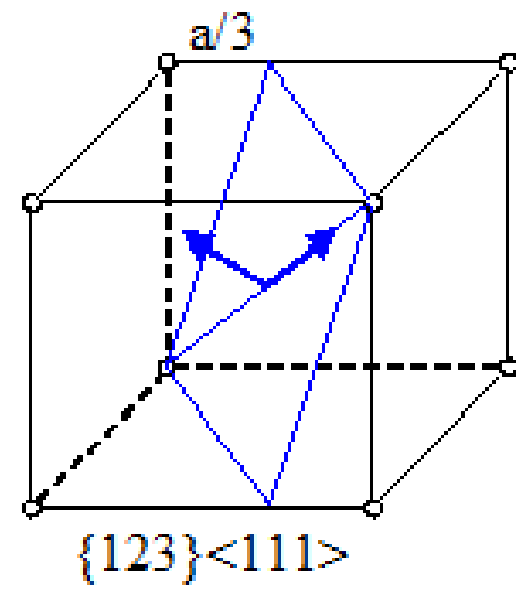
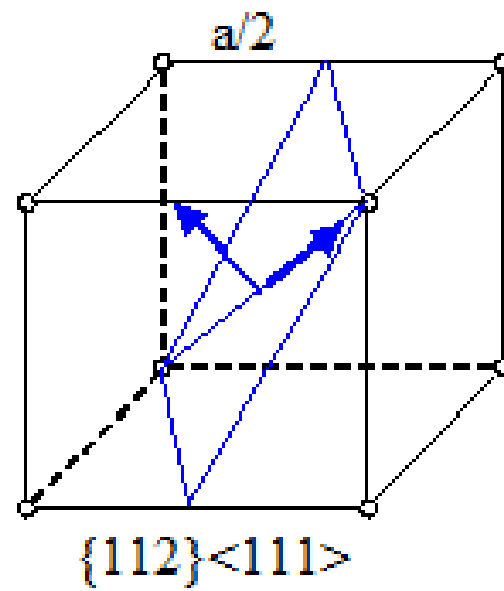
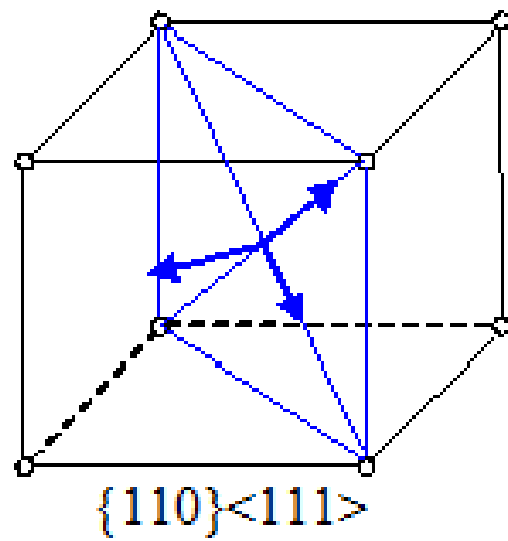
Burgers vectors of dislocations in cubic crystals

Monoatomic FCC	$\frac{1}{2}\langle 110 \rangle$	
Monoatomic BCC	$\frac{1}{2}\langle 111 \rangle$	
Monoatomic SC	$\langle 100 \rangle$	
NaCl type structure	$\frac{1}{2}\langle 110 \rangle$	
CsCl type structure	$\langle 100 \rangle$	
DC type structure	$\frac{1}{2}\langle 110 \rangle$	

.... **c)**

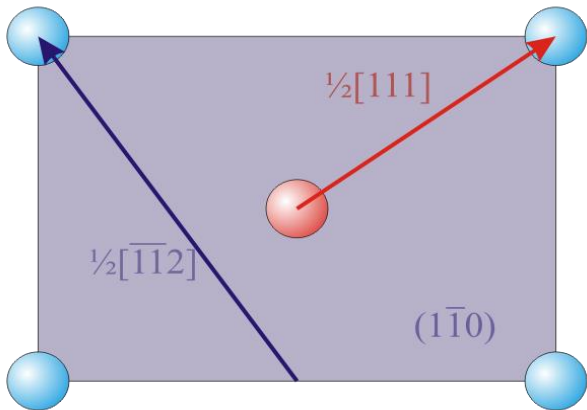
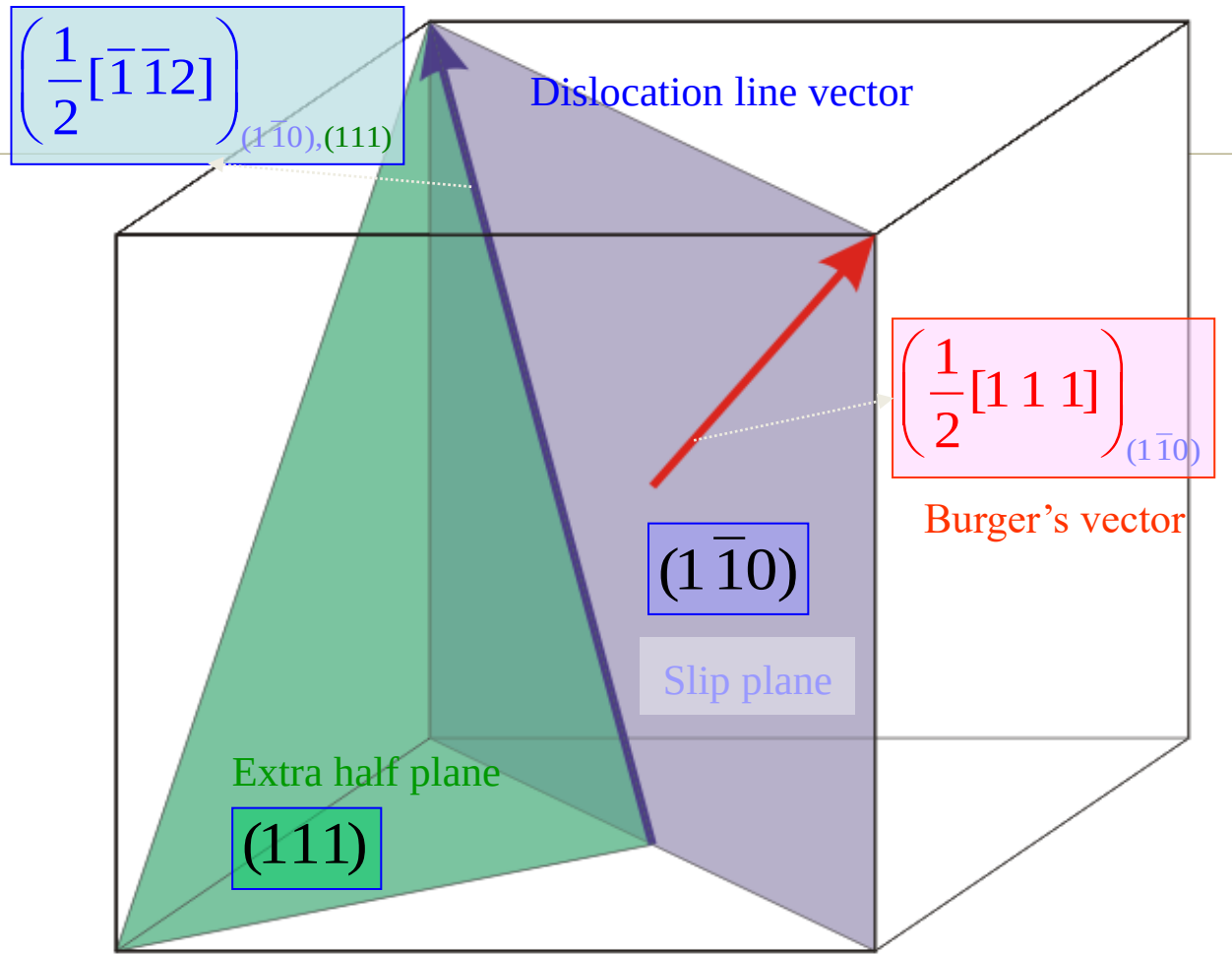
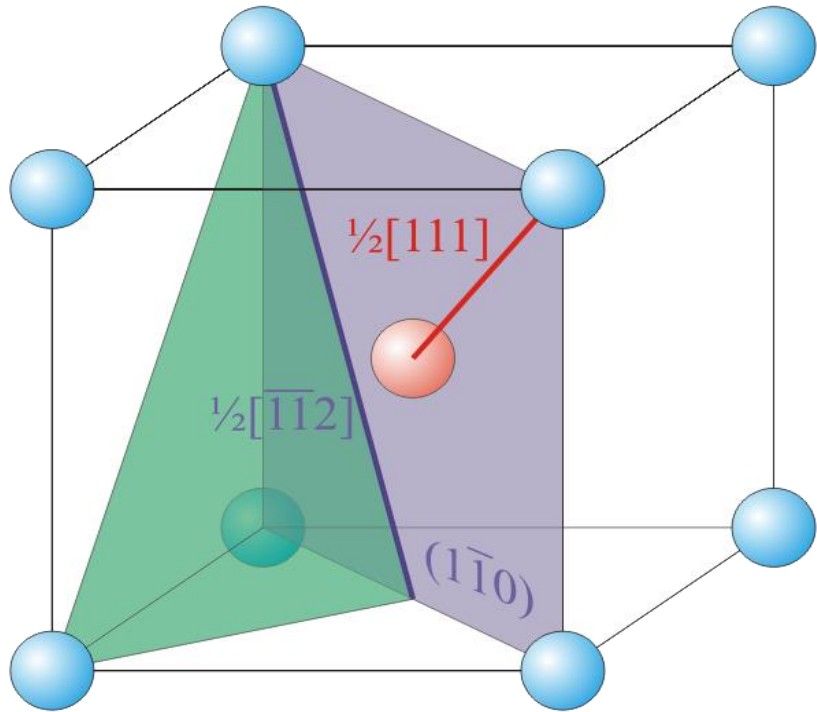


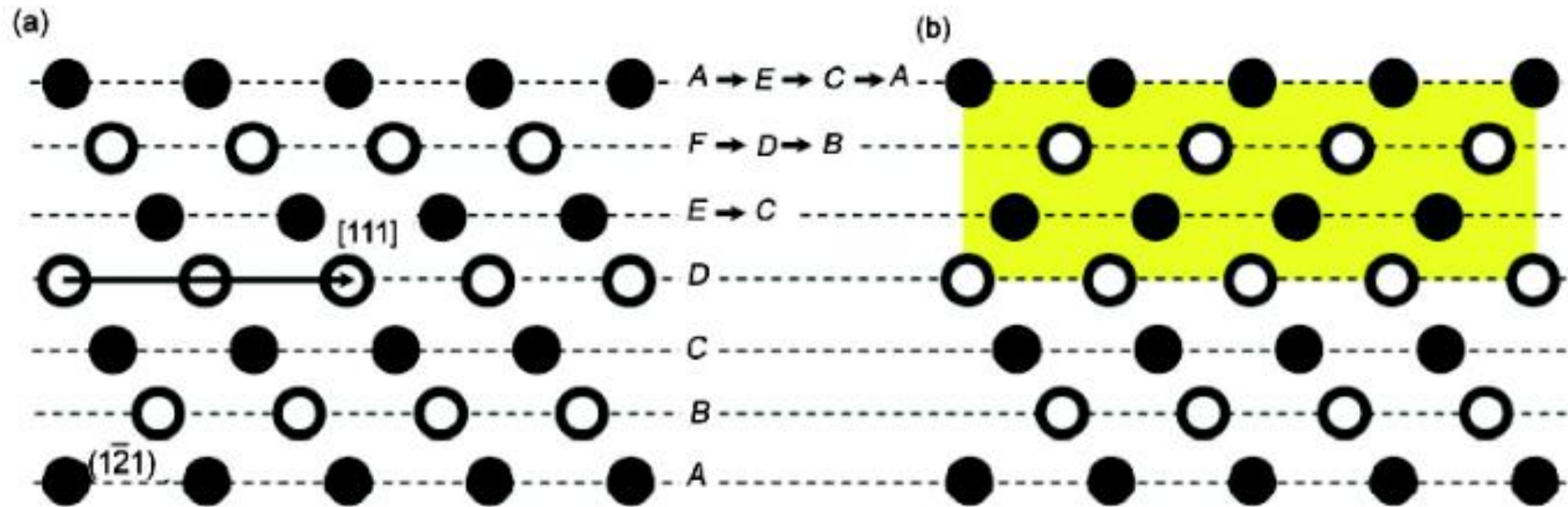
- ☐ Due to accidents in crystal growth from the melt.*
- ☐ Mechanical deformation of the crystal.
- ☐ Nucleation of dislocation.
 - Homogenous nucleation of a dislocation required high stresses ($\sim G/10$).
 - Stress concentrators in the crystal can aid the process.
- ☐ Dislocation density increases due to plastic deformation mainly by multiplication of pre-existing dislocations.



BCC

Pure edge dislocation





Color online

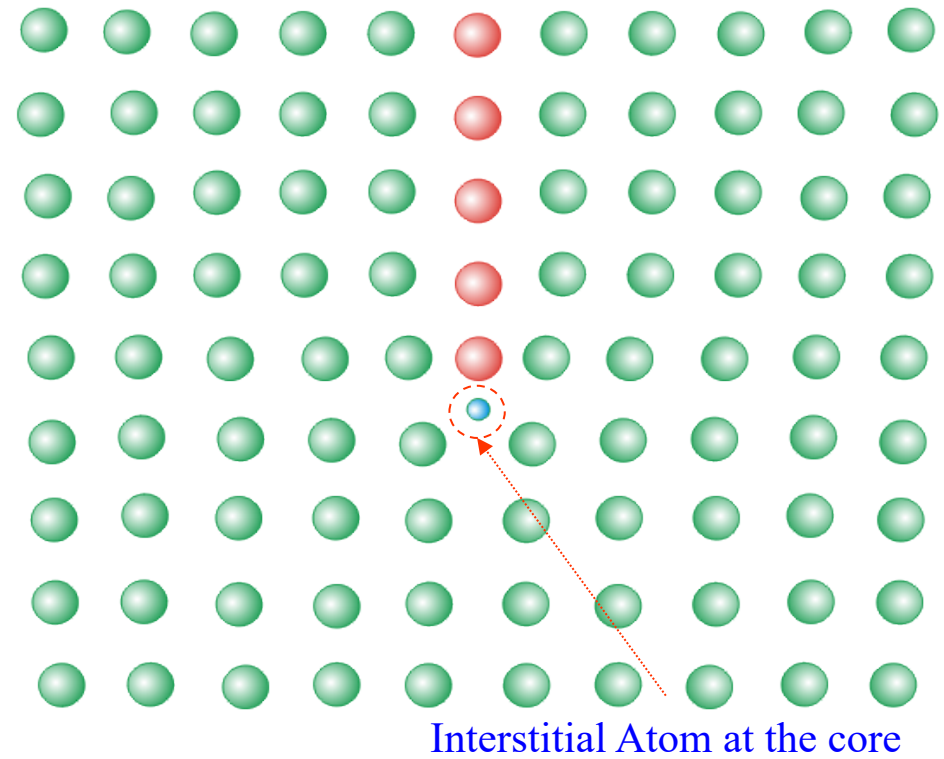
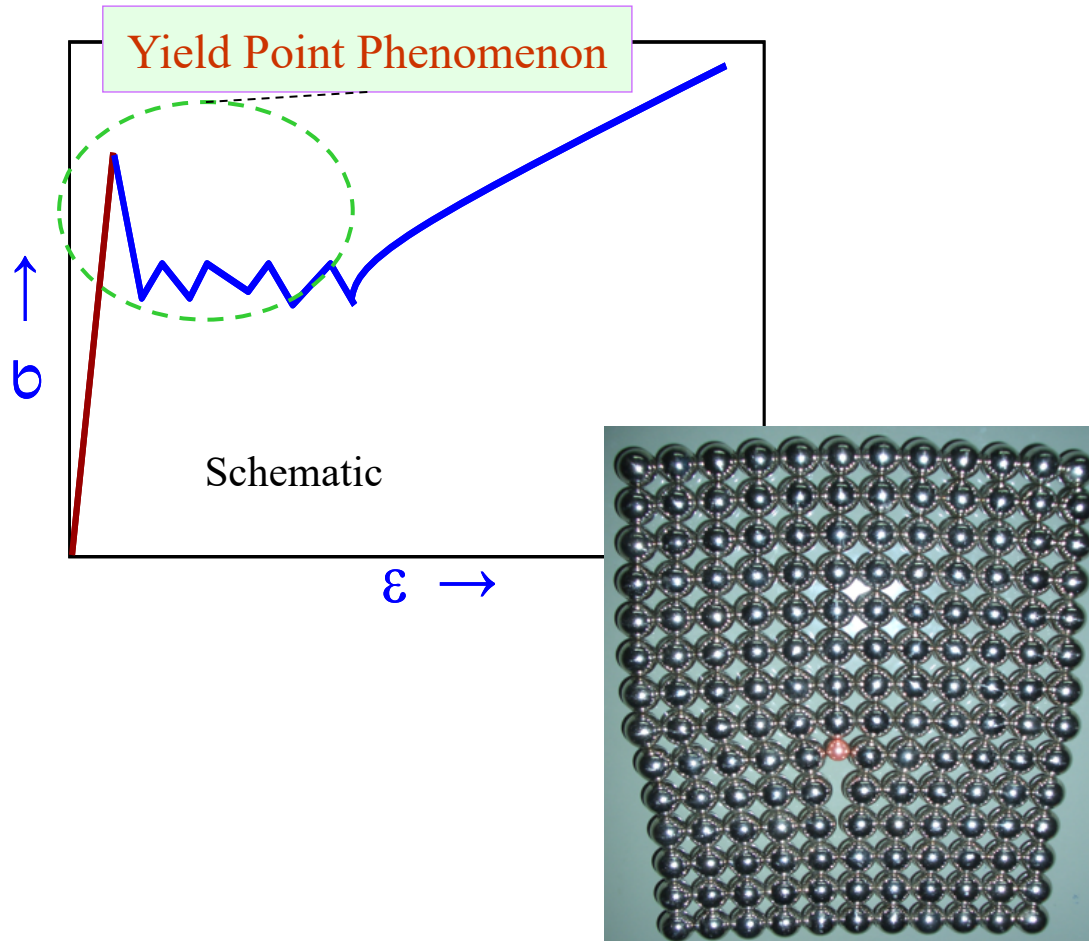
(a) a bcc lattice structure projected along $[10\bar{1}]$, and $(1\bar{2}1)$ plane stacking sequence was $\cdots ABCDEF A \cdots$

(b) $\{112\} \langle 111 \rangle$ -type twin formed after layers E , F and A continued to shear by $1/6[111]$ (Yellow zone stands for the twin lattice)

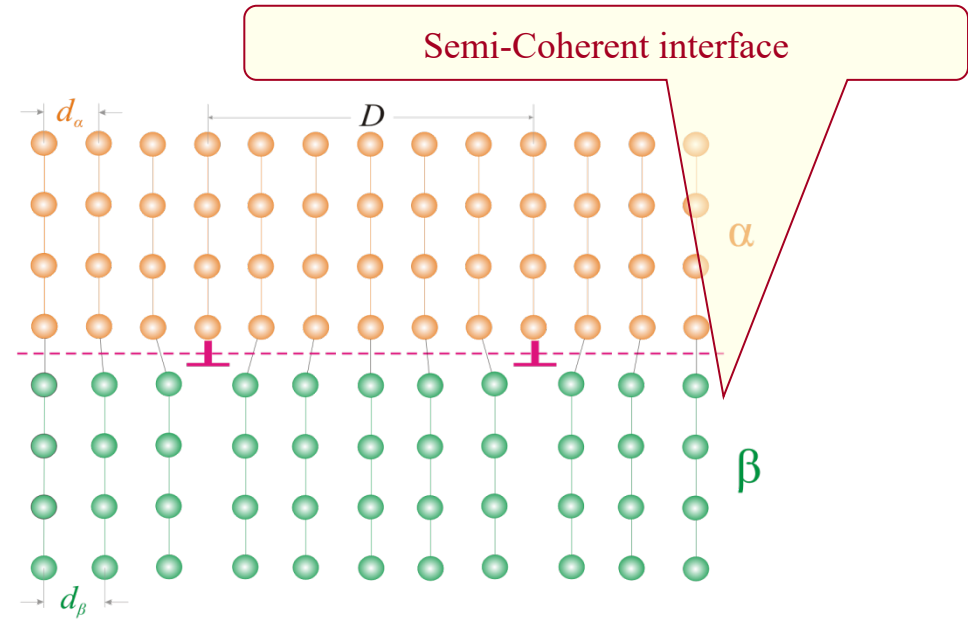
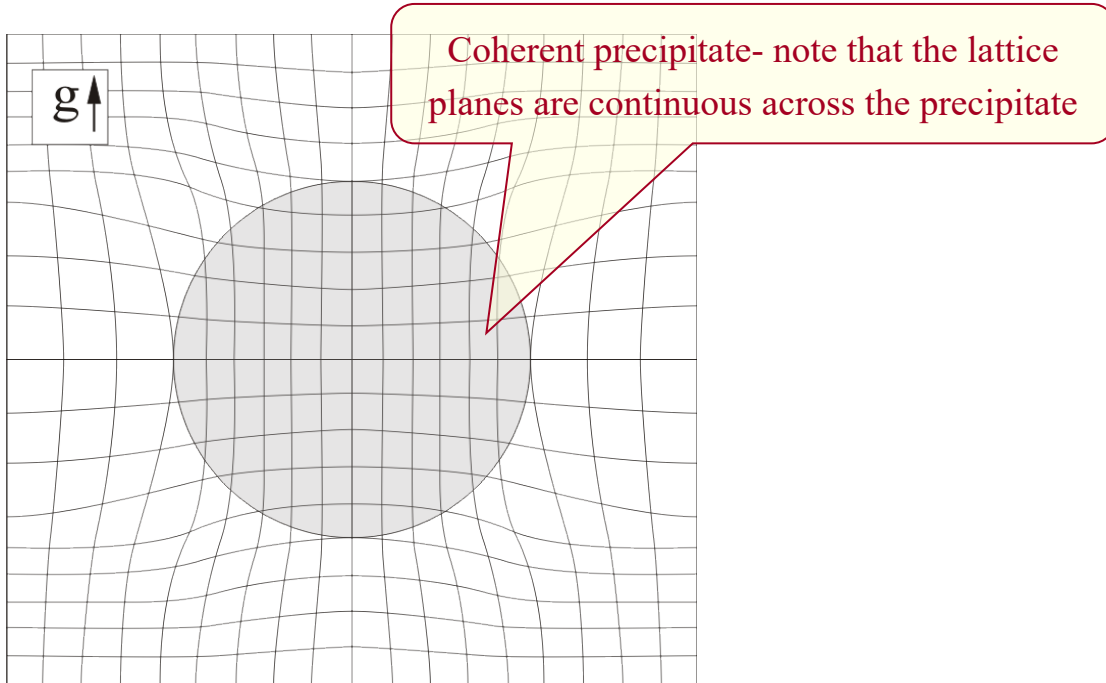
Edge dislocation - point defect interactions

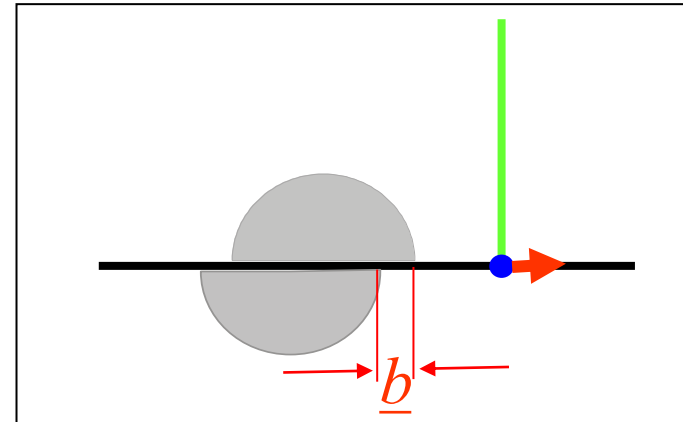
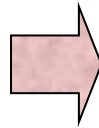
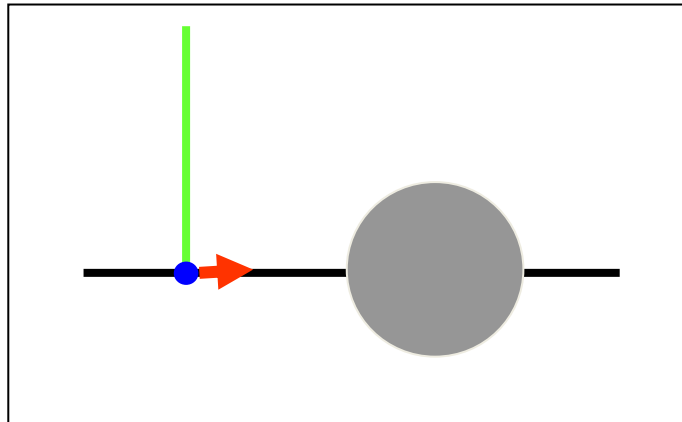
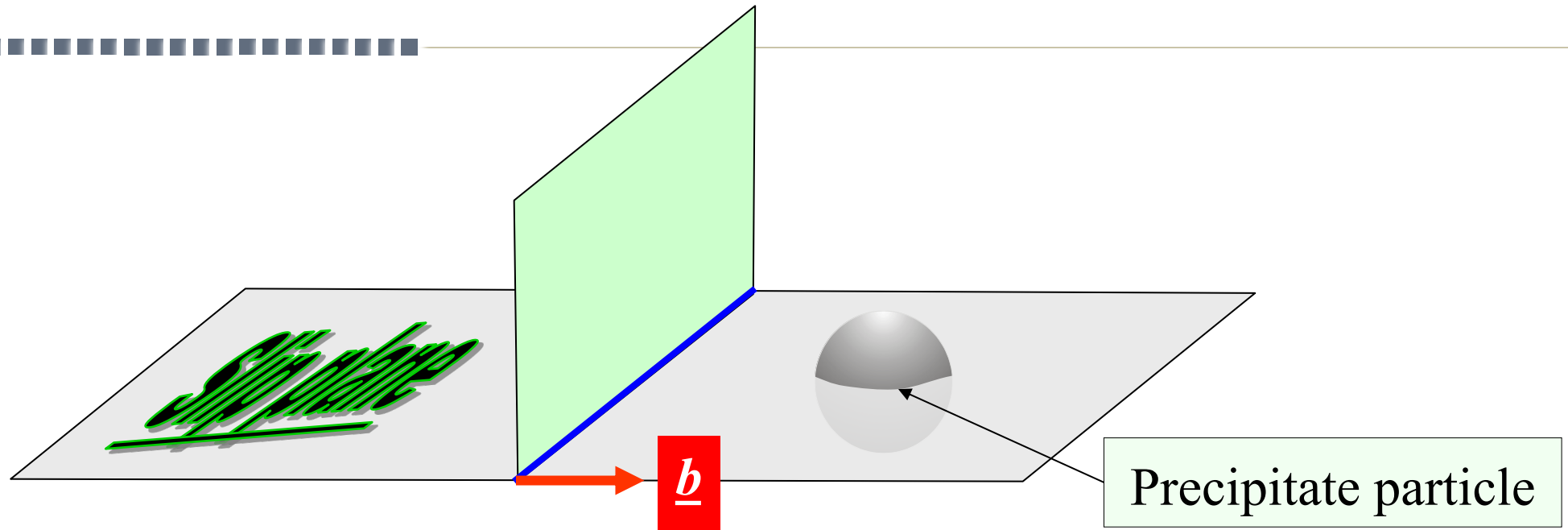
Point Defect	Tensile Region	Compressive Region
Vacancy	Repelled	Attracted
Interstitial <i>(of size larger than the void)</i>	Attracted	Repelled
Smaller substitutional atom	Repelled	Attracted
Larger Substitutional atoms	Attracted	Repelled

Yield Point Phenomenon

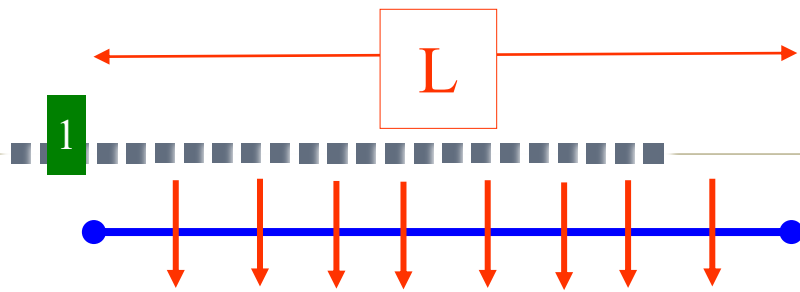


Dislocation-Precipitate Interactions



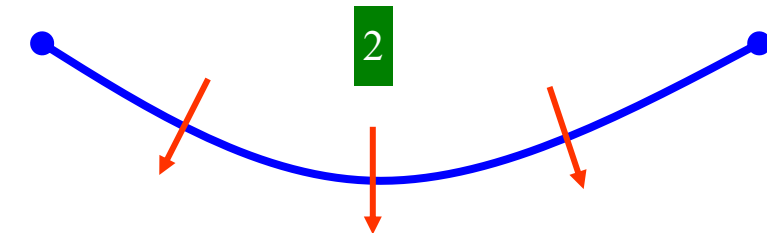


Increasing stress

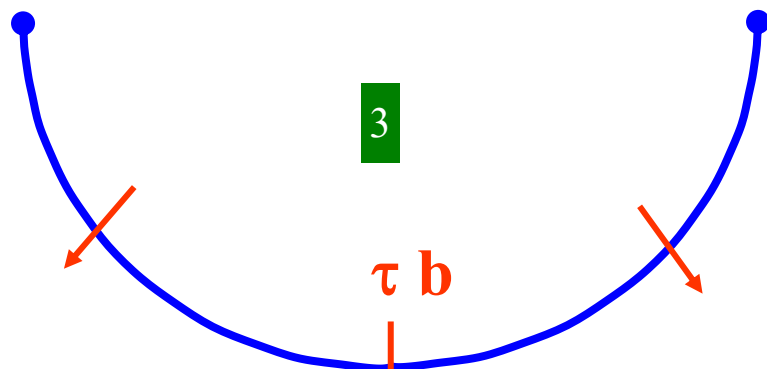


τb

Direction of dislocation motion
is $\perp$ to the dislocation line
(except at A and B)



τb



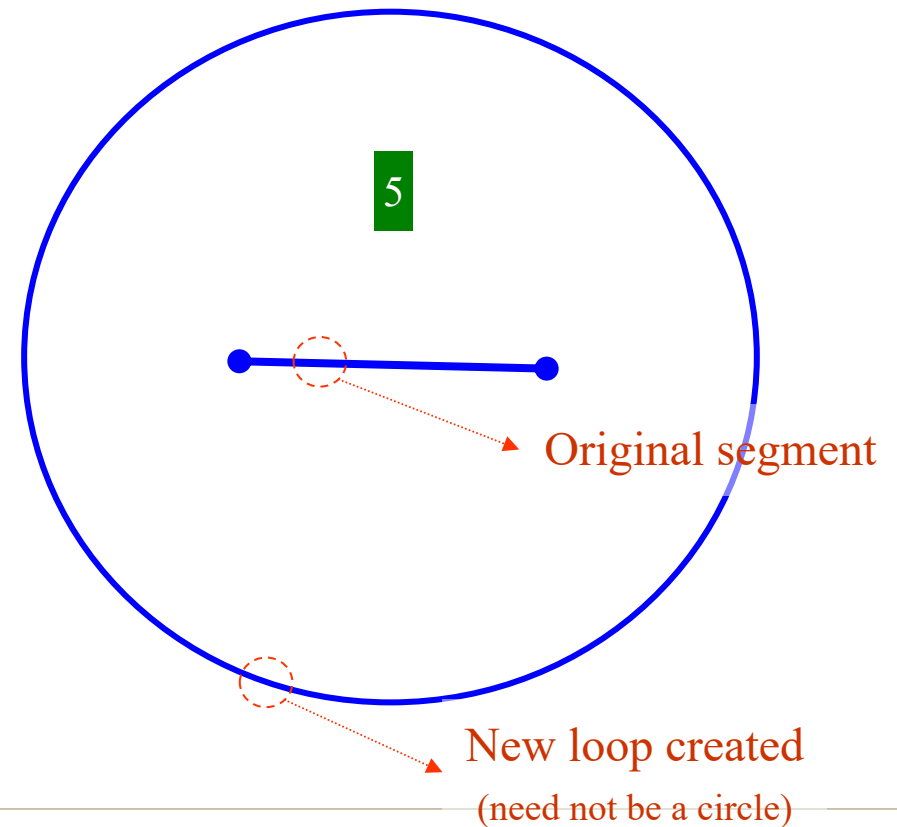
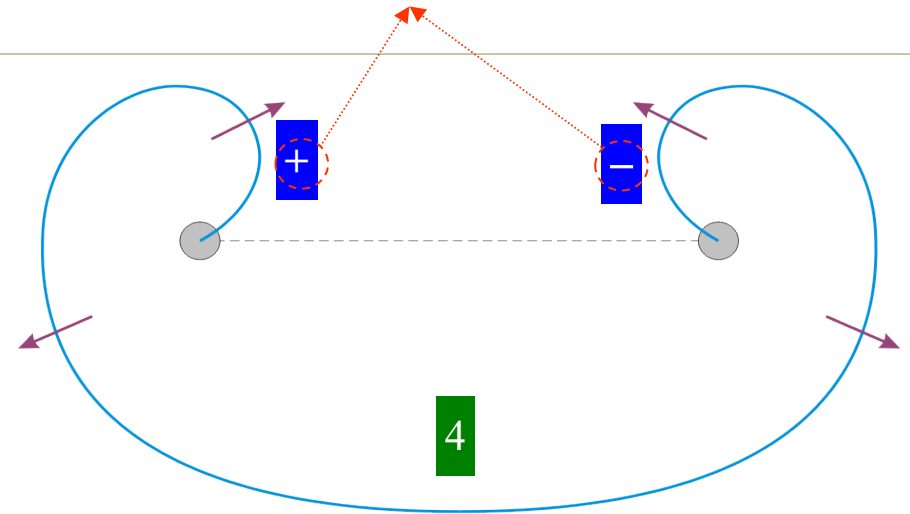
τb

semicircle $\rightarrow$

corresponds to **maximum stress** required
to expand the loop

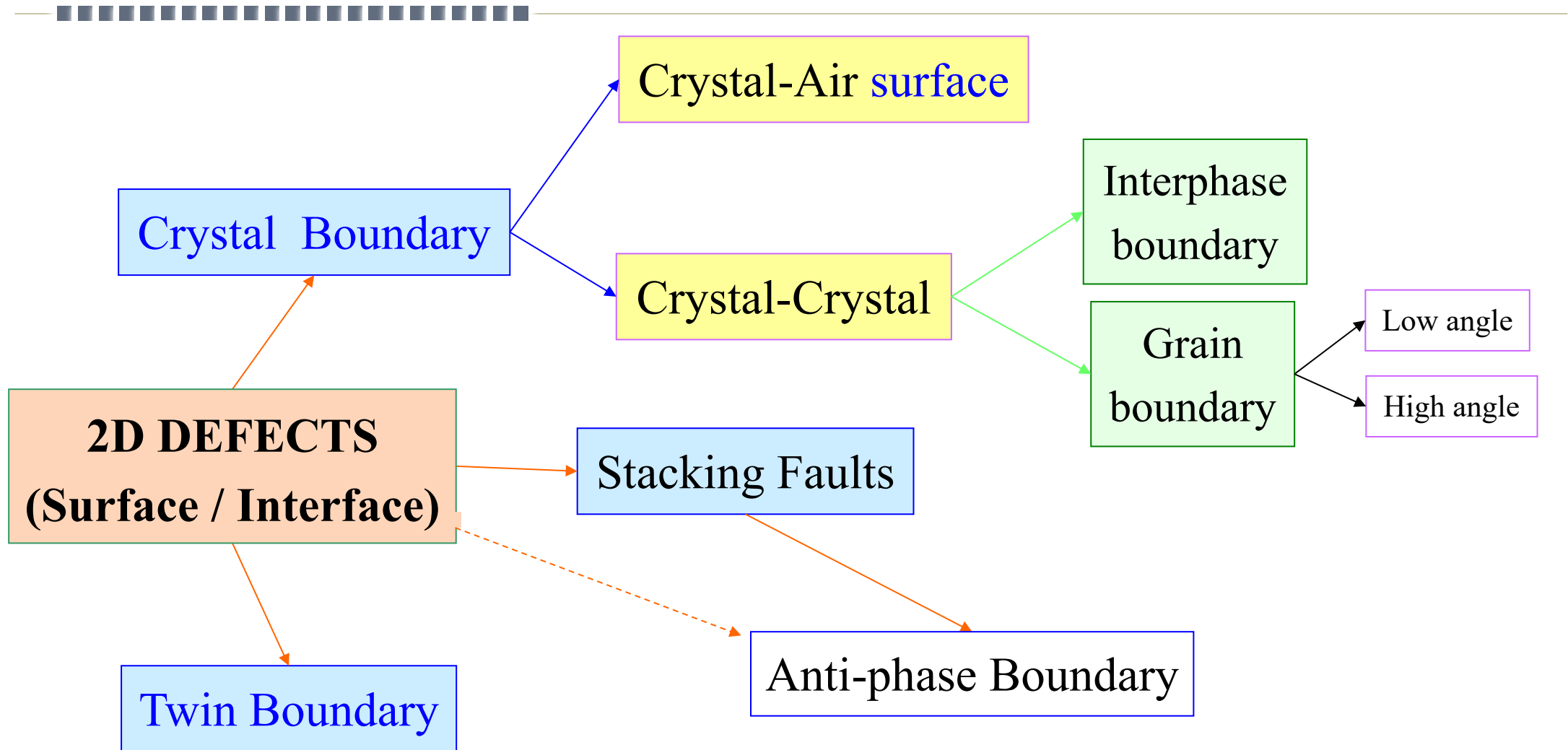
After this decreasing stress is required to expand the loop

+ and - segments come together and annul each other

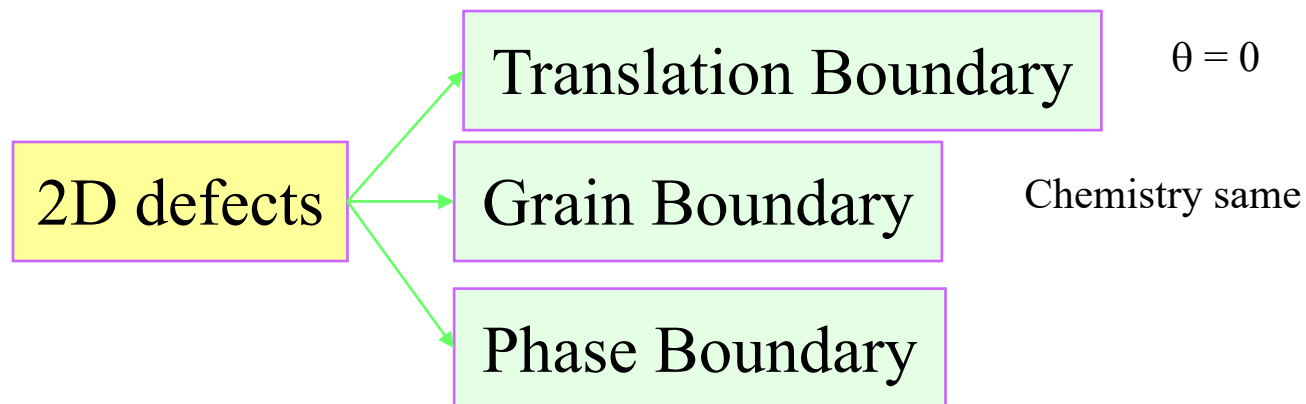
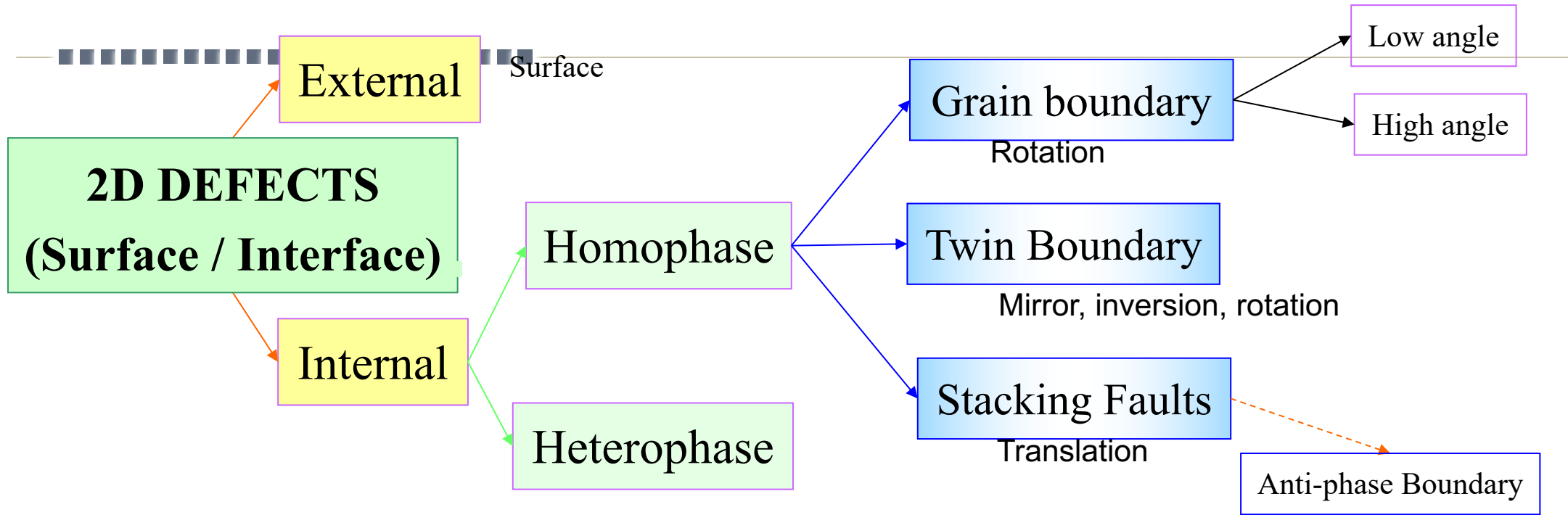


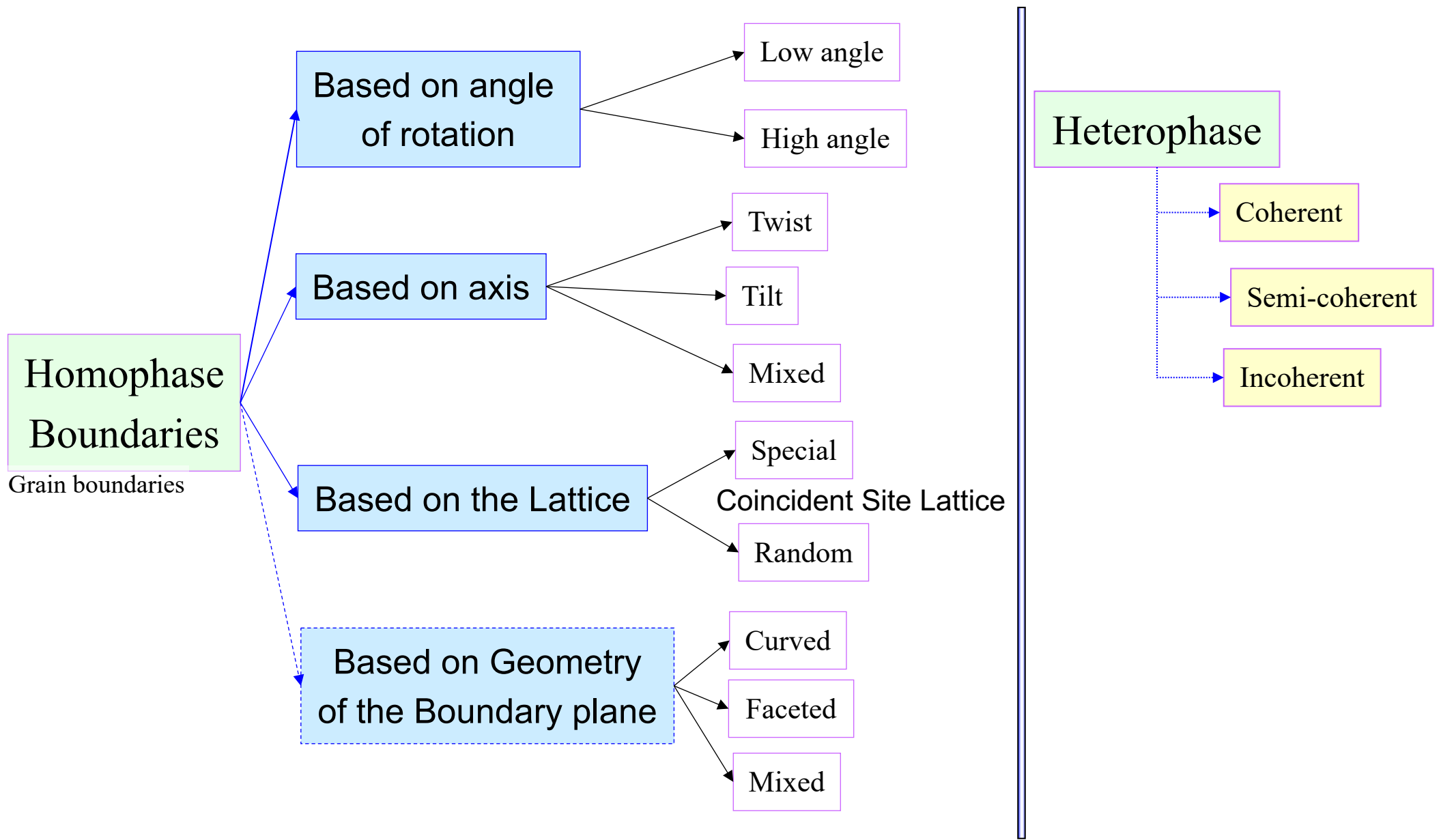
Interfacial Defects

- **Interfacial defects** are boundaries that have **two dimensions** and normally separate regions of the materials that have different crystal structures and/or crystallographic orientations.
- These imperfections include ***external surfaces, grain boundaries, phase boundaries, twin boundaries, and stacking faults.***



Alternate views





Interfacial Defects

External Surfaces

- One of the most obvious boundaries is the external surface, along which the crystal structure terminates.
- Surface atoms are not bonded to the maximum number of nearest neighbors, and are therefore in a higher energy state than the atoms at interior positions. The bonds of these surface atoms that are not satisfied give rise to a surface energy, expressed in units of energy per unit area (J/m^2).
- To reduce this energy, materials tend to minimize, if at all possible, the total surface area. For example, liquids assume a shape having a minimum area—the droplets become spherical. Of course, this is not possible with solids, which are mechanically rigid.

$$\sigma = \frac{\text{Force}}{\text{Length}} = \frac{F}{L} \rightarrow \frac{[N]}{[m]} \quad \dots \quad \gamma = \frac{\text{Energy}}{\text{Area}} = \frac{E}{A} \rightarrow \frac{[J]}{[m^2]} = \frac{[Nm]}{[m^2]} = \frac{[N]}{[m]}$$

- Dimensionally γ and σ are identical $\rightarrow$ Physically they are different type of quantities.

LIQUIDS

Surface Energy $\Leftrightarrow$ Surface Tension

In a liquid: surface energy and surface tension are equivalent descriptions of the surface

SOLIDS

Surface Energy $\neq$ Surface Tension

Except in certain circumstances*

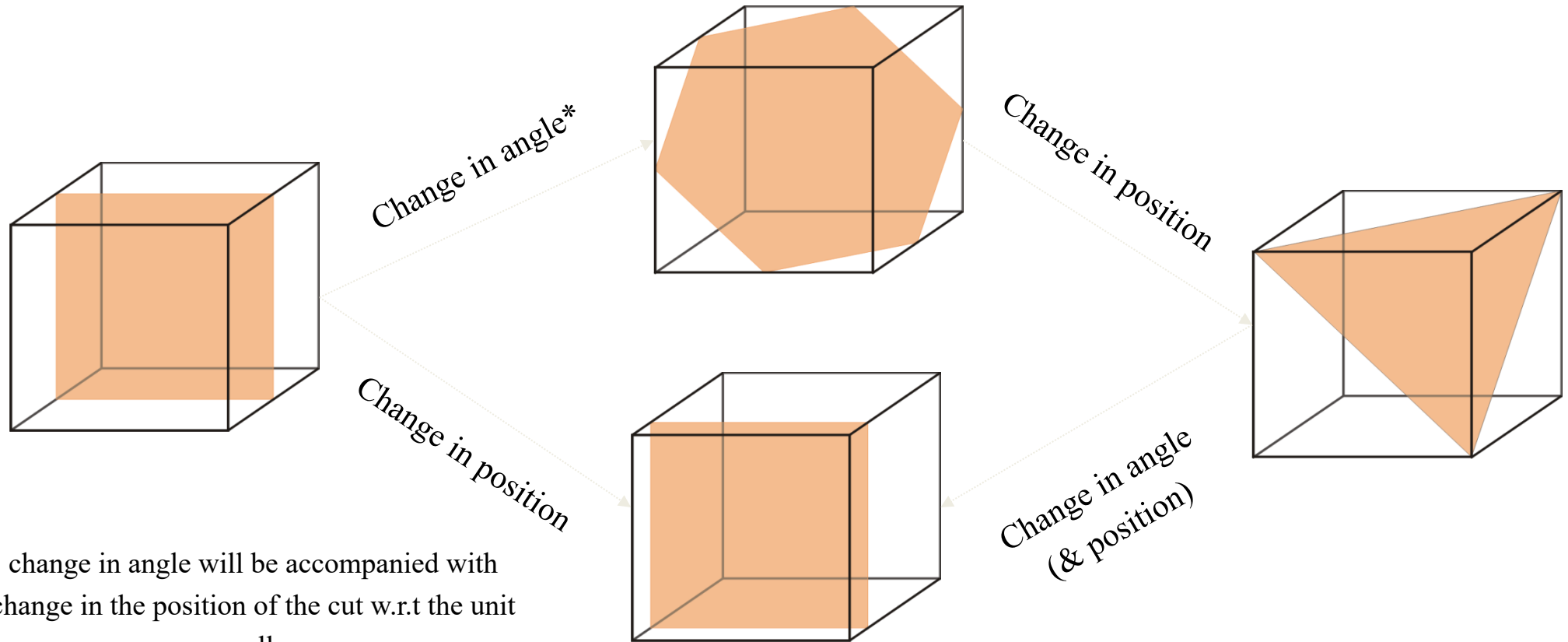
+ Surface energy is Anisotropic

in a solid

Surface Energy (γ) $\rightarrow$ is the reversible work required to create an unit area of surface
(at constant V, T & μ_i)

Surface Tension (σ) $\rightarrow$ is the average of surface stresses in two mutually perpendicular directions

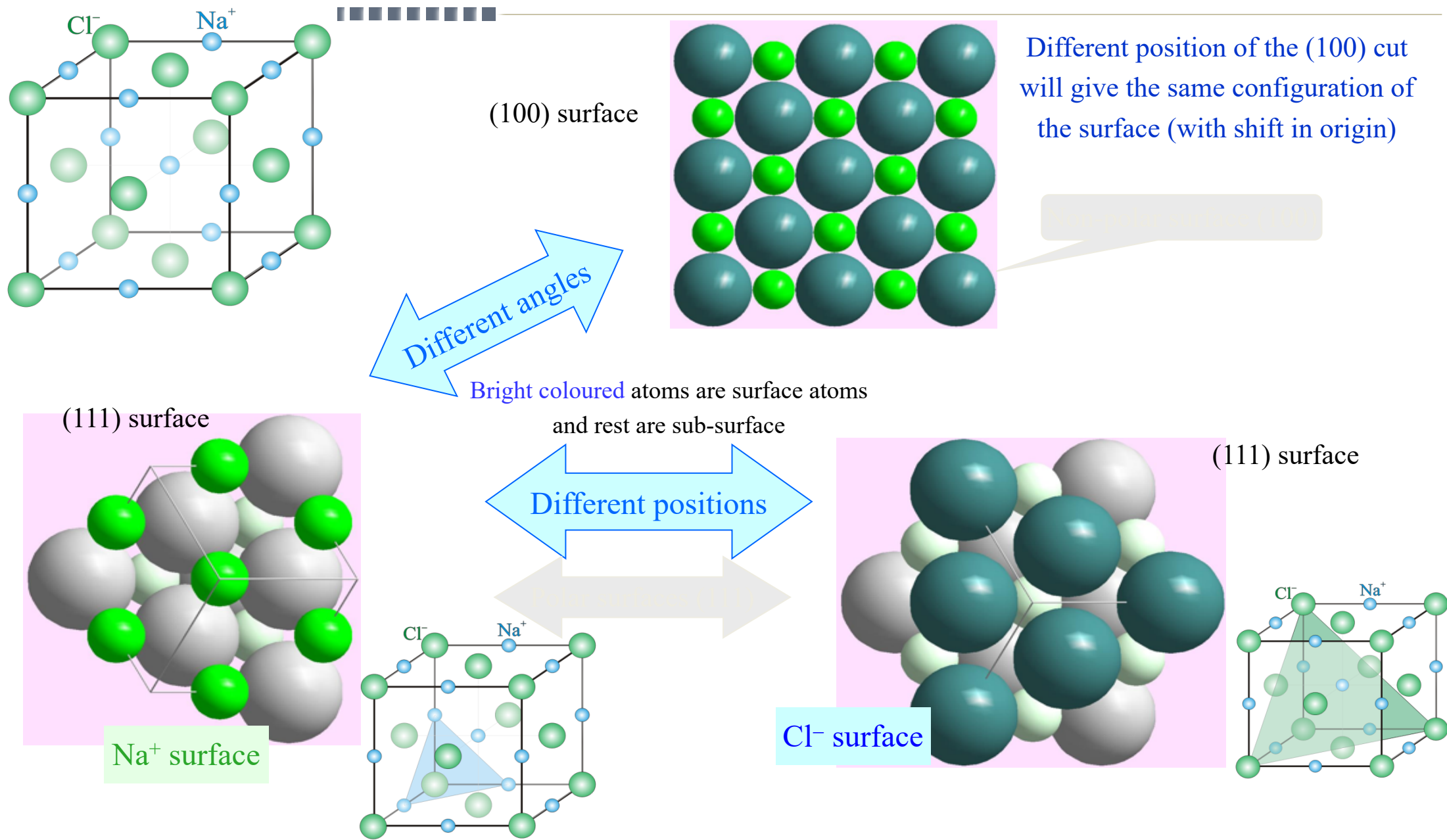
External surface of the crystal



* change in angle will be accompanied with change in the position of the cut w.r.t the unit cell

** More about this shortly

E.g. NaCl

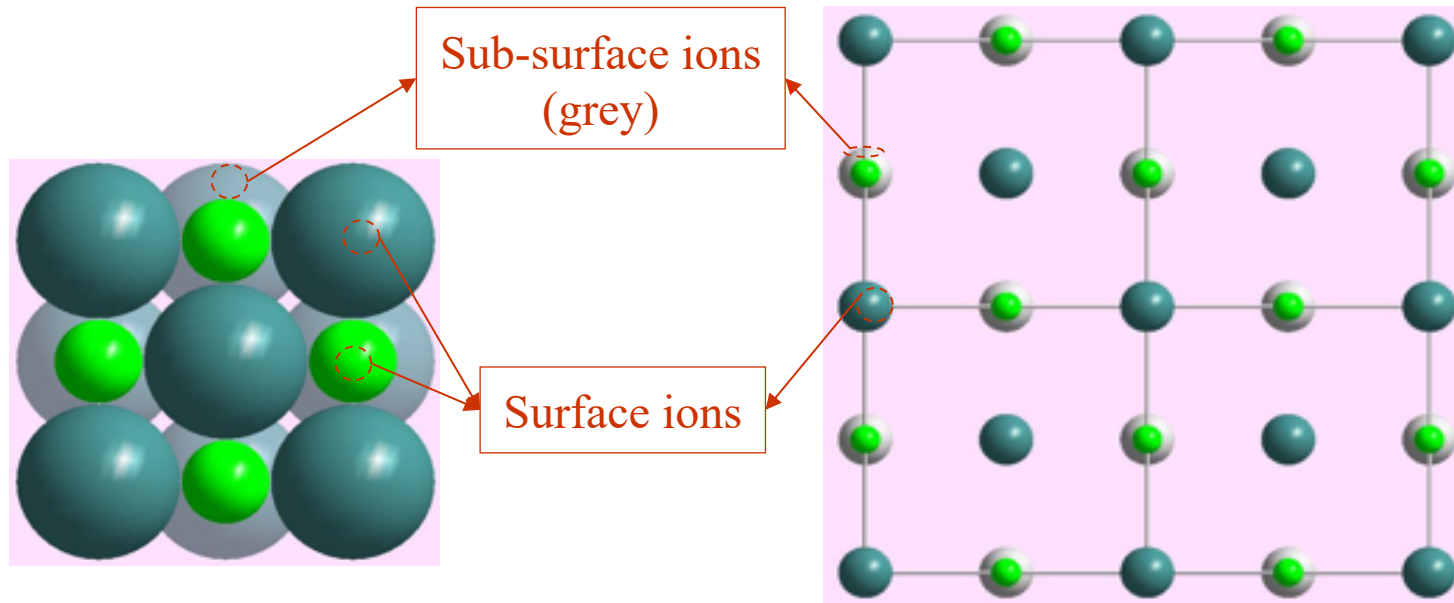


In fact a single cut with a (111) plane will give both these surfaces: one on each side of the cut



More views & aspects: (100) surface

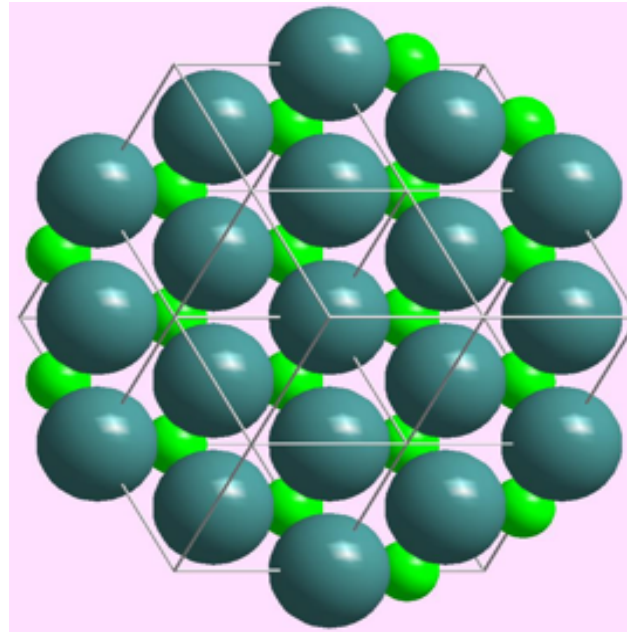
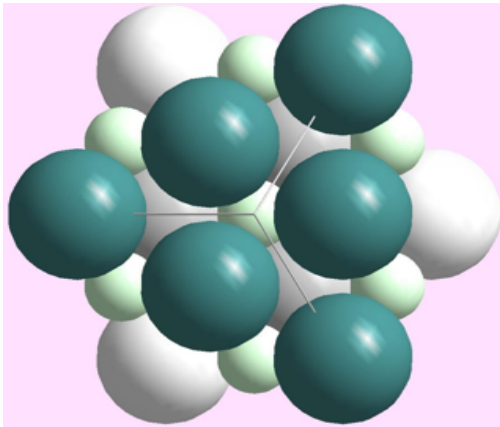
- The (100) surface is electrically neutral and is 'non polar' → equal number of Na^+ and Cl^- ions



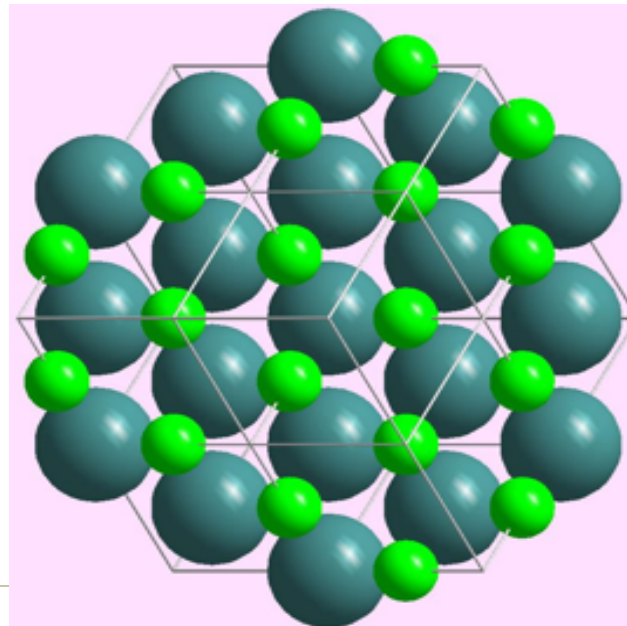
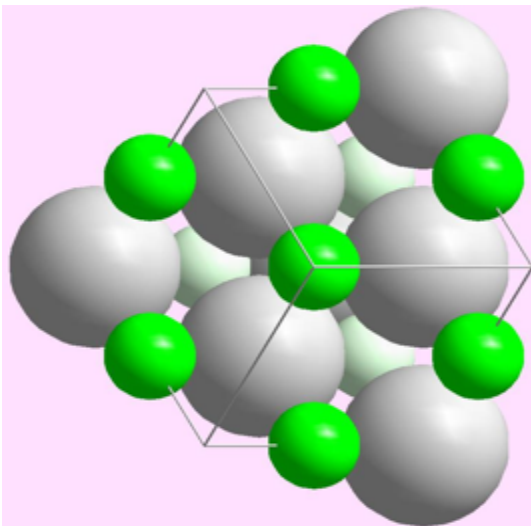
The non-polar (110) surface

More views & aspects: (111) surface

- ❑ The (111) surface is electrically charged and is 'Polar'
→ The Na^+ terminated surface is +vely charged and the Cl^- terminated surface is -vely charged.

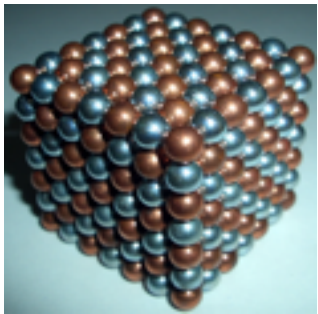


Two layers of ions:
(layers below that are hidden for clarity)
dark green (Cl^-) ions are on the surface

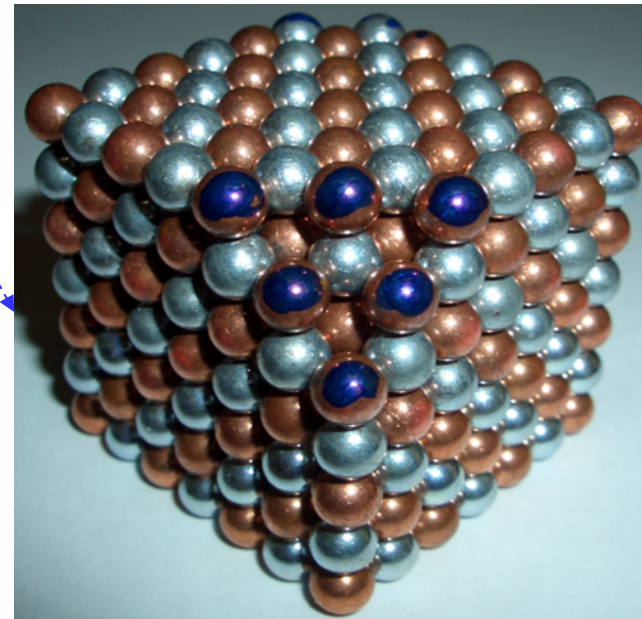


Two layers of ions:
(layers below that are hidden for clarity)
bright green (Na^+) ions are on the surface

Bright green are surface ions and rest
are sub-surface

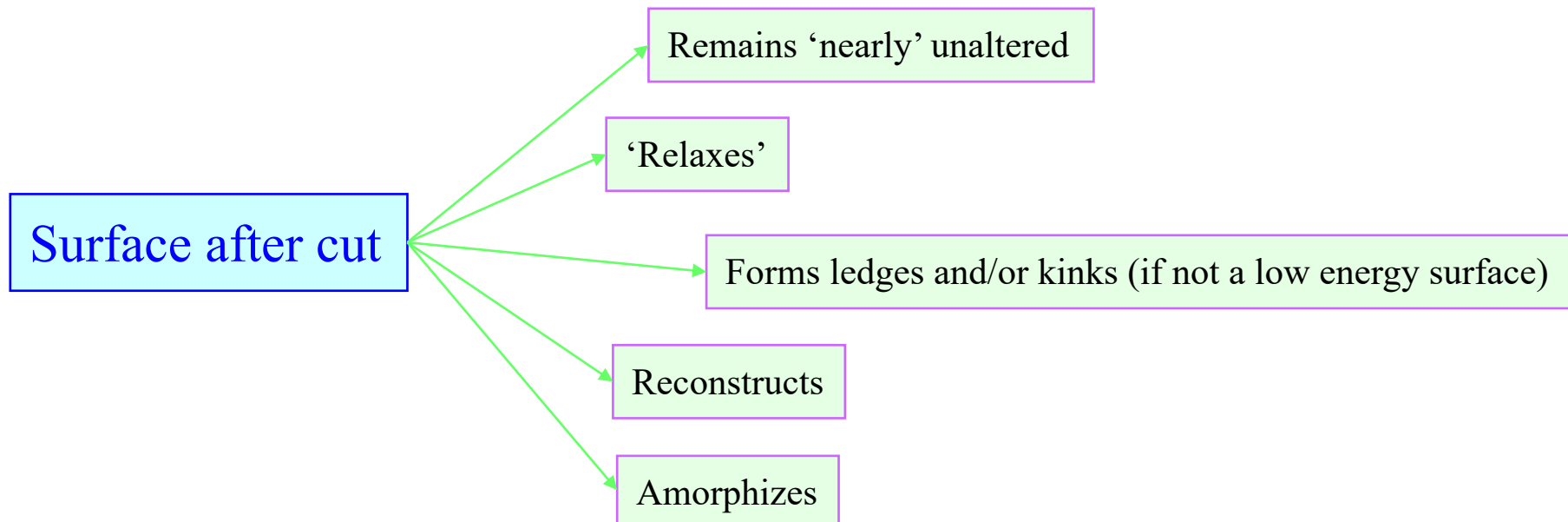


Two (111) cuts



What can happen after the cut?

- ❑ After the cut is made, the surface usually does not remain merely as a termination of the bulk.
- ❑ The surface can alter its structure in a few ways so as to reduce the energy. Some of the possibilities are shown below.
- ❑ First we treat the surface as unaltered before considering briefly surface relaxation and reconstruction.



Calculation of the surface energy (case of unaltered surface)

Surface Energy/
unit area (J/m²)

$$\gamma = \frac{n_a \cdot n_b \cdot E_b}{2}$$

No. of atoms/unit area
 No. of bonds broken/atom
 Bond energy/bond

As two surfaces are created / bond broken

Understanding
the units

$$LHS \rightarrow \frac{[J]}{[m^2]} = \frac{[Nm]}{[m^2]} = \frac{[N]}{[m]}$$

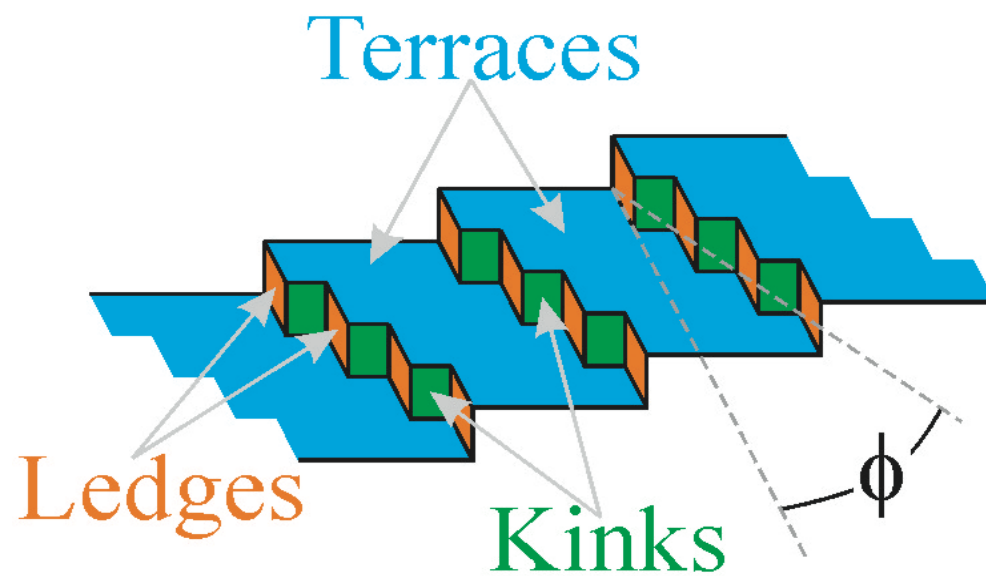
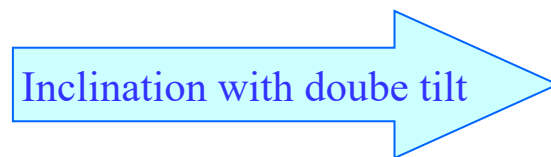
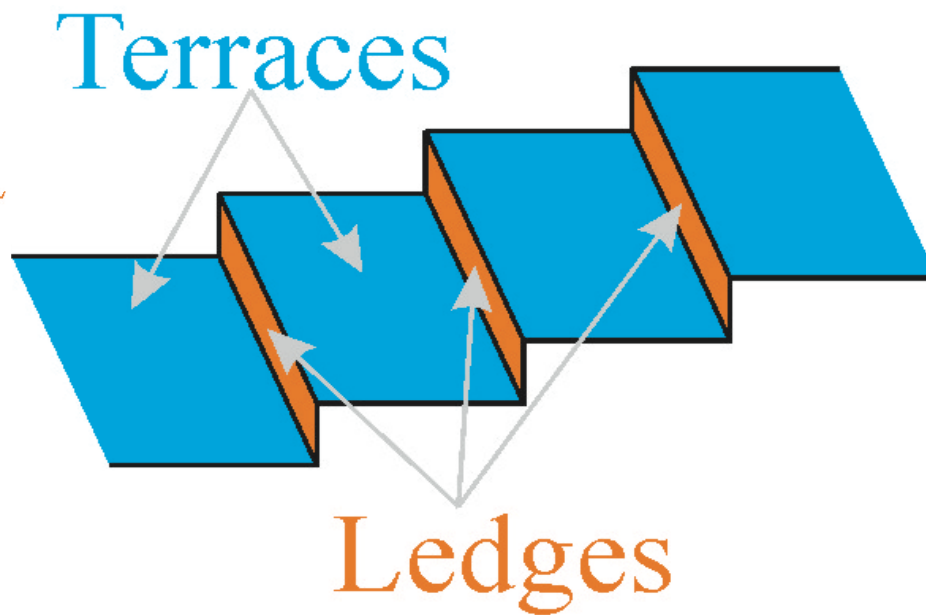
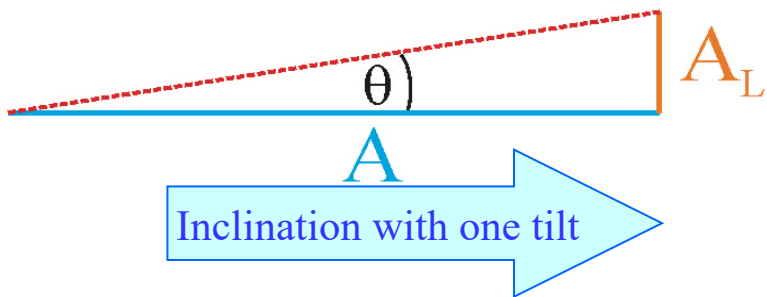
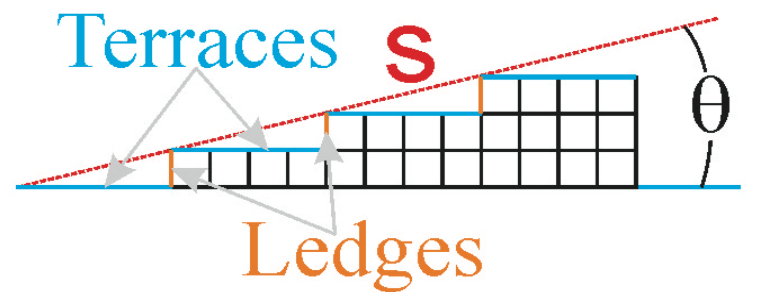
No of bonds broken

$$RHS \rightarrow \left(\frac{\#atoms}{[m^2]} \right) \left(\frac{\#bonds}{atom} \right) \left(\frac{[J]}{bond} \right) = \frac{[J]}{[m^2]} = \frac{[N]}{[m]}$$

Surface free energies (γ) of some crystals (J/m²)

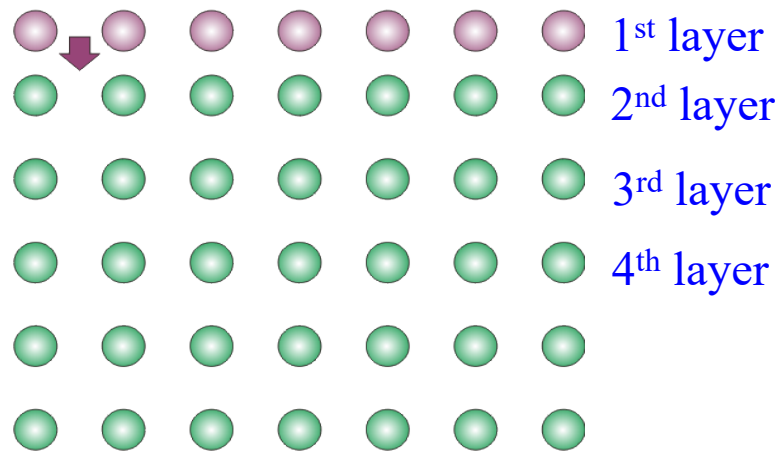
NaCl	LiF	CaF ₂	MgO	Si	Ag	Fe	Au	Cu
0.30	0.34	0.45	1.2	1.24	1.14	1.4	1.4	1.65

Note the values are ~ 1 J/m²

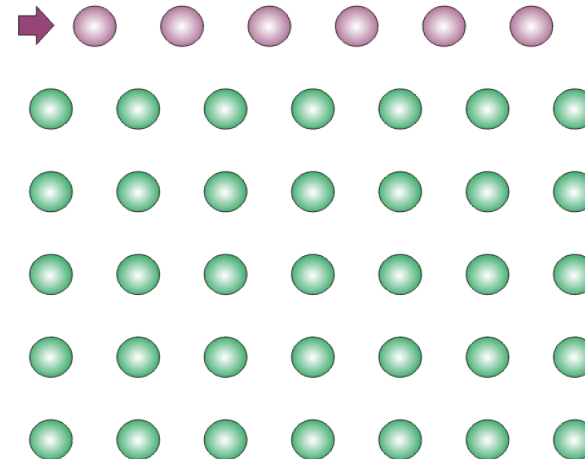


Surface Relaxation

Normal Relaxation



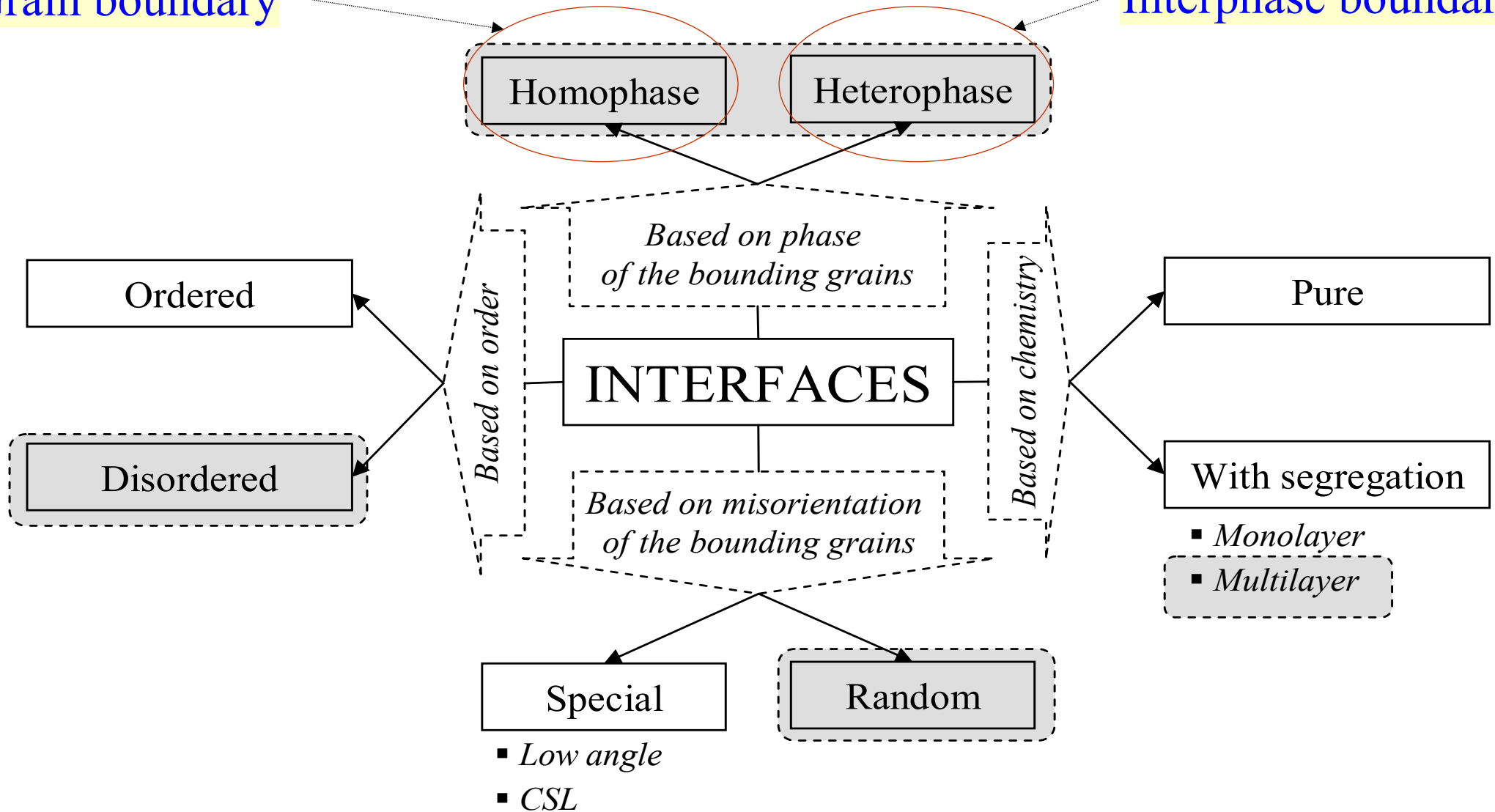
Lateral Relaxation



A classification of interfaces based on multiple viewpoints

Grain boundary

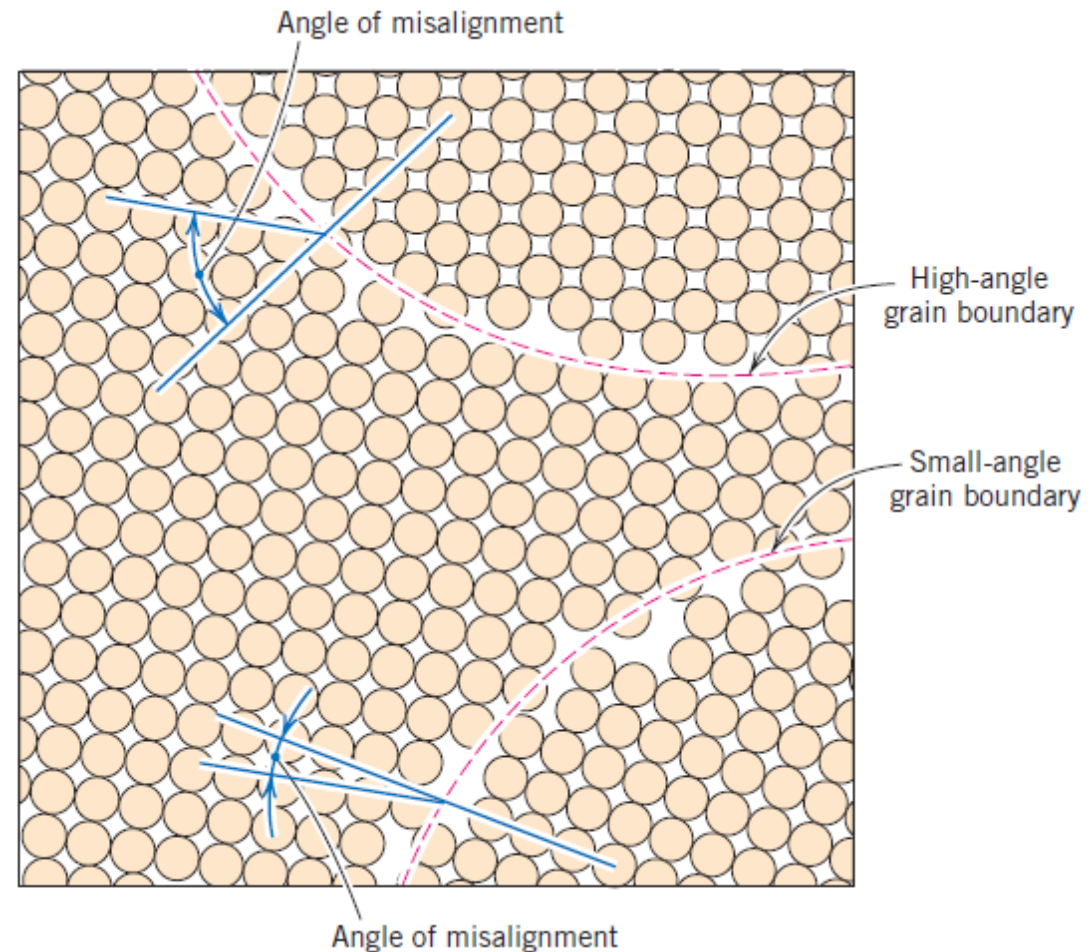
Interphase boundary



Interfacial Defects

Grain Boundaries

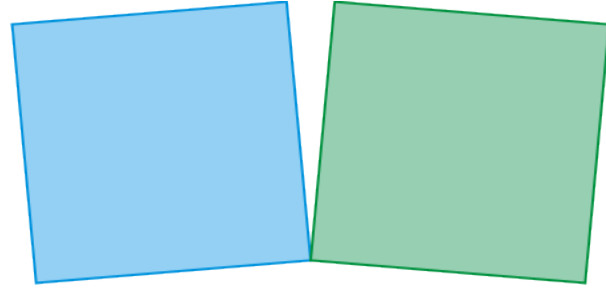
- In polycrystalline material some atomic mismatch in a transition from the crystalline orientation of one grain to that of an adjacent one.
- Various degrees of crystallographic misalignment between adjacent grains are possible.
- When this orientation mismatch is slight, on the order of a few degrees, then the term small- (or low-) angle grain boundary is used



- low density in grain boundaries
leading high mobility, high diffusivity,
high chemical reactivity

Structural dislocations can accommodate angular (tilt, twist or a combination) or linear misfits

Accommodated by an array of edge dislocations



Tilt

Angular misfit

Twist

Linear misfit

Misfit

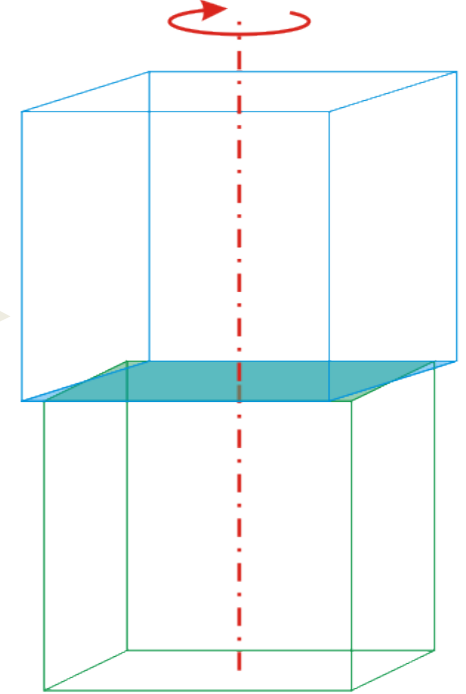
Structural dislocations can
accommodate

between two crystals



Stretch and fit

Axis of Twist



Accommodated by an array of
screw dislocations

Low angle grain boundaries (misorientation angle $< 10^\circ$)

Two extremes

TILT

An array of edge dislocations

TWIST

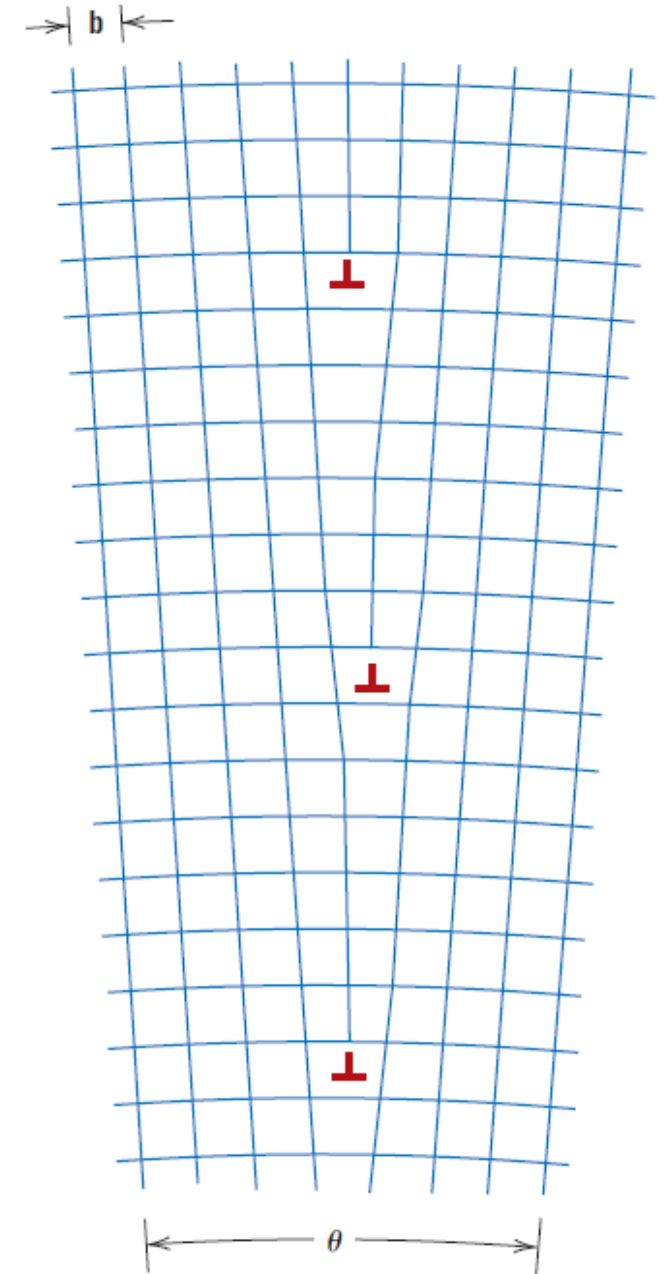
An array of screw dislocations

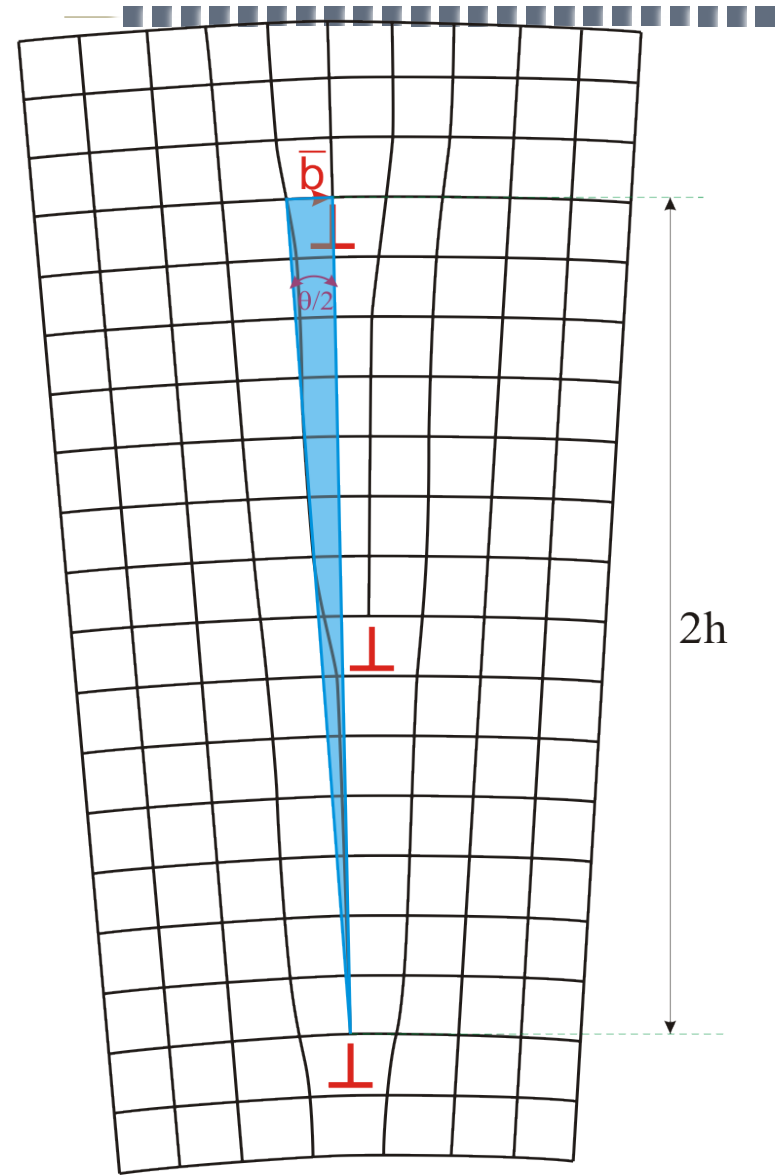
- Rotation axis lies on the boundary plane

- Rotation axis lies $\perp$ to the boundary plane
-

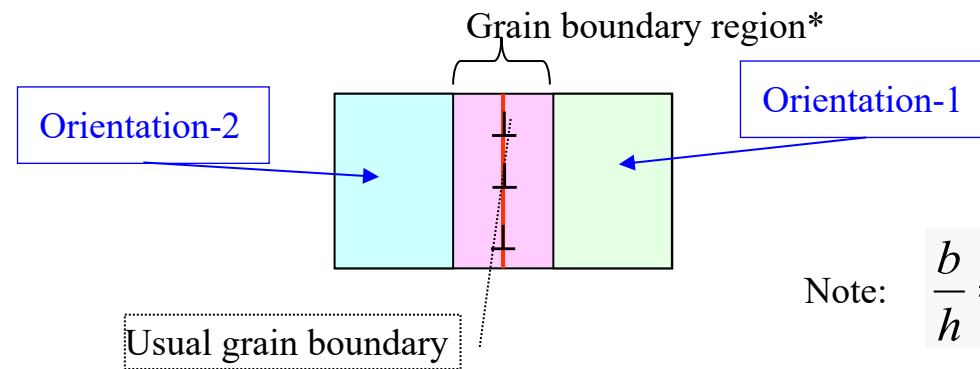
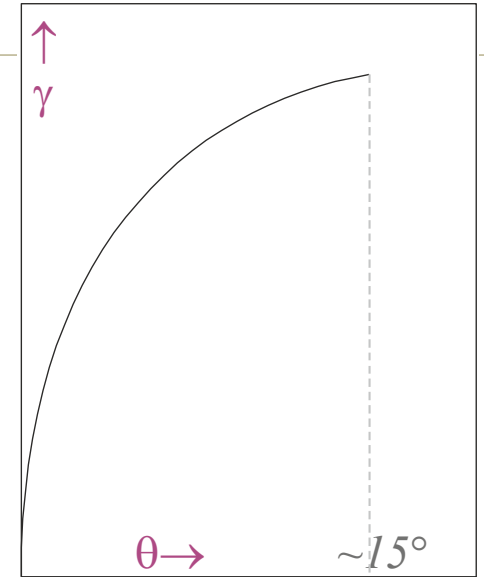
Interfacial Defects

- One simple small-angle grain boundary is formed when edge dislocations are aligned in the manner of Figure.
- This type is called a **tilt boundary**; the angle of misorientation (θ), is also indicated in the figure.
- When the angle of misorientation (θ), is parallel to the boundary, a **twist boundary** results, which can be described by an array of screw dislocations.





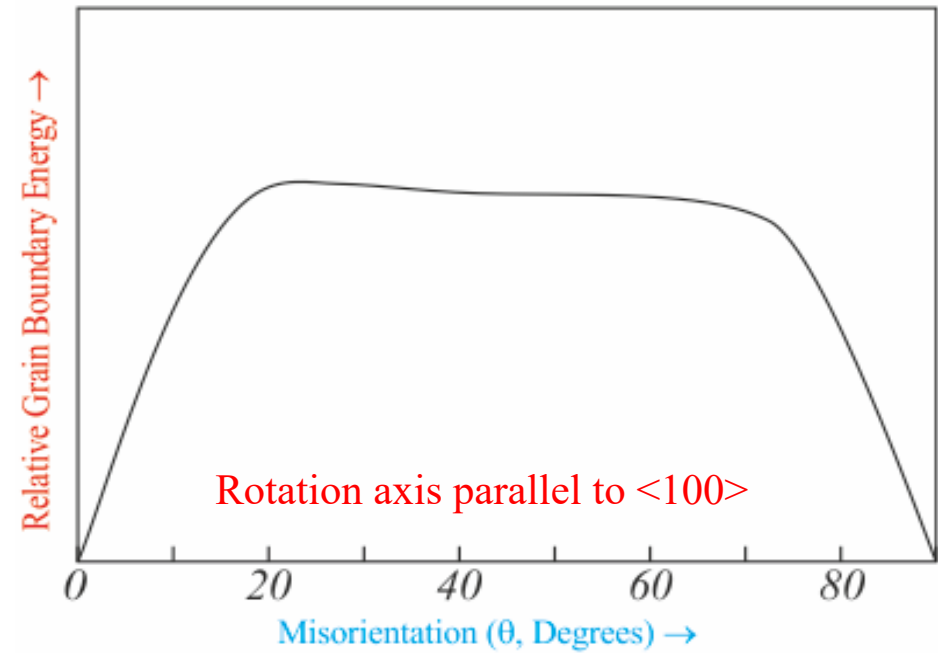
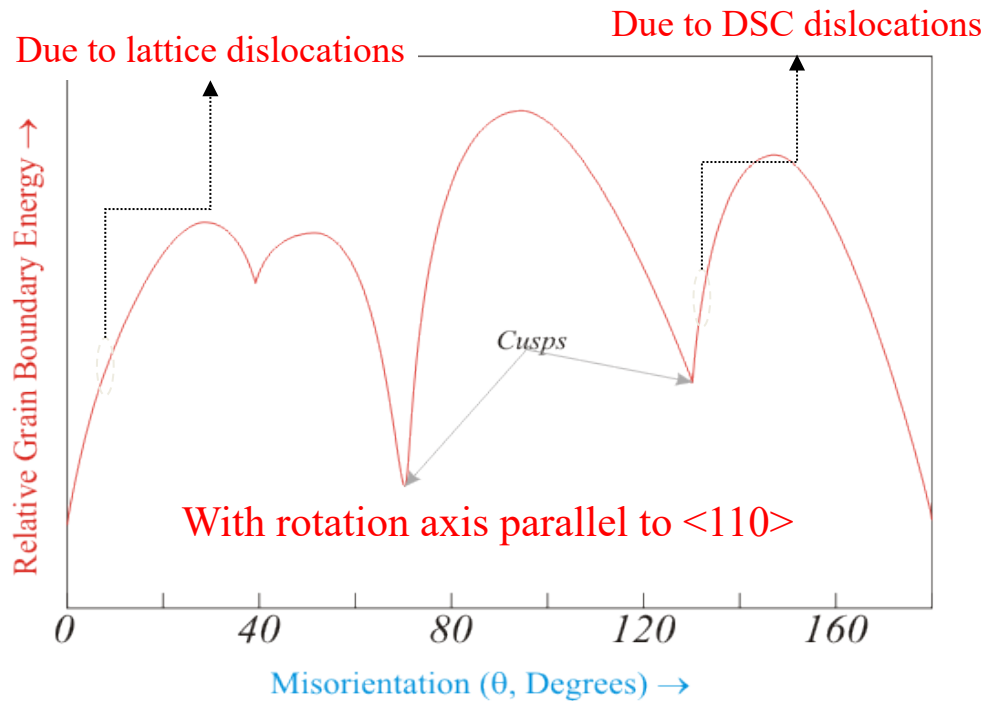
$$\frac{b}{2h} = \sin\left(\frac{\theta}{2}\right) \quad \frac{b}{h} \sim \theta$$



Note: $\frac{b}{h} = \tan \theta$

Mentioned in some texts- no physical basis

Grain boundary energy and its variation with misorientation



Interfacial Defects

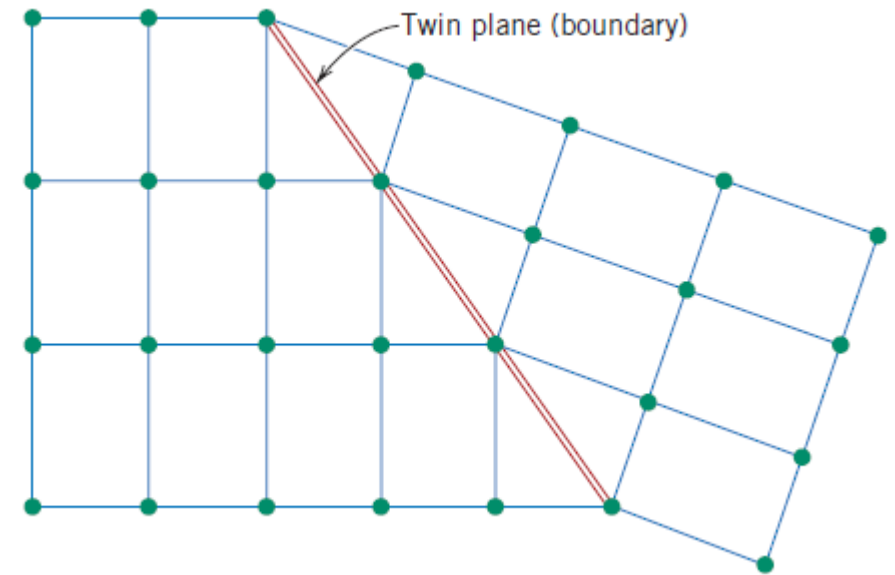
Phase Boundaries

- exist in multiphase materials, wherein a different phase exists on each side of the boundary;
- furthermore, each of the constituent phases has its own distinctive physical and/or chemical characteristics.
- phase boundaries play an important role in determining the mechanical characteristics of some multiphase metal alloys.

Interfacial Defects

Twin Boundaries is a special type of grain boundary across which there is a specific mirror lattice symmetry; that is, atoms on one side of the boundary are located in mirror-image positions of the atoms on the other side.

The region of material between these boundaries is appropriately termed a **twin**.



- Twins result from atomic displacements that are produced
 - from applied mechanical shear forces (mechanical twins), and
 - also during annealing heat treatments following deformation (annealing twins)

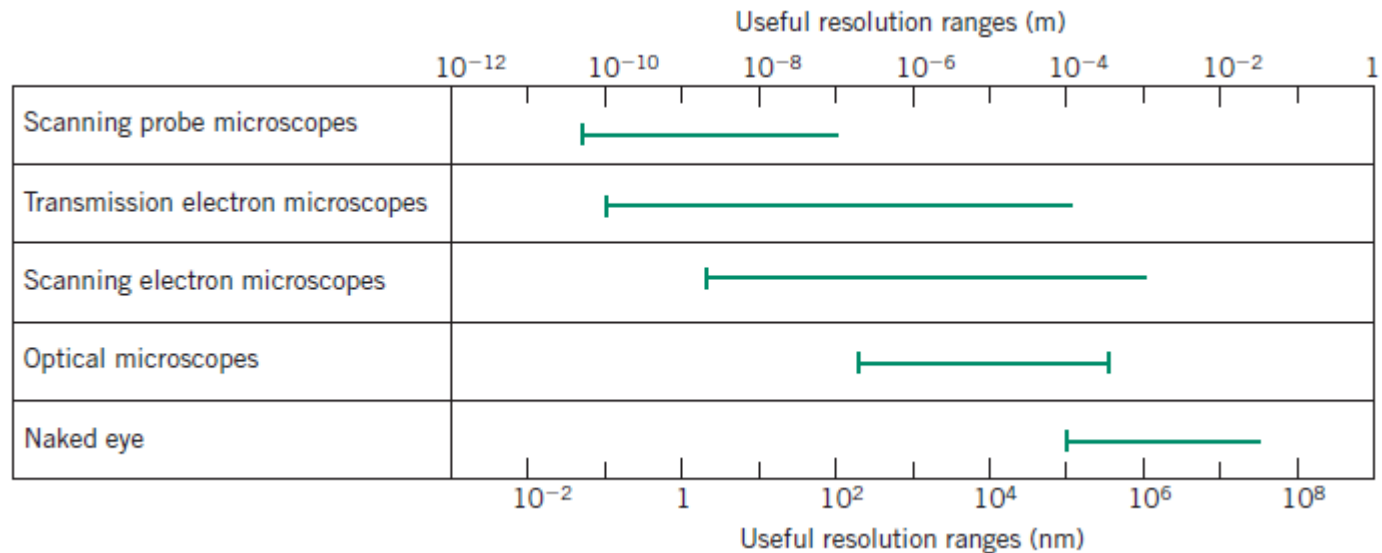
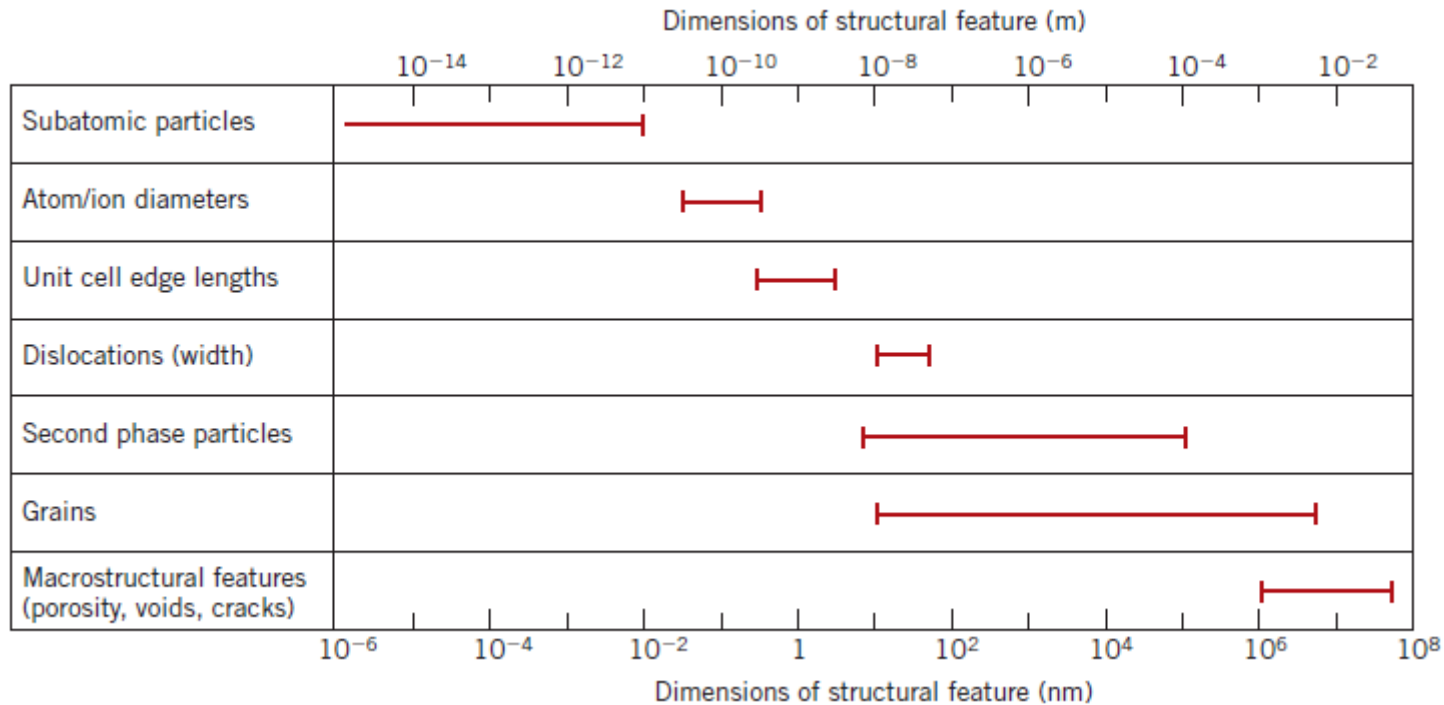
Bulk or Volume Defects

- Other defects exist in all solid materials that are much larger.
 - include pores,
 - cracks,
 - foreign inclusions, and
 - other phases.

They are normally introduced during processing and fabrication steps.

- Casting blow holes, porosity – Gas entrapment during melting and pouring. Improper welding parameters/practice
- Shrinkage cavity due to improper risering
- Non-metallic inclusions – Slag, oxide particles or sand entrapment
- Cracks – Uneven heating/cooling, thermal mismatch, constrained expansion/contraction all leading to stress development

Microscopic Examination



Grain Size Determination

- The grain size is often determined when the properties of a polycrystalline material are under consideration.
- Grains are classified based on the size (i) Coarse Grain (ii) Fine grain
- The **ASTM (American Society for Testing and Materials)** has prepared several standard comparison charts, all having different average grain sizes. To each is assigned a number ranging from 1 to 10, which is termed the ***grain size number***. (Coarse 1-5 and fine 5-8.)
- The number is derived from the formula

$$n = 2^{N-1}$$

where n = the number of grains per square inch at 100X magnification

N = ASTM Grain Size number

Grain Size Determination

EXAMPLE PROBLEM 4.4

Computations of ASTM Grain Size Number and Number of Grains per Unit Area

- (a) Determine the ASTM grain size number of a metal specimen if 45 grains per square inch are measured at a magnification of 100×.
(b) For this same specimen, how many grains per square inch will there be at a magnification of 85×?

Solution

(a) In order to determine the ASTM grain size number (n) it is necessary to employ Equation 4.16. Taking logarithms of both sides of this expression leads to

$$\log N = (n - 1) \log 2$$

Solving for n yields

$$n = \frac{\log N}{\log 2} + 1$$

From the problem statement, $N = 45$, and therefore

$$n = \frac{\log 45}{\log 2} + 1 = 6.5$$

(b) At magnifications other than 100×, use of the following modified form of Equation 4.16 is necessary:

$$N_M \left(\frac{M}{100} \right)^2 = 2^{n-1} \quad (4.17)$$

In this expression N_M = the number of grains per square inch at magnification M . In addition, the inclusion of the $(M/100)^2$ term makes use of the fact that, while magnification is a length parameter, area is expressed in terms of units of length squared. As a consequence, the number of grains per unit area increases with the square of the increase in magnification.

Solving Equation 4.17 for N_M , realizing that $M = 85$ and $n = 6.5$, leads to

$$\begin{aligned} N_M &= 2^{n-1} \left(\frac{100}{M} \right)^2 \\ &= 2^{(6.5-1)} \left(\frac{100}{85} \right)^2 = 62.6 \text{ grains/in.}^2 \end{aligned}$$

Summary

Vacancies and Self-Interstitials

- Point defects are those associated with one or two atomic positions; these include vacancies (or vacant lattice sites) and self-interstitials (host atoms that occupy interstitial sites).
- The equilibrium number of vacancies depends on temperature according to Equation.

Impurities in Solids

- An alloy is a metallic substance that is composed of two or more elements.
- A solid solution may form when impurity atoms are added to a solid, in which case the original crystal structure is retained and no new phases are formed.
- For substitutional solid solutions, impurity atoms substitute for host atoms.
- Interstitial solid solutions form for relatively small impurity atoms that occupy interstitial sites among the host atoms.
- For substitutional solid solutions, appreciable solubility is possible only when atomic diameters and electronegativities for both atom types are similar, when both elements have the same crystal structure, and when the impurity atoms have a valence that is the same as or less than the host material.

Summary

Specification of Composition

- Composition of an alloy may be specified in weight percent (on the basis of mass fraction) or atom percent (on the basis of mole or atom fraction).
- Expressions were provided that allow conversion of weight percent to atom percent and vice versa.
- Computation of average density and average atomic weight for a two-phase alloy are possible using other equations.

Dislocations—Linear Defects

- Dislocations are one-dimensional crystalline defects of which there are two pure types: **edge and screw**.
 - An edge may be thought of in terms of the lattice distortion along the end of an extra half-plane of atoms.
 - A screw is as a helical planar ramp.
 - For mixed dislocations, components of both pure edge and screw are found.
- The magnitude and direction of lattice distortion associated with a dislocation are specified by its Burgers vector.
- The relative orientations of Burgers vector and dislocation line are (1) perpendicular for edge, (2) parallel for screw, and (3) neither perpendicular nor parallel for mixed.

Summary

Interfacial Defects

- Within the vicinity of a grain boundary (which is several atomic distances wide), there is some atomic mismatch between two adjacent grains that have different crystallographic orientations.
- For a high-angle grain boundary, the angle of misalignment between grains is relatively large; this angle is relatively small for small-angle grain boundaries.
- Across a twin boundary, atoms on one side reside in mirror-image positions of atoms on the other side.

Equation Summary

<i>Equation</i>	<i>Solving For</i>
$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$	Number of vacancies per unit volume
$N = \frac{N_A \rho}{A}$	Number of atomic sites per unit volume
$C_1 = \frac{m_1}{m_1 + m_2} \times 100$	Composition in weight percent
$C'_1 = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100$	Composition in atom percent
$C'_1 = \frac{C_1 A_2}{C_1 A_2 + C_2 A_1} \times 100$	Conversion from weight percent to atom percent
$C_1 = \frac{C'_1 A_1}{C'_1 A_1 + C'_2 A_2} \times 100$	Conversion from atom percent to weight percent
$C''_1 = \left(\frac{\frac{C_1}{\rho_1}}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}} \right) \times 10^3$	Conversion from weight percent to mass per unit volume
$\rho_{ave} = \frac{100}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}}$	Average density of a two-component alloy
$A_{ave} = \frac{100}{\frac{C_1}{A_1} + \frac{C_2}{A_2}}$	Average atomic weight of a two-component alloy
$N = 2^{n-1}$	Number of grains per in. ² at 100× magnification